LESSON 1

# 1. What is Economics?

## Lesson overview

This lesson introduces economics as the study of how societies use limited resources to meet unlimited wants. It explains the concepts of scarcity, choice, and the basic way economists think about problems.

## Learning outcomes

- Define economics in your own words
- Explain why scarcity forces people to make choices
- Identify examples of trade■offs in everyday life
- Describe the basic steps of economic reasoning

## 1.1 What Is Economics?

Economics is the study of how people, families, businesses, and governments make choices when they have limited resources but many wants.

Think of a **resource** as anything that can be used to satisfy a need or desire - money, time, land, labor, or even a favorite video■game console. A **want** is simply something people would like to have, such as a new phone, a vacation, or a bigger paycheck. Because resources are limited while wants are endless, we must decide **what to do with what we have**. That decision■making process is what economics tries to understand.

> **Definition:** *Economics = the study of choices about scarce resources to satisfy unlimited wants.*

Why does this matter? If we can see the pattern behind everyday decisions, we can predict outcomes, avoid waste, and improve well■being. For example, a family with a fixed monthly income must decide how much to spend on groceries, rent, entertainment, and savings. By looking at those choices, economics can tell us whether the family is likely to run out of money, save enough for emergencies, or feel satisfied with their lifestyle.

### *A simple comparison*

Imagine you have a small bag of candy that you want to share with three friends. The candy is the **resource**; the desire of each friend to get candy is the **want**. You cannot give everyone the whole bag, so you must choose how many pieces each friend receives. Your decision depends on factors like how hungry each friend is, whether someone will trade you a favor later, or if you want to keep a piece for yourself. This tiny scenario mirrors the larger world: societies allocate land, labor, and capital to produce food, clothing, and services for their citizens.

### *Breaking down new words*

- **Scarcity** - the condition of having fewer resources than the total amount of wants. It is not about poverty alone; even wealthy societies face scarcity because they cannot produce everything they desire.
- **Choice** - the act of selecting one option over another when more than one is possible.
- **Trade■off** - what you give up when you make a choice. In the candy example, giving a larger piece to one friend means giving a smaller piece to another.

When economists talk about "the problem of scarcity," they are really saying: *Because we cannot have everything, we must make choices, and every choice has a cost.* That cost is called an **opportunity cost**, which is the value of the next best alternative you give up.

> **Note:** *Opportunity cost is not always measured in money; it can be time, enjoyment, or any other benefit you forgo.*

### *Building the idea step by step*

- **Identify the resources** - What do you have? Money, time, skills, natural assets?
- **List the wants** - What do you or others desire? More goods, better services, leisure?
- **Recognize scarcity** - Compare the amount of resources to the total of wants. If resources < wants, scarcity exists.
- **Make a choice** - Decide how to allocate resources among the wants.
- **Consider the trade■off** - Ask, "What am I giving up by choosing this option?"

Each step follows logically from the previous one; skipping any would leave the reasoning incomplete.

> **In daily life:** *When you decide to study for a test instead of watching a movie, the movie time you miss is the opportunity cost of studying.*

### *Why economists care about these steps*

Economists use the same logic to analyze big■scale issues like why a country imports oil, why a company raises prices, or why a government builds a new highway. By treating every decision as a choice under scarcity, they can create models that predict how changes in prices, technology, or policies will shift behavior.

For instance, if the price of gasoline rises, drivers may choose to drive less, car■pool, or buy more fuel■efficient cars. The **equation** that captures this relationship in a simple form is:

\text{Quantity Demanded} = f(\text{Price})

where the function f shows that quantity demanded usually **decreases** as price **increases**. This basic idea stems directly from the definition of economics as the study of choices under scarcity.

> **Remember:** *Economics starts with a simple observation - resources are limited - but it grows into a powerful tool for understanding almost every decision we make.*

[Figure to be added: Land, labor, and capital used to produce goods and services.]

Figure 1.1. Land, labor, and capital used to produce goods and services.

[Figure to be added: Limited pizza slices versus many hungry friends.]

Figure 1.2. Limited pizza slices versus many hungry friends.

[Figure to be added: Choosing between a video game and studying for a test.]

Figure 1.3. Choosing between a video game and studying for a test.

[Figure to be added: Three Main Resources]

Figure 1.4. Three Main Resources

[Figure to be added: Key Idea of Scarcity]

Figure 1.5. Key Idea of Scarcity

[Figure to be added: Elements of a Choice]

Figure 1.6. Elements of a Choice

[Figure to be added: Basic Economic Thinking Process]

Figure 1.7. Basic Economic Thinking Process

## 1.2 Resources and Wants

Resources are the things we use to make the things we want.

In economics we usually talk about three kinds of resources:

- **Land** - all natural gifts such as soil, water, minerals, forests and the air we breathe.
- **Labor** - the effort that people put into work, whether it is a farmer planting crops or a programmer writing code.
- **Capital** - the tools, machines, buildings and technology that help labor turn land into products.

> **Definition:** *Resources* are the inputs (land, labor, capital) that are used to produce goods and services.

A **want** is simply something a person would like to have. Wants can be as small as a cup of coffee in the morning or as large as a new smartphone, a college education, or a safe neighbourhood. The key idea in economics is that **human wants are unlimited** - we can always imagine something else we would like to enjoy or improve.

Why do we say wants are unlimited? Think about a child who just learned to ride a bike. After mastering the first bike, the child soon wants a faster bike, then a helmet, then a bike lock, then a bike for a sibling. There is always a new desire that appears once the previous one is satisfied. This pattern repeats for adults too: after buying a comfortable sofa, we might want a bigger TV; after getting the TV, we might want a better sound system, and so on. Because people can always imagine a better or different option, the list of wants never ends.

Why is this important? If wants were limited, we could simply give everyone everything they asked for and there would be no need to decide how to use resources. Because wants keep growing while resources stay the same, we must make choices about who gets what, when, and how much. This is the heart of economics: **scarcity** (limited resources) forces us to **prioritize** (choose among unlimited wants).

### *An everyday comparison*

Imagine a small kitchen with only one stove, two pots, and a single chef. The chef wants to prepare breakfast, lunch, and dinner for the whole family today. The stove and pots are the **resources** - there are only so many of them, and they can be used for only one dish at a time. The meals the family hopes to eat are the **wants** - they are many and varied. The chef must decide the order of cooking, perhaps preparing a quick omelet for breakfast, a soup for lunch, and a stew for dinner. The decision is forced by the limited kitchen tools, not by a lack of desire to eat.

> **In daily life:** When you have a limited amount of money (a resource) but many things you would like to buy (wants), you must decide which items to purchase first.

### *Building the idea step■by■step*

- **Identify the resources** - list the land, labor, and capital you have.
- **List the wants** - write down everything you would like to have or achieve.
- **Compare quantities** - notice that the total amount of each resource is a fixed number (e.g., 5 hours of work time, 20, one garden plot).
- **Notice the gap** - the number of wants usually exceeds the amount of resources. This gap is the **scarcity gap**.

When the scarcity gap exists, we must make **trade■offs**: choosing one option means giving up another. For example, spending 20 on a movie ticket means you cannot spend that same20 on a new book. The concept of trade■offs follows directly from the distinction between limited resources and unlimited wants.

> **Remember:** Unlimited wants + limited resources = the need to make choices.

A common mistake is to think that scarcity only matters for big things like national budgets. In reality, scarcity shows up in tiny daily decisions, such as how you allocate your study time between math and history, or whether you use the last slice of pizza for yourself or share it with a friend. Recognizing the scarcity of resources helps you become a more thoughtful decision■maker.

> **Common mistake:** Assuming that because a resource feels abundant today (e.g., a full pantry), it will stay abundant forever. Resources can become scarce if demand rises or supply falls.

Understanding that resources are limited while wants are endless sets the stage for all later economic ideas: pricing, markets, and policies that try to allocate resources efficiently. By keeping this simple distinction in mind, you can see the hidden economics in every choice you make.

## 1.3 Scarcity Explained

Scarcity means that the things we need or want are not unlimited. In plain words, **scarcity** is the condition that there is not enough of a resource - such as food, time, money, or even clean air - to give everyone exactly what they would like.

Why does this matter? If everything were abundant, we would never have to decide how to use it. No choices would be necessary, and the whole idea of economics - studying how people allocate limited resources - would disappear. Because scarcity is real, we must constantly

**choose** what to do with what we have.

> **Definition:** *Scarcity* is the gap between limited resources and unlimited human wants.

Think of a simple everyday comparison: imagine a small pizza that you and three friends want to share. The pizza is a **resource** (something that can be used to satisfy a want). Your friends' desire for pizza is a **want** (something you would like to have). Since the pizza is only one whole, it cannot give each person a full slice unless you cut it. The fact that the pizza is not enough to give everyone exactly what they want illustrates scarcity.

Let's break down the new words:

- **Resource** - any material, service, or ability that can be used to satisfy a need or want (e.g., money, time, land, labor).
- **Want** - a desire for something that is not currently possessed; wants are virtually endless because human preferences keep changing.
- **Choice** - the act of selecting one option over another when faced with limited resources.

When scarcity exists, we face **trade■offs**: getting more of one thing means giving up some of another. Economists often write this relationship as an **opportunity cost** equation:

\text{Opportunity Cost} = \text{Value of the best alternative forgone}

In the pizza example, if you decide to give yourself two larger slices, the **opportunity cost** is the smaller slices your friends receive. The value of those missed slices is the cost of your choice.

Why is it useful to recognise scarcity? First, it forces us to think about priorities. Second, it helps us understand why prices exist: when a resource is scarce, people are willing to pay more to obtain it, and sellers can charge higher prices. Third, it explains why governments and businesses must plan - if they ignored scarcity, they would waste resources and create shortages.

> **In daily life:** When you have 20 and want to buy both a new book and a movie ticket, you must decide which purchase gives you more satisfaction. The20 is scarce; the two items are wants. Your decision shows scarcity at work.

A common mistake is to think that scarcity only applies to money. In reality, **time** is a classic scarce resource. Everyone has only 24 hours each day, yet we have countless activities we would like to do - studying, exercising, playing video games, sleeping, and hanging out with friends. Because time cannot be increased, we must schedule it, showing scarcity in a non■monetary form.

> **Common mistake:** Assuming that "plenty" means "no scarcity." Even abundant items become scarce when the number of people who want them grows faster than the supply (e.g., a popular concert ticket).

Finally, remember that scarcity does not mean "nothing is available." It simply means **there is not enough of a resource to satisfy every desire completely**. This gap creates the need for **allocation** - the process of deciding who gets what, how much, and when. Allocation can be done by markets (through prices), by governments (through rules), or by families (through agreements).

> **Remember:** Scarcity is the engine of economics; it turns wants into choices, and choices into trade■offs. Recognising scarcity helps you make better decisions, whether you are budgeting pocket money, planning a study schedule, or voting on public policies.

## 1.4 Choice and Trade■offs

Choosing one thing means giving up something else.

When you decide to spend your time, money, or effort on one option, the next best thing you could have done is left behind. Economists call that "the cost of the next best alternative" **opportunity cost**.

> **Definition:** *Opportunity cost* is the value of the best alternative that you forgo when you make a choice.

Why does this idea matter?

Every decision uses a limited resource - usually time, money, or a physical good. Because resources are limited, you cannot do everything you want. Recognizing what you give up helps you compare options more clearly and avoid hidden regrets.

### *Step■by■step illustration*

- **Identify the resource** - Suppose you have 10 to spend on a snack.
- **List the possible uses** - You could buy a chocolate bar, a fruit cup, or save the money for later.
- **Pick one option** - You choose the chocolate bar.
- **Ask the question:** What did you give up? The fruit cup (or the saved money) is the next best thing you could have done.
- **Assign a value** - If the fruit cup would have made you just as happy as the chocolate, its value is the opportunity cost of buying the chocolate.

In symbols, if you choose option A and the next best alternative is B, the opportunity cost (OC) is

OC = \text{Value of B}

If you think of "value" as satisfaction (often called *utility* in economics), the formula becomes

OC = U_B, where U_B is the utility you would have received from B.

### *Everyday comparison*

Imagine you have a Saturday afternoon free. You could (A) study for an upcoming test, (B) watch a movie, or (C) play basketball with friends. If you decide to study, the opportunity cost is the enjoyment you would have gotten from the movie **or** the fun of playing basketball - whichever you liked more.

> **In daily life:** When you choose to study, the opportunity cost might be the extra points you could have earned by watching the movie and relaxing, which could improve your mood later.

### *Why the concept is useful*

- **Prioritising** - By comparing opportunity costs, you can see which choice gives you the greatest net benefit.
- **Budgeting** - Money is a classic scarce resource. Knowing the opportunity cost of a purchase prevents impulse buying.
- **Time management** - Time cannot be stored; every hour spent on one activity is an hour lost for another.

### *Common mistake*

> **Common mistake:** Thinking that "cost" only means the price you pay in dollars.

Many people forget that the true cost includes what you give up, not just the money you spend. For example, buying a cheap fast■food meal may seem inexpensive, but the opportunity cost could be poorer health or lower energy later.

### *A simple numeric example*

You have 4 hours of free time. Option 1: work a part■time job paying 15 per hour. Option 2: volunteer at a local charity, gaining experience valued at20 per hour (in future job prospects).

If you work, you earn 60.

If you volunteer, you earn 0 now but gain80 worth of future benefit.

The opportunity cost of working is the 80 of future benefit you miss, while the opportunity cost of volunteering is the60 you could have earned today.

> **Remember:** Always ask, "What am I giving up?" after you pick an option. That answer is the opportunity cost, and it tells you whether the choice is worth it.

### *Summing up*

- Every choice uses a limited resource.
- The next best alternative you do not take is the **opportunity cost**.
- Calculating this cost helps you make smarter, more transparent decisions.

By habitually thinking about what you sacrifice, you become a more careful planner of your money, time, and effort - a skill that will serve you well in school, work, and everyday life.

## 1.5 Basic Economic Thinking

Economists solve problems the way a detective solves a mystery: they start with a clear question, gather clues, weigh the evidence, and then choose the best answer.

The "detective method" in economics is called **basic economic thinking**. It is a simple, repeatable process that helps anyone make smarter choices about money, time, or any limited resource.

> **Definition:** *Basic economic thinking* is the step■by■step way economists (and anyone) decide what to do when resources are scarce. The steps are:

> 1. Identify the problem.

> 2. List possible alternatives.

3. Compare the costs and benefits of each alternative.

4. Choose the alternative with the greatest net benefit.

### *1. Identify the problem*

First, you must know exactly what you are trying to solve. A vague problem like "I need more money" is hard to tackle. Instead, turn it into a specific question: *How can I afford a new bike this month?*

Why this matters: if the problem is unclear, you may waste time on solutions that don't help. Think of a doctor who tries to treat a fever without first checking whether it's caused by an infection or heatstroke. The treatment could be wrong.

### *2. Consider alternatives*

Next, write down every realistic way you could address the problem. For the bike example, alternatives might be:

- Save part of your weekly allowance.
- Do extra chores for extra pay.
- Borrow money from a friend.
- Wait and buy a cheaper second■hand bike later.

Listing alternatives forces you to look beyond the first idea that pops into your head. It also reveals options you might have missed, like a discount sale.

### *3. Weigh costs and benefits*

Now you compare each alternative by looking at **costs** (what you give up) and **benefits** (what you gain). Economists call this the **cost■benefit analysis**.

- *Cost* includes money, time, effort, and any other sacrifice.
- *Benefit* includes the positive outcome you expect, such as enjoyment from riding the bike.

You can even write a simple equation for each option:

Net Benefit = Total Benefits - Total Costs

If the net benefit is positive, the option is worth considering; the larger the net benefit, the more attractive the option.

**Example:**

Saving allowance:

- Benefit = bike price 200 (future enjoyment).
- Cost = 20 per week for 10 weeks =200 (no spending on other treats).
- Net Benefit = 200 -200 = 0 (you get the bike but give up all other fun for 10 weeks).

Doing extra chores:

- Benefit = 200 bike.
- Cost = 5 extra hours per week x 5/hour =25 per week x 4 weeks = 100 (lost leisure time).
- Net Benefit = 200 -100 = 100 (you still have100 worth of free time left).

### *4. Make a decision*

Finally, pick the alternative with the highest net benefit, *provided* it is realistic and fits your values. In the example, doing extra chores gives a net benefit of 100, which is better than the zero net benefit of just saving allowance. So the logical decision is to earn extra money by working.

**In daily life:** When you decide whether to study for a test or watch a movie, you are using basic economic thinking. The "problem" is getting a good grade. The "alternatives" are studying now or later, or not studying at all. You weigh the cost (lost entertainment) against the benefit (higher grade, future opportunities) and choose the option with the best overall payoff.

**Common mistake:** Forgetting to count *opportunity cost*. Opportunity cost is the value of the best thing you give up when you choose one option. In the bike case, the opportunity cost of saving allowance is the fun you miss out on during those weeks. Ignoring this cost can make an option look better than it really is.

**Remember:** Economic thinking is not about being greedy; it is about using limited resources wisely. By following the four steps - identify, list, compare, decide - you turn a confusing choice into a clear, logical plan. This habit works for big decisions like choosing a college major and for tiny ones like deciding whether to buy a snack now or wait for lunch.

## 1.6 Everyday Examples of Economic Decisions

Imagine you have a small amount of money, a limited amount of time, and many things you would like to do. Economics is the study of how you decide what to get with those limited resources.

First, think of **resources** as anything you can use - cash, minutes in a day, or even your own energy. **Wants** are the things you would like to have or do, such as buying a new video game, learning a new language, or hanging out with friends. Because resources are limited while wants are unlimited, you must make choices. Every choice means giving up something else; this is called a **trade■off**.

### *Step■by■step example: spending pocket money*

- **Identify the resources** - you have 10 of pocket money this week.
- **List your wants** - you could buy a comic book (6), a snack (2), or save the money for a bigger purchase later.
- **Consider the trade■offs** - if you buy the comic, you will have only 4 left for snacks or saving. If you spend all on snacks, you won't have the comic.

> **Definition:** *Trade■off* - the loss of one benefit when you gain another because resources are limited.

- **Make a decision** - you decide to buy the comic and a small snack, leaving 2 to save. This decision reflects **opportunity cost**, the value of the next best alternative you gave up (in this case, the extra snack you could have bought).

> **Note:** *Opportunity cost* is not always measured in money; it can be time, enjoyment, or any other benefit you miss out on.

### *Choosing a school subject*

Suppose you must pick one elective for the next term. Your resources are **time** (you can only attend one class each period) and **energy** (you feel tired after a long day). Your wants might be:

- Learning a new language (interest in travel)
- Taking a computer■coding class (future job prospects)
- Joining a sports team (fun and fitness)

You cannot do all three because each class meets at the same time. You weigh the benefits:

- Language class: improves cultural knowledge, but you enjoy it less.
- Coding class: high future value, but it is challenging.
- Sports: great for health, but you already play after school.

After comparing, you choose the coding class because its **future benefit** outweighs the immediate fun of sports.

> **In daily life:** When you pick a subject, you are using **cost■benefit analysis** - weighing the advantages (benefits) against the disadvantages (costs) of each option.

### *Deciding how to spend free time*

Your weekend has 12 free hours. You could:

- Study for an upcoming test (2 hours)
- Watch a movie with friends (3 hours)
- Play a video game (2 hours)
- Help a neighbor with chores (1 hour)

Each activity gives you a different **utility** - a word economists use for the satisfaction or happiness you get from something. You might value studying the most because it helps your grades, but you also need relaxation.

A simple way to think about it is to assign a personal "happiness score" from 0 to 10 for each activity:

- Study: 8
- Movie: 7
- Game: 6
- Chores: 4

Now, multiply each score by the hours you plan to spend and add them up. The combination that gives the highest total score while staying within 12 hours is your optimal schedule.

Total utility = sum_i=1^n (happiness score_i times hours_i)

> **Remember:** The goal is not always to maximize total utility; sometimes you must respect other limits, like needing at least 7 hours of sleep.

### *Why this matters*

Understanding these everyday decisions shows how economics is not just about markets or governments; it is a way of thinking about any situation where you have limited resources and many possible wants. By recognizing trade■offs, opportunity costs, and utility, you become a more deliberate decision■maker.

> **Common mistake:** Assuming that "more is always better." In reality, adding one more activity can reduce overall satisfaction if it crowds out something you value more.

When you next decide whether to buy a snack, pick a class, or schedule your weekend, remember you are practicing economics. The same principles that guide nations also guide your pocket money. Use them, and you'll find it easier to make choices that feel right for you.

## 1.7 Summary and Reflection

We have just talked about what economics really means. In the simplest words, **economics is the study of how people use limited resources to satisfy unlimited wants**. This idea matters because every decision you make - what to eat for lunch, whether to buy a new video game, or how much time to spend studying - depends on what you have (money, time, skills) and what you would like to have.

Think of a **resource** as anything you can use: cash, a bike, an hour of free time, or even a piece of information. A **want** is any desire you have, such as wanting a tasty snack, a good grade, or a fun weekend outing. Because resources are **scarce** - there is not enough of them to give everyone everything they want - you must choose. Choosing one thing means giving up something else; this is called a **trade■off**. The value of the thing you give up is the **opportunity cost**. For example, if you spend 5 on a movie ticket, the opportunity cost might be the pizza you could have bought with that5.

> **Definition:** *Scarcity* means that the total amount of a resource is limited compared to the total amount of wants for that resource.

> **Note:** When we say "limited," we do not mean "zero." Even a tiny amount of a resource can be useful, but it can only be used in one way at a time.

> **In daily life:** If you have two hours free after school, you can either practice the piano or finish your homework. The time you spend on one activity is the **opportunity cost** of not doing the other.

> **Remember:** Economics does not tell you what to choose; it gives you a way to think about the consequences of each choice.

The basic economic thinking steps are:

- **Identify the resources** you have (money, time, skills).
- **List your wants** that compete for those resources.
- **Compare alternatives** and think about what you must give up for each option.
- **Choose** the option whose benefits you value most compared to its opportunity cost.

A useful shortcut is the simple equation that many economists write:

\text{Opportunity Cost} = \text{Value of the Next Best Alternative}

You do not need to solve complex math; just remember that the "cost" of a decision is what you lose by not picking the next best thing.

Now, let's turn the ideas inward. Think of a recent personal decision - perhaps you decided to skip a video■game session to study for a test, or you chose to buy a snack instead of saving the money for a concert ticket. Answer the following questions in a short paragraph (about 5■7 sentences):

- **What resources** did you have for this decision? (e.g., money, time, energy)
- **What were the competing wants**? List at least two things you wanted at the same time.
- **What was the trade■off** you faced? Identify the option you gave up.
- **What was the opportunity cost**? Explain why the thing you gave up mattered to you.
- **How does thinking like an economist help you see the decision differently?**

Write your reflection in a notebook or on a computer, and be honest about the feelings behind each choice. Seeing the hidden cost of a decision often makes future choices clearer, because you can weigh the benefits and the sacrifices more deliberately.

When you finish, read your paragraph aloud. Does it feel like you have captured the main ideas - scarcity, trade■offs, and opportunity cost? If something feels missing, add a sentence that links the decision back to the idea that **resources are limited, so every choice matters**. This practice will help you use economic thinking not just in class, but in everyday life.

### Terminology at a glance

| Term | Meaning |
|---|---|
| Economics | The study of choices about scarce resources to satisfy unlimited wants. |
| Resource | Anything used to satisfy a want, such as money, time, land, labor, or capital. |
| Want | A desire for something that is not essential but would improve well■being. |
| Scarcity | The condition that resources are limited while human wants are unlimited. |
| Opportunity Cost | The value of the best alternative that is given up when a choice is made. |
| Trade■off | Choosing one option means giving up another; the sacrifice involved in a decision. |
| Basic Economic Thinking | A step■by■step process: identify the problem, list alternatives, compare costs and benefits, choose the best net benefit. |
| Capital | Man■made tools, machines, buildings, and technology used to help labor turn land into products. |

### Practical application

Exercise 1.1:

Exercise 1.2:

Exercise 1.3:

Exercise 1.4:

Exercise 1.5:

Exercise 1.6:

### Checklist

- [ ] Define economics in your own words
- [ ] Explain why scarcity forces people to make choices
- [ ] Identify examples of trade■offs in everyday life
- [ ] Describe the basic steps of economic reasoning

LESSON 2

# 2. Basic Economic Problems

### Lesson overview

This lesson explains the three fundamental economic problems that every society faces: what to produce, how to produce, and for whom. Understanding these questions helps students see why resources are limited and choices must be made.

### Learning outcomes

- Identify the three basic economic problems
- Explain how scarcity creates the need to decide what to produce
- Describe factors that influence how goods are produced
- Discuss criteria that determine for whom goods are produced

### 2.1 What is a Basic Economic Problem?

Imagine you have a big box of crayons. The box holds only 24 colors, but you would love to draw a rainbow, a forest, a city skyline, and a portrait all at the same time. Because the crayons are **limited** - there are only so many of them - you cannot draw everything you want. This simple picture shows the **basic economic problem**:

> **Definition:** A basic economic problem is a question that appears whenever the resources we have (like crayons, time, money, land, or workers) are limited while our wants or needs are unlimited.

***Why does this happen?***

Resources are anything that can be used to produce goods and services:

- **Natural resources** such as water, oil, and timber.
- **Human resources** meaning the labor and skills of people.
- **Capital resources** like machines, factories, and computers.

All of these are **finite** - there is only a certain amount available at any moment. In contrast, human desires are **unlimited**. People always want more food, better schools, newer phones, cleaner air, and so on. Because we cannot have everything, we must make choices.

> **Note:** The word *scarcity* does not mean that something is rare in a dramatic sense; it simply means that the supply of a resource is less than the total amount people would like to have.

### *Turning the problem into a question*

When we recognize scarcity, we can phrase the situation as a series of questions:

- **What to produce?** - Which goods or services should we use our limited resources to create?
- **How to produce?** - What methods or techniques should we use to turn resources into those goods?
- **For whom to produce?** - Who will receive the goods and services we make?

These three questions together capture the whole **basic economic problem**. Every society, from a small village to a large nation, must answer them.

### *An everyday comparison*

Think about a family's weekly grocery budget. Suppose the family has 100 to spend and a list of wants: fresh fruit, meat, snacks, and a new bottle of juice. The money is limited, but the desires are not. The family must decide:

- **What to buy?** Maybe they choose fruit and meat but skip the snack.
- **How to buy?** They could shop at a discount store (cheaper but less convenient) or a premium store (more convenient but more expensive).
- **For whom?** Perhaps they buy more fruit for the children and less for the adults.

The family's decisions illustrate the same three questions that a whole economy faces, just on a smaller scale.

> **In daily life:** Whenever you plan a budget, schedule your time, or choose between activities, you are confronting the basic economic problem.

### *Why is it useful to name the problem?*

Labeling this situation as a "basic economic problem" helps us think systematically. Instead of feeling overwhelmed by endless wants, we can break the issue into manageable parts:

- Identify the **resource constraints** (money, time, materials).
- List the **desired outcomes** (goods, services, experiences).
- Compare alternatives using simple tools like a **cost■benefit analysis**, which weighs the advantages of one choice against its disadvantages.

A simple equation often appears in this analysis:

Benefit of Choice A - Cost of Choice A > Benefit of Choice B - Cost of Choice B

If the left■hand side is larger, Choice A is preferred.

> **Remember:** Economics is not just about money; it is the study of how we allocate *any* scarce resource to satisfy *any* set of wants.

### *Common misunderstanding*

A frequent mistake is to think that scarcity only applies to "poor" societies or to "rare" items. In reality, **scarcity exists everywhere**, even in wealthy nations. For example, clean air is abundant in some places but limited in heavily industrialized cities. Recognizing this prevents the error of assuming that more money automatically solves all problems.

> **Common mistake:** Believing that "more resources will eliminate the basic economic problem." Even if a country discovers a new oil field, new wants will arise, keeping the problem alive.

### *Summing up*

The basic economic problem is the starting point for all economic thinking. It tells us that because resources are limited and wants are unlimited, we must make choices about what to produce, how to produce it, and who gets it. By framing everyday decisions in this way, we

develop a habit of thoughtful, efficient decision■making that will serve us in personal life, school projects, and future careers.

[Figure to be added: A limited box of 24 crayons shows scarcity of resources.]

Figure 2.1. A limited box of 24 crayons shows scarcity of resources.

[Figure to be added: Choosing between wheat fields and shoe workshops illustrates the production trade■off.]

Figure 2.2. Choosing between wheat fields and shoe workshops illustrates the production trade■off.

[Figure to be added: A monthly family budget shows how limited money is allocated to different needs.]

Figure 2.3. A monthly family budget shows how limited money is allocated to different needs.

[Figure to be added: Scarcity vs Unlimited Wants]

Figure 2.4. Scarcity vs Unlimited Wants

[Figure to be added: Factors of Production]

Figure 2.5. Factors of Production

[Figure to be added: Three Basic Economic Decisions]

Figure 2.6. Three Basic Economic Decisions

[Figure to be added: Production vs Distribution]

Figure 2.7. Production vs Distribution

## 2.2 Why Societies Must Choose: The Concept of Scarcity

Societies face the **basic economic problem** because the resources they own - land, labor, capital, and entrepreneurship - are limited, while the wants and needs of people are virtually unlimited.

When we say *resources are limited*, we mean there is only a certain amount of each thing that can be used at any one time. For example, a country may have only 10 million hectares of arable land, 5 million workers, and a fixed amount of factories and machines. These are called **factors of production**.

> **Definition:** *Scarcity* is the condition that resources are insufficient to satisfy all the desires of a society.

Because of scarcity, a society cannot simply produce everything it wants. It must decide **what** to make, **how** to make it, and **for whom** to make it. These three decisions are the core of economic planning and are often called the *allocation* of resources.

### *Why scarcity forces choices*

Imagine you have a small garden and only enough water to water either tomatoes or carrots each day. If you water the tomatoes, the carrots will wilt; if you water the carrots, the tomatoes will not grow. You cannot have both thriving plants at the same time because the water (a resource) is scarce. The same logic works for whole economies: when a resource is scarce, using it for one purpose means it cannot be used for another.

This idea is useful because it helps us understand why prices exist, why budgets are prepared, and why governments must set priorities. When resources are scarce, the *opportunity cost* of a decision becomes important. Opportunity cost is the value of the next best alternative that is given up when a choice is made.

Opportunity Cost = Value of the best forgone alternative

### *Everyday comparison*

Think of your weekly allowance of 20. You want to buy a new video game that costs15 and also go to the movies, which costs 8. You cannot afford both because15 + 8 =23, which exceeds your 20 limit. You must choose. If you buy the game, the opportunity cost is the movie you miss; if you go to the movies, the opportunity cost is the game you forgo. This simple budgeting exercise mirrors the larger decisions societies make with far larger sums and more complex goods.

> **In daily life:** When you decide how to spend limited time - studying for a test or practicing a sport - you are facing scarcity of time and experiencing opportunity cost.

### *Building the idea step by step*

- **Identify the resource:** First, recognize what is limited (land, labor, money, time).
- **List the possible uses:** Write down the different ways the resource could be employed (e.g., building schools vs. building factories).
- **Compare benefits:** Estimate the benefit each use would bring to society (jobs created, education improved, etc.).
- **Choose the best use:** Select the option that gives the greatest net benefit, remembering that the other options are now forgone and represent the opportunity cost.

Because societies differ in values, cultures, and stages of development, the *best* use may vary. A country rich in natural resources might prioritize exporting raw materials, while another with a highly educated workforce might focus on technology services.

> **Note:** The concept of scarcity does **not** mean that some people will always be poor; it simply means choices must be made about how to use what we have.

### *Balancing the three problems*

The three fundamental questions - *what to produce*, *how to produce*, and *for whom to produce* - are interlinked. If a society decides to produce more of a scarce good, it must allocate more resources (labor, capital) to that production, which may reduce the resources available for other goods. This creates a chain of trade■offs that policymakers must constantly evaluate.

> **Remember:** Scarcity is the engine that drives economic decision■making; every choice creates an opportunity cost, and understanding that cost helps societies allocate resources more wisely.

## 2.3 What to Produce?

When a community decides **what to produce**, it is answering the question: *Which goods and services should we make with the resources we have?*

The answer is not random; it follows a few clear criteria that help the society use its limited resources wisely.

> **Definition:** *Consumer demand* is the amount of a good or service that people want to buy at a given price.

First, producers look at consumer demand. If many people want smartphones, but few want typewriters, the market will signal a higher demand for phones. High demand usually means that selling the product will bring more money, which can be used to pay workers, invest in new ideas, and keep the business alive.

Why does this matter? Imagine you have a small garden and only enough space to grow either tomatoes or carrots. If your family loves tomatoes and eats them every day, planting tomatoes will satisfy a strong need and reduce the chance of waste. If you chose carrots instead, you might end up with more carrots than anyone wants, and the tomatoes you could have grown would be missing from the table. The same logic works on a much larger scale for whole economies.

> **In daily life:** Think of a school cafeteria that must decide whether to serve pizza or sushi. They look at what students have ordered in the past (demand) and choose the meal that most students will enjoy and actually eat.

Second, **resource availability** limits what can be produced. Resources are the inputs needed to create goods: land, labor, capital (machines, tools), and entrepreneurship (the ability to combine the other three). If a country has abundant sunshine but little oil, it will find it easier and cheaper to produce solar panels than to refine petroleum. Trying to make a product that needs scarce resources would raise costs and could lead to shortages of those resources for other important uses.

> **Note:** *Scarcity* means that there is not enough of a resource to satisfy all possible wants at the same time.

Because resources are scarce, producers compare the amount of a resource needed for one product with the amount needed for another. This comparison is often expressed with an **opportunity cost** equation:

\text{Opportunity Cost of Good A} = \frac{\text{Resources used for Good A}}{\text{Resources that could have produced Good B}}

The lower the opportunity cost, the more attractive the product becomes, assuming demand is similar.

Third, societies may have **social goals** that influence production choices. Social goals are objectives that a community values beyond profit, such as protecting the environment, improving public health, or preserving cultural heritage. For example, a government might subsidize the production of electric buses even if demand for them is currently lower than for diesel buses, because reducing air pollution is a social goal. These goals can shift the balance: a product with moderate demand may still be produced if it helps achieve a larger societal benefit.

> **Common mistake:** Assuming that the most profitable product is always the best to produce, ignoring environmental or health impacts.

Putting the three criteria together, the decision process looks like this:

- **Check demand:** Is there a strong desire for the product?
- **Assess resources:** Do we have enough of the necessary inputs, and at what cost?
- **Align with social goals:** Does the product help meet broader community objectives?

If the answer to all three is "yes," the product is a good candidate for production. If any answer is "no," planners may either adjust the product (e.g., make a cheaper version) or choose a different one.

> **Remember:** The "what to produce" question is a balancing act among consumer wants, resource limits, and the values a society wants to uphold. By weighing these factors, economies aim to create the right mix of goods and services that improve overall well■being.

## 2.4 How to Produce?

When we talk about **how** to produce goods and services, we are asking: *what mix of resources and methods should we use to turn raw materials into the things people want?*

The answer depends on three main groups of resources, called **factors of production**:

- **Labor** - the work done by people.
- **Capital** - machines, tools, factories, and other man■made equipment.
- **Technology** - the knowledge and techniques that tell us how to combine labor and capital most effectively.

> **Definition:** *Factors of production* are the inputs (labor, capital, technology, land) that are combined to create output.

### *Why the Choice Matters*

Resources are limited, so we cannot use the most expensive machines for every product, nor can we rely only on cheap hand■crafting. Each method has a **trade■off** between **efficiency** (producing a lot with few resources) and **equity** (making sure many people have a chance to work and earn).

- **Efficiency** means lower cost per unit, higher output, and often lower prices for consumers.
- **Equity** means spreading jobs and income more evenly, even if that raises the cost of production.

A society must decide where it wants to stand on this spectrum.

> **Note:** *Trade■off* means that improving one goal (like efficiency) usually makes another goal (like equity) harder to achieve.

### *Simple Example: Baking Bread*

Imagine a small bakery that can bake bread in two ways:

- Method - Resources Used - Output per hour - Cost per loaf - Workers needed
- Hand■mixing - Simple bowls, one baker - 10 loaves - 0.80 - 1

- Machine■mixing - Mixer, electricity, two bakers - 30 loaves - 0.50 - 2

The machine■mixing method is **more efficient**: it produces three times as many loaves for a lower cost per loaf. However, it requires buying a mixer (capital) and hiring an extra worker, which may be harder for a community that wants to keep jobs spread among many small families.

> **In daily life:** Choosing between a hand■crafted product and a mass■produced one is a real■world version of the efficiency■equity trade■off.

### *The Role of Technology*

Technology can shift the whole picture. Suppose a new, cheaper mixer is invented. The cost of the machine■mixing method falls, making it both **more efficient** and **more accessible**. In economics we write this as:

\text{Cost per unit} = \frac{\text{Total cost of inputs}}{\text{Number of units produced}}

If the denominator (units produced) rises faster than the numerator (total cost), the cost per unit drops. This is called **economies of scale**.

> **Remember:** *Economies of scale* happen when larger production reduces the average cost of each item.

### *Balancing Efficiency and Equity*

Policymakers use several tools to influence the "how to produce" decision:

- **Subsidies** - payments that lower the cost of using certain technologies, encouraging firms to adopt them.
- **Taxes** - charges on high■efficiency methods that might displace many workers, used to fund retraining programs.
- **Regulations** - rules that require a minimum share of labor■intensive production, protecting employment.

These tools try to move the economy toward a point where enough goods are made cheaply **and** enough people have decent jobs.

> **Common mistake:** Assuming that the most efficient method is always the best choice, ignoring the social impact on employment and income distribution.

### *Summing Up*

- **How to produce** is about selecting the right mix of labor, capital, and technology.
- Every choice creates a trade■off between producing cheaply (efficiency) and providing work and income to many people (equity).
- Advances in technology can improve both goals, but they may also require new policies to share the benefits fairly.

Understanding these ideas helps us see why governments and businesses constantly debate the best production methods, and why the answers differ from one country to another.

## 2.5 For Whom to Produce?

When we ask **"for whom should the economy produce?"** we are really asking how the goods and services that are made get divided among the people who live in a society. This is called the *distribution* question.

> **Definition:** *Distribution* is the way a society decides who receives the output of production - the food, clothing, phones, schools, hospitals, and everything else that is created.

Why does this matter? Resources are limited, so the choices we make about *what* to produce must be matched with choices about *who* gets it. If we ignore distribution, we might end up with plenty of goods that only a small group can enjoy, while many others have none. Understanding distribution helps us see whether an economy is fair, whether people's basic needs are met, and how policies can change the outcome.

### *Preferences and Income*

Two main forces shape who gets what:

- **Preferences** - what people want. If many people want smartphones, producers will try to make more of them.
- **Income** - how much money people have to spend. Those with higher income can afford more goods, while those with low income can buy only a few.

Think of a school cafeteria. The menu (what is prepared) is decided by the kitchen staff, but who actually eats each dish depends on two things: the students' taste (some prefer pizza, others prefer salad) and the amount of money they have on their lunch cards. A student with a full card can buy a pizza, while a student with only a few dollars may have to choose a cheaper sandwich.

### *Market Mechanism*

In many market economies, the *price system* is the main tool that links income and preferences to distribution. Prices tell producers what people are willing to pay. When a good is expensive, only those with enough money can buy it; when it is cheap, more people can afford it.

\text{Quantity Demanded} = f(\text{Price}, \text{Income}, \text{Preferences})

The equation says that the amount of a good people want to buy depends on its price, their income, and their personal likes. If the price goes up, the *quantity demanded* usually falls, especially for people with low income. This is why luxury items (expensive watches, sports cars) are mostly bought by wealthier households.

> **Note:** The market does **not** automatically guarantee that everyone gets enough of the things they need. It only allocates goods to those who can pay the market price.

### *Government Policies*

Because markets can leave some people without essential goods, governments often step in. They use policies such as:

- **Taxes** - taking money from higher■income earners and using it to fund public services.
- **Subsidies** - giving money to producers or consumers to lower the price of certain goods (e.g., food stamps, school meal programs).
- **Direct provision** - the government itself produces or purchases goods and gives them away for free or at a low price (public schools, hospitals).

These tools change the effective price that different groups face, reshaping the distribution. For example, a subsidy on wheat lowers the price of bread for everyone, helping low■income families afford a staple food.

> **In daily life:** When you receive a discount coupon for a grocery store, the store is using a *subsidy*■like tool to make the product cheaper for you, altering the usual market distribution.

### *Equity vs. Efficiency*

Economists often talk about a trade■off between *equity* (fairness) and *efficiency* (getting the most output from limited resources). Policies that heavily redistribute income (high taxes, large welfare programs) can improve equity but may reduce incentives for people to work or invest, potentially lowering overall production. Finding a balance is a central challenge for any society.

> **Remember:** Distribution decisions are about **who gets what**, and they involve both market forces (prices, income) and government actions (taxes, subsidies, public services).

Understanding this helps us see why debates about taxation, welfare, and public spending are really debates about *for whom* we are producing. By examining preferences, income levels, and policy tools, we can evaluate whether an economy is meeting the needs of all its members or favoring only a few.

## 2.6 Balancing the Three Problems

Imagine a small town that has only two resources: a farm field and a workshop. The town can use the field to grow wheat or the workshop to make shoes. If the town decides to grow more wheat, it must use more of the field and fewer workers for shoes, so shoe production falls. This simple picture shows the **core idea** of balancing the three basic economic problems: **what to produce, how to produce, and for whom to produce**. Every choice in one area pushes the other areas in a new direction, creating trade■offs that planners must constantly manage.

> **Definition:** *Trade■off* - a situation in which gaining more of one thing automatically means having less of another because resources are limited.

Why does this happen? Resources such as land, labor, capital (machines, tools) and technology are **scarce** - there is not enough of them to satisfy every possible want. Because of scarcity, a society cannot simultaneously achieve the maximum of every goal. Instead, it must allocate resources in a way that reflects its priorities. The three problems are linked by the same pool of resources, so a decision that solves one problem partly solves the others, but also creates new constraints.

### *The Production Possibility Frontier (PPF)*

Economists often draw a curve called the **production possibility frontier** to illustrate the trade■offs. The PPF shows the maximum combinations of two goods that an economy can produce with its current resources and technology. If the town produces Q_w units of wheat and Q_s pairs of shoes, the frontier might be described by a simple equation:

\frac{Q_w}{100} + \frac{Q_s}{50} = 1

This equation says that producing 100 units of wheat uses up all the resources, leaving none for shoes, while producing 50 pairs of shoes uses all the resources, leaving none for wheat. Any point **inside** the curve (e.g., Q_w=40, Q_s=20) means resources are not fully used; any point **outside** is impossible with current resources.

When the town moves along the curve - say, from 60 wheat and 10 shoes to 50 wheat and 15 shoes - it **reallocates** labor and land. The decision "produce more shoes" (changing **what** to produce) automatically changes **how** those shoes are made (perhaps using more labor and less machinery) and influences **for whom** the extra shoes are available (maybe they become affordable for poorer families). Thus, a single shift touches all three questions.

> **In daily life:** When you decide to spend your allowance on a video game instead of a book, you are choosing *what* to consume, *how* you enjoy your leisure (playing versus reading), and *for whom* the benefit goes (you versus a friend who might have liked the book).

### *Interdependence of the Three Problems*

- **What to Produce ↔ How to Produce**

Choosing to make more of a good often requires a different production method. If the town wants more wheat, it might adopt a high■yield seed (changing the **technology** used). That technology may be expensive, affecting the **cost** and therefore the price paid by consumers, which ties into the "for whom" question.

- **How to Produce ↔ For Whom to Produce**

A method that uses cheap labor can keep prices low, making the product affordable for low■income households. Conversely, a high■tech, capital■intensive method may raise prices, limiting the product to wealthier buyers.

- **For Whom to Produce ↔ What to Produce**

If a society decides to prioritize basic food security for all, it may allocate more land to staple crops (what) and less to luxury items. This reduces the variety of goods available, affecting overall welfare.

> **Note:** Trade■offs are not failures; they are the natural result of limited resources. Good planning tries to find a balance that matches a society's values and goals.

### *Why Balancing Matters*

If planners ignore the connections, they may create **inefficiencies**. For example, producing a huge surplus of shoes while wheat is scarce leads to waste (unsold shoes) and hunger. By looking at the PPF, decision■makers can see the **opportunity cost** - the amount of wheat forgone when more shoes are produced. Recognizing opportunity cost helps societies decide whether the extra shoes are worth the lost wheat.

> **Common mistake:** Thinking that solving one problem automatically solves the others. In reality, each solution reshapes the other two problems, so continuous adjustment is required.

In summary, balancing the three basic economic problems means understanding that every choice about **what**, **how**, and **for whom** is linked through the same limited resources. Trade■offs are inevitable, but by visualizing these links - using tools like the production possibility frontier - societies can make informed decisions that reflect their priorities and improve overall well■being.

## 2.7 Real■World Examples and Discussion

When we talk about the three basic economic problems - what to produce, how to produce, and for whom to produce - we are really describing the choices that any group of people must make when resources are limited.

Think of a small family that has a fixed amount of money each month. That money is a **resource** (something useful but not unlimited). The family must decide:

- **What** to buy (food, clothing, entertainment).
- **How** to buy it (cheap bulk items, organic brands, or a mix).
- **Who** gets it (parents, children, or maybe a relative who is staying over).

Each decision influences the others, just like the three economic problems.

> **In daily life:** *A family budget is a miniature economy.*

> The total amount of money the family earns, M, must be split among all expenses. A simple way to see this is with the equation

> M = C + H + S

> where C is consumption (food, clothes), H is housing costs, and S is savings. The equation shows that if the family wants to save more (S goes up), they must reduce consumption or housing costs. This trade■off is the essence of scarcity.

### *Example 1: Choosing a Dinner Menu*

Imagine a school cafeteria that can serve either pizza or a salad bar for lunch. The cafeteria has a limited amount of ingredients and a fixed kitchen staff schedule.

- **What to produce?** The cafeteria must pick a menu that satisfies most students.
- **How to produce?** They could make pizza in large batches (low cost per slice) or prepare fresh salads (higher labor, higher cost).
- **For whom to produce?** If many students prefer pizza, serving only salad would leave many hungry, while only pizza might disappoint health■conscious students.

The cafeteria manager might decide to serve pizza on Mondays and salads on Tuesdays, balancing the three problems over the week. This shows that "balancing" does not mean solving everything at once, but arranging choices over time.

> **Definition:** *Balancing the three problems* means arranging production, methods, and distribution so that the most important needs are met given the limited resources.

### *Example 2: A Community Garden*

A neighborhood decides to turn a vacant lot into a community garden. The land, water, and volunteers are scarce resources.

- **What to produce?** The gardeners can grow vegetables, flowers, or a mix.
- **How to produce?** They could use organic methods (no chemicals, more labor) or conventional methods (cheaper, faster growth).
- **For whom to produce?** The harvest could be shared equally among all households, or given first to families with children.

If the garden chooses to grow mostly vegetables using organic methods and shares the produce equally, it reflects a decision that values health and fairness, even though it may yield less total food than a conventional, cash■crop approach. The community's values shape how the three problems are answered.

> **Note:** *Values and preferences* are not the same as "needs." They guide the "for whom" decision, showing that economics is not only about numbers but also about what people care about.

### *Example 3: A Teen's After■School Time*

Even a single teenager faces the three problems. Suppose Alex has 10 hours after school each week.

- **What to produce?** Alex can study, work a part■time job, or practice a sport.
- **How to produce?** Studying can be done alone (low cost) or with a tutor (higher cost). Working can be at a fast■food restaurant (easy hours) or a tutoring center (higher pay).
- **For whom to produce?** The benefits of studying go to Alex's future self, while earnings help Alex's family now.

Alex decides to spend 4 hours studying alone, 3 hours working at the tutoring center, and 3 hours playing sport. The allocation shows a personal version of the three economic problems, where time is the scarce resource.

> **Common mistake:** *Thinking "more is always better."*

> Adding more of one good (like extra pizza slices) does not automatically improve welfare if it means cutting back on something else important (like a healthy salad or study time). Scarcity forces trade■offs.

### *Bringing It All Together*

Everyday situations - family budgets, school meals, community projects, or a teen's schedule - illustrate the same three questions that economists ask about whole societies. By recognizing that resources (money, time, land, labor) are limited, we can see why choices must be made, why trade■offs exist, and how values influence who gets what.

When you notice a decision in your own life, ask yourself:

- **What** am I trying to get?
- **How** will I get it?
- **For whom** is it intended?

Answering these simple questions helps you understand the basic economic problems and prepares you to think like an economist, even in the smallest daily choices.

## Terminology at a glance

| Term | Meaning |
|---|---|
| Basic economic problem | The situation where limited resources cannot satisfy unlimited wants. |

| Term | Meaning |
|---|---|
| Scarcity | The condition that resources are insufficient to meet all desires. |
| Factors of production | Inputs used to create goods and services: land, labor, capital, and technology. |
| Consumer demand | The amount of a good or service people want to buy at a given price. |
| Trade■off | A situation where gaining more of one thing means having less of another because resources are limited. |
| Allocation | The process of deciding how resources are distributed among competing uses. |
| Distribution | How the output of production is divided among members of society. |
| Equity | Fairness in the way resources and goods are shared among people. |

### Practical application

Exercise 2.1:

Exercise 2.2:

Exercise 2.3:

Exercise 2.4:

Exercise 2.5:

Exercise 2.6:

### Checklist

- [ ] Identify the three basic economic problems
- [ ] Explain how scarcity creates the need to decide what to produce
- [ ] Describe factors that influence how goods are produced
- [ ] Discuss criteria that determine for whom goods are produced

LESSON 3

# 3. Supply and Demand Fundamentals

### Lesson overview

This lesson explains the basic laws of demand and supply, how they interact to create market equilibrium, and how prices adjust. Students will see how these ideas shape everyday market decisions.

### Learning outcomes

- Define the law of demand and the law of supply
- Draw and interpret demand and supply curves
- Identify the market equilibrium point
- Explain how price changes signal shortages or surpluses
- Calculate equilibrium price and quantity from simple data

### 3.1 1. The Law of Demand

The law of demand says that, **ceteris paribus**, when the price of a good goes up, the amount people want to buy (the quantity demanded) goes down.

*"Ceteris paribus"* is a Latin phrase that means "all other things being equal." It tells us to hold everything else constant - income, tastes, the price of other goods, etc. - so we can see the pure effect of the price change.

> **Definition:** *Law of Demand* - a higher price leads to a lower quantity demanded, assuming all other factors stay the same.

Why does this happen?

- **Budget constraint** - People have limited money. If a product becomes more expensive, they can afford to buy less of it.
- **Substitution effect** - When the price of one good rises, consumers tend to switch to cheaper alternatives.
- **Diminishing marginal utility** - The more of a good you already have, the less extra satisfaction (utility) each additional unit gives you. When the price rises, the extra satisfaction is not worth the higher cost, so you buy less.

Think of a simple everyday comparison: imagine you love buying chocolate bars for 1 each. One day the shop raises the price to2. With the same pocket money, you can only buy half as many bars, so you either buy fewer bars or look for a cheaper snack. The change in your buying behaviour illustrates the law of demand.

> **In daily life:** When gasoline prices increase, many drivers drive fewer miles or choose public transport, showing a lower quantity demanded for fuel.

### *From words to a simple formula*

Economists often write the relationship between price (P) and quantity demanded (Q_d) as:

Q_d = f(P)

where f is a decreasing function: as P rises, f(P) falls. In a straight■line (linear) approximation we can write:

Q_d = a - bP

Here:

- a is the quantity demanded when price is zero (the intercept).
- b is a positive number that tells us how much quantity falls when price rises by one unit (the slope).

Because b>0, the slope -b is negative, confirming that the curve slopes downward.

### *Worked example - coffee shop latte*

**Example 1 (Easy): Latte price change**

A coffee shop raises the price of a latte from \3 to4 and sells 20 fewer lattes per day.

How much does the quantity demanded change for each \1 increase in price?

- **Identify the change in price** - subtract the old price from the new price.

Delta P = 4 - 3 = 1 dollar

- **Identify the change in quantity demanded** - subtract the new quantity from the old quantity (the problem tells us the drop is 20 lattes).

Delta Q_d = -20 lattes

- **Compute the slope b** - the change in quantity per one■dollar change in price.

b = -fracDelta Q_dDelta P = -frac-201 = 20

**Answer:** For each \1 increase in the latte price, the coffee shop sells 20 fewer lattes per day. The negative sign in\Delta Q_d shows a decrease, matching the law of demand.

### *Interpreting the result*

The slope b = 20 tells us the **price sensitivity** of latte buyers at this shop. If the shop raised the price by \2, we would expect roughly2 \times 20 = 40 fewer lattes sold, assuming the linear relationship holds and no other factors change.

> **Note:** The law of demand describes a *relationship*, not a fixed rule. Real markets may show exceptions (e.g., luxury goods where higher price can increase prestige and demand), but those cases involve other factors changing, so the "ceteris paribus" condition is violated.

### *Graphical intuition (without drawing)*

Imagine a graph with price on the vertical axis and quantity demanded on the horizontal axis. The line Q_d = a - bP starts high on the quantity axis (when price is low) and slopes down to the right. Moving rightward (higher quantity) forces you down to a lower price point; moving upward (higher price) forces you leftward to a lower quantity. This visual slope is exactly what the equation -b tells us.

### *Why the law matters*

Understanding that higher prices reduce quantity demanded helps businesses set prices, governments predict tax impacts, and consumers anticipate how market changes affect their spending. For instance, if a city imposes a tax that raises the price of movie tickets, the law of demand predicts fewer tickets will be sold, which can affect cinema revenues and even local employment.

> **Remember:** The law of demand holds only when *all other influences* (income, tastes, prices of related goods) stay the same. If any of those change, the observed quantity may move for reasons other than price.

### *Quick check*

Suppose a bakery lowers the price of a muffin from \2 to1.5 and observes that sales rise from 80 muffins to 100 muffins per day. Using the same steps as the latte example, the slope would be:

- Delta P = 1.5 - 2 = -0.5 dollars

- Delta Q_d = 100 - 80 = 20 muffins
- b = -frac20-0.5 = 40

The positive b (40) again confirms that a price drop leads to a quantity increase, consistent with the law of demand.

By mastering this simple relationship, you are ready to explore how supply works and how the two together determine market outcomes.

[Figure to be added: Buyers and sellers negotiating prices at a farmer's market.]

Figure 3.1. Buyers and sellers negotiating prices at a farmer's market.

[Figure to be added: A factory increasing production after a price rise.]

Figure 3.2. A factory increasing production after a price rise.

[Figure to be added: How a per■unit tax creates a gap between buyer price and seller receipt.]

Figure 3.3. How a per■unit tax creates a gap between buyer price and seller receipt.

[Figure to be added: Demand vs. Supply]

Figure 3.4. Demand vs. Supply

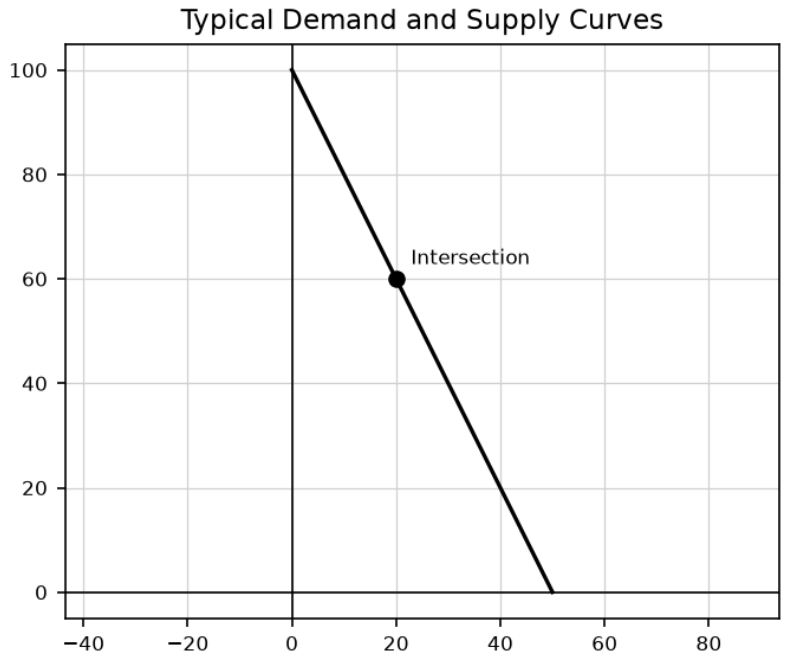



Figure 3.5. Downward■sloping demand (Q = 100■2P) and upward■sloping supply (Q = 20+3P) intersect at price 20, quantity 60.




Figure 3.6. Finding Market Equilibrium




Figure 3.7. Movement vs. Shift

## 3.2 2. The Law of Supply

The law of supply says that, **ceteris paribus** (*all other things being equal*), producers are willing to offer more of a good when its price rises, and less when its price falls.

> **Definition:** *Law of Supply* - a positive relationship between the price of a product and the quantity that producers are ready to sell, holding everything else constant.

Why does this happen?

- **Higher revenue potential** - If the market price goes up, each unit sold brings in more money. That extra revenue can cover higher costs (like labor or raw materials) and still leave a profit.
- **Incentive to use resources** - Producers have limited resources (land, machines, workers). A higher price makes it worthwhile to shift those resources toward the profitable good and away from less■profitable activities.
- **Covering higher marginal costs** - As a firm produces more, the cost of producing each additional unit (the *marginal cost*) usually rises. A higher price is needed to make those extra units profitable.

Think of a lemonade stand. When the price is 1 per cup, the student running the stand might sell 20 cups a day and earn20. If the price rises to 2 per cup, the same student can earn40 by selling the same 20 cups, or they might decide to make more lemonade to earn even more. The extra profit motivates them to buy more lemons, sugar, and maybe a larger pitcher.

> **In daily life:** A baker sees the price of bread go up from \2 to3 per loaf. Because each loaf now brings in an extra \1, the baker decides to bake 20% more loaves each morning.

### *From words to a simple formula*

Economists often write the supply relationship as

Q_s = f(P)

where

- Q_s = quantity supplied (how many units producers are ready to sell)
- P = market price of the good
- f = a function that shows how Q_s changes when P changes, holding other factors constant.

If we assume a straight■line (linear) supply curve, the function can be written as

Q_s = a + bP

- a is the quantity supplied when price is zero (often a negative number, showing that producers would not supply anything at a zero price).
- b is the *slope* of the supply curve; because the law of supply says the curve slopes upward, b is a positive number.

### *Worked example - how a price rise changes a farmer's planting decision*

**Example 2 (Easy): Tomato farmer reacts to a price increase**

A farmer receives \2 more per kilogram of tomatoes. As a result, the farmer decides to plant 15 % more acres for tomatoes.

- **Identify the percentage change in quantity supplied.**

The farmer's "quantity supplied" in this case is the area planted. A 15 % increase means

Delta Q_s% = 15%

- **Relate the percentage change to the price change.**

The price increase is \2 per kilogram. To see the *price elasticity of supply* (how responsive supply is), we use

Elasticity = fracDelta Q_s%Delta P%

First find the percentage change in price. Suppose the original price was \4/kg, so

Delta P% = frac24times 100% = 50%

- **Calculate the elasticity.**

Elasticity = frac15%50% = 0.30

**Answer:** The farmer's supply of tomatoes is relatively inelastic (elasticity = 0.30). A \2 price rise leads to a modest 15 % increase in planted acres.

The steps show how a higher price encourages the farmer to use more land, but the response is not huge because other factors (like limited land or labor) restrict how much more can be supplied.

#### *What the example tells us*

- The positive sign of the elasticity confirms the law of supply: quantity supplied rises when price rises.
- The magnitude (0.30) tells us the response is small; producers may be constrained by resources, technology, or risk.

> **Note:** Elasticity is a *ratio*; it tells us the size of the response, not just its direction.

#### *When the law does not hold*

The phrase *ceteris paribus* is crucial. If something else changes at the same time, the simple upward■sloping relationship can break down. Examples:

- **Input price rise** - If the cost of fertilizer doubles, the farmer may cut back even though tomato prices are higher.
- **Regulatory limits** - A government may cap the amount of land that can be used for a crop, preventing the farmer from expanding.

These are called *shifts* of the supply curve (the whole curve moves) rather than *movements* along the curve (a change in quantity supplied because of a price change).

#### *Quick check - think of a coffee shop*

Suppose a coffee shop sells a cup of coffee for \3 and sells 100 cups a day. If the price rises to4, the shop could earn an extra \100 per day *if* it sold the same 100 cups. Because the extra profit looks attractive, the shop may decide to open earlier, hire an extra barista, and try to sell 120 cups. The increase from 100 to 120 cups is a *movement* along the supply curve caused by the higher price.

> **Common mistake:** Thinking that a higher price always means producers will supply *infinitely* more. In reality, the upward slope is limited by resources, technology, and other costs.

#### *Summarizing the law*

- **Higher price -> higher quantity supplied** (ceteris paribus).
- The relationship is captured by an upward■sloping supply curve, often written Q_s = a + bP with b>0.
- The *elasticity of supply* measures how strong the response is; values greater than 1 are "elastic," less than 1 are "inelastic."
- Real■world constraints can shift the whole curve, turning a price change into a smaller or larger effect.

Understanding the law of supply helps us predict how producers will react to market signals, and it sets the stage for later topics like market equilibrium and policy analysis.

## 3.3 3. Drawing Demand and Supply Curves

The first thing to understand is **what a graph does**: it turns numbers into a picture that we can read at a glance.

When we talk about markets, the two numbers we care about most are **price** (how much money is paid for one unit of a good) and **quantity** (how many units are bought or sold).

> **Definition:** *Price* is the amount of money required to purchase one unit of a product.

> **Definition:** *Quantity* is the number of units of a product that are bought or sold.

#### *1. Setting up the axes*

- **Choose the vertical axis (the y■axis) for price.**
- We put price on the vertical line because we usually think of price as the "height" of a market condition - higher price means the line goes up.

- **Choose the horizontal axis (the x■axis) for quantity.**
- Quantity runs left■to■right, just like distance on a road map.

The point where the two axes cross is called the **origin** (0, 0). It represents a situation where both price and quantity are zero - a market that does not exist yet.

> **Note:** The scale on each axis can be chosen freely, but it must be consistent (e.g., 1 increments for price, 10■unit increments for quantity).

### *2. Plotting a demand curve*

The **law of demand** tells us that, all else equal, consumers buy more when the price is lower. This creates a **downward■sloping** line on our graph.

**Steps to plot a demand curve**

- List a few price■quantity pairs that follow the demand rule.
- Mark each pair as a point on the graph: the price tells you how high to go, the quantity tells you how far to the right.
- Connect the points with a smooth line; the line should slope downwards from left to right.

### *Worked Example 3 (Medium): Coffee demand*

*Question:* A coffee shop finds that when the price is 2 per cup, 120 cups are sold each day. If the price rises to4, only 80 cups are sold. Plot the demand curve using these two points.

- **Write the two price■quantity pairs.**

$(P_1, Q_1) = (2, 120)$

$(P_2, Q_2) = (4, 80)$

- **Mark the points on the graph.**
- For (2,120) go up to the 2 mark on the vertical axis, then move right to the 120 mark on the horizontal axis.
- For (4,80) go up to 4 on the vertical axis, then right to 80 on the horizontal axis.
- **Draw a straight line through the two points.** Because demand is assumed linear in this simple example, the line connecting them represents the whole demand curve.

**Answer:** The demand curve is a straight line that passes through (2,120) and (4,80), sloping downwards as price increases.

### *3. Plotting a supply curve*

The **law of supply** says producers are willing to sell more when the price is higher. This creates an **upward■sloping** line.

**Steps to plot a supply curve**

- Gather price■quantity pairs that follow the supply rule.
- Plot each pair on the same axes as before.
- Connect the points; the line should rise from left to right.

### *Worked Example 4 (Medium): Tomato supply*

*Question:* A farmer can bring 30 kilograms of tomatoes to market when the price is 1 per kilogram, and 70 kilograms when the price is3 per kilogram. Plot the supply curve.

- **Write the two price■quantity pairs.**

$(P_1, Q_1) = (1, 30)$

$(P_2, Q_2) = (3, 70)$

- **Mark the points on the graph.**
- For (1,30) go up to 1 on the vertical axis, then right to 30 on the horizontal axis.
- For (3,70) go up to 3 on the vertical axis, then right to 70 on the horizontal axis.
- **Draw a straight line through the two points.** Because we assume a linear supply relationship, the line connecting them is the supply curve.

**Answer:** The supply curve is a straight line through (1,30) and (3,70), sloping upwards as price rises.

### *4. Putting demand and supply together*

Now we place **both** curves on the same set of axes. The point where they intersect is called the **market equilibrium** - the price and quantity where what consumers want to buy exactly equals what producers want to sell.

> **In daily life:** Think of a busy farmers' market. If the price of strawberries is too low, sellers run out quickly (shortage). If it is too high, many strawberries stay unsold (surplus). The price that clears the market is the equilibrium.

#### *Worked Example 5 (Medium): Plotting both curves together*

*Question:* Using the coffee demand points (2,120) and (4,80) and the tomato supply points (1,30) and (3,70), draw both curves on the same graph and identify the intersection (if any).

- **Draw the axes** with price on the vertical axis and quantity on the horizontal axis.
- **Plot the two demand points** and connect them with a downward■sloping line (as done in Example 3).
- **Plot the two supply points** and connect them with an upward■sloping line (as done in Example 4).
- **Locate the intersection.** Because the demand points are at higher quantities (80-120) than the supply points (30-70), the two lines do **not** cross within the plotted range. This tells us that, with these particular numbers, the coffee market and the tomato market are unrelated; we would need additional points or a different good to find a true equilibrium.

**Answer:** The demand and supply lines do not intersect in the displayed range, indicating no common price■quantity pair for these two separate goods. In practice, we would plot demand and supply for the *same* product to find equilibrium.

#### *5. Why the graph matters*

A graph lets us see instantly:

- **How sensitive** quantity is to price changes (the steepness of the line).
- **Where shortages or surpluses** would appear (areas above or below the equilibrium).
- **What happens** if something shifts the curve (e.g., a new tax moves the supply curve upward).

> **Remember:** The vertical axis is always price, the horizontal axis is always quantity. Keeping this convention avoids confusion when you compare many markets.

By mastering the simple act of drawing these two lines, you gain a visual tool that economists use to explain almost every market phenomenon.

## 3.4 4. Market Equilibrium

When buyers and sellers meet in a market, each side has its own wishes.

Buyers want to purchase a certain amount of a good at a price they are willing to pay - this is the **quantity demanded**.

Sellers are ready to offer a certain amount at a price they are willing to accept - this is the **quantity supplied**.

If the price is too high, sellers will want to sell more than buyers want to buy, creating a **surplus**.

If the price is too low, buyers will want more than sellers are ready to provide, creating a **shortage**.

The market solves this problem by moving the price until the two quantities become exactly the same. The point where **quantity demanded equals quantity supplied** is called **market equilibrium**.

> **Definition:** *Market equilibrium* is the price■quantity pair at which the amount buyers want to buy equals the amount sellers want to sell.

Why does this matter?

When the market is at equilibrium, there is no pressure for the price to change. No one is left with unsold goods, and no one is turned away because the good is too expensive. This makes the market efficient - resources are used where they are most valued.

#### *Visualising equilibrium on a graph*

- **Draw the axes** - the vertical axis (y■axis) shows price (P), the horizontal axis (x■axis) shows quantity (Q).
- **Plot the demand curve** - it slopes downward because, all else equal, a lower price makes buyers want more (the law of demand).
- **Plot the supply curve** - it slopes upward because a higher price encourages sellers to produce more (the law of supply).
- **Find the intersection** - the point where the two lines cross is the equilibrium. The price coordinate of this point is the **equilibrium price (Pe)**, and the quantity coordinate is the **equilibrium quantity (Qe)**.

At this crossing, the market "balances" - the two forces are equal and opposite, just like a seesaw that is perfectly level.

#### *Finding equilibrium algebraically*

When the demand and supply relationships are given as equations, we can solve for the equilibrium by setting them equal to each other:

$Q_d(P) = Q_s(P)$

where Q_d(P) is the demand function and Q_s(P) is the supply function. Solving this equation gives the equilibrium price P_e. Substituting P_e back into either the demand or supply equation yields the equilibrium quantity Q_e.

> **Note:** The algebraic method works for any linear or non■linear functions, as long as you can solve the equation for P.

### *Worked example*

**Example 4 (Medium): Solving for equilibrium price and quantity**

Given the demand function Q_d = 100 - 5P and the supply function Q_s = 20 + 3P, find the equilibrium price and quantity.

- **Set demand equal to supply.**

100 - 5P = 20 + 3P

- **Collect the P terms on one side and constants on the other.**

Subtract 3P from both sides:

100 - 8P = 20

Then subtract 20 from both sides:

80 - 8P = 0

- **Solve for P.**

Add 8P to both sides:

80 = 8P

Divide by 8:

P = 10

This P is the equilibrium price P_e.

- **Find the equilibrium quantity.**

Substitute P = 10 into either the demand or supply equation (they give the same result). Using demand:

Q_e = 100 - 5(10) = 100 - 50 = 50

**Answer:** The market equilibrium is a price of **10** and a quantity of **50 units**. At this price, buyers want exactly 50 units and sellers are willing to provide exactly 50 units.

### *What equilibrium tells us about the market*

- **Stability:** If the price is above P_e, a surplus appears. Sellers lower the price to clear excess stock, moving the market back toward P_e.
- **Instability:** If the price is below P_e, a shortage appears. Buyers compete for the limited goods, pushing the price up toward P_e.

> **In daily life:** Think of a farmer's market where the price of strawberries settles at 3 per basket. At that price, the farmer brings exactly the number of baskets that shoppers are ready to buy. If the farmer tried4 per basket, baskets would sit unsold (a surplus). If the price were 2, shoppers would line up and the farmer would run out (a shortage). The3 price is the equilibrium.

### *Quick checklist for finding equilibrium*

- Write down the demand and supply equations.
- Set them equal to each other and solve for P.
- Plug the resulting P back into either equation to get Q.
- Verify that the same Q comes from both equations - this confirms the solution.

> **Remember:** Equilibrium is a *point*, not a range. It is the unique price■quantity pair where the two curves intersect (unless the curves are parallel, in which case no equilibrium exists).

Understanding equilibrium gives you a powerful tool: you can predict how changes - like a new tax, a subsidy, or a shift in consumer taste - will move the price and quantity away from the original balance, and then see how the market will respond to restore a new equilibrium. This foundation will be essential when we later discuss how policies affect real markets.

## 3.5 5. Price Mechanisms: Shortages and Surpluses

When the price in a market is **not** at the point where the amount that buyers want to buy equals the amount that sellers want to sell, the market experiences either a **shortage** or a **surplus**.

A **shortage** happens when the price is **too low**. At that price, many consumers want to buy the good, but producers are not willing to supply enough of it.

A **surplus** happens when the price is **too high**. At that price, producers are eager to sell a lot, but consumers do not want to buy as much.

Why does this matter? Prices act like signals. When a shortage appears, the signal tells producers "you could earn more if you raised the price," and it tells consumers "there isn't enough of the good, you may have to wait or look for alternatives." When a surplus appears, the signal tells producers "you are making too many units, lower the price," and it tells consumers "there is plenty available, you can buy more if you want." These adjustments push the market back toward **equilibrium**, the price where quantity demanded equals quantity supplied.

### *Everyday comparison*

Think of a popular concert ticket. If the ticket price is set very low, more people want tickets than there are seats - a shortage. The organizer will quickly raise the price or add extra shows. If the price is set very high, many seats stay empty - a surplus. The organizer will lower the price or offer discounts. In both cases the price moves in the direction that reduces the mismatch.

### *Step■by■step reasoning*

- **Identify the current price** and compare it with the **equilibrium price** (the price where the demand curve meets the supply curve).
- **Calculate quantity demanded** at that price using the demand equation.
- **Calculate quantity supplied** at that price using the supply equation.
- **Compare the two quantities**:
- If *quantity demanded* > *quantity supplied*, the market has a **shortage** of size

Shortage = Q_D - Q_S.

- If *quantity supplied* > *quantity demanded*, the market has a **surplus** of size

Surplus = Q_S - Q_D.

> **Definition:** *Shortage* = excess demand; *Surplus* = excess supply.

> **Note:** The size of a shortage or surplus is measured in the same units as the good (e.g., units, kilograms, tickets).

### *Worked Example 5 (Hard): Shortage or surplus at a given price*

**Question:** In the previous lesson we had the demand function Q_D = 20 - 2P and the supply function Q_S = 4P. If the price is set at 6, calculate the resulting shortage or surplus.

- **Find quantity demanded at P = 6**

Q_D = 20 - 2(6) = 20 - 12 = 8.

- **Find quantity supplied at P = 6**

Q_S = 4(6) = 24.

- **Compare the two quantities**

Since Q_S > Q_D, there is a surplus.

Surplus = Q_S - Q_D = 24 - 8 = 16.

**Answer:** At a price of 6, the market has a surplus of **16 units**.

### *How the market reacts*

When a surplus appears, sellers lower their price to attract more buyers. As the price falls, the demand curve tells us that more consumers will want the good (quantity demanded rises), while the supply curve tells us that producers will be willing to sell fewer units (quantity supplied falls). The price keeps moving until the surplus disappears and the market reaches equilibrium again.

### *Worked Example 6 (Hard): Price ceiling creates a shortage*

**Question:** Suppose the government imposes a price ceiling of 1 on the same market (demandQ_D = 20 - 2P, supplyQ_S = 4P). Show how this creates a shortage and compute its size.

- **Apply the price ceiling:** the maximum legal price is P = 1.
- **Calculate quantity demanded at P = 1**

Q_D = 20 - 2(1) = 20 - 2 = 18.

- **Calculate quantity supplied at P = 1**

Q_S = 4(1) = 4.

- **Determine the mismatch**

Because Q_D > Q_S, a shortage occurs.

Shortage = Q_D - Q_S = 18 - 4 = 14.

**Answer:** The 1 price ceiling creates a shortage of **14 units**.

> **In daily life:** A price ceiling is like a rent■control rule that says landlords cannot charge more than a set amount. If the ceiling is below the market rent, there are more people wanting apartments than apartments available - a shortage.

### *Why the price moves automatically*

If the price is above equilibrium, sellers notice unsold inventory and lower the price to clear it. If the price is below equilibrium, buyers find the good hard to obtain, and sellers can raise the price because buyers are willing to pay more. This "price mechanism" is the invisible hand that coordinates millions of individual decisions without any central planner.

> **Remember:** The market always pushes the price toward the point where *quantity demanded = quantity supplied*. Shortages push prices up; surpluses push prices down.

### *Quick checklist for identifying shortages and surpluses*

- Step - Action - What you look for
- 1 - Write down the current price P - Given or imposed by policy
- 2 - Compute Q_D using the demand equation - How much buyers want
- 3 - Compute Q_S using the supply equation - How much sellers offer
- 4 - Compare Q_D and Q_S - Q_D > Q_S -> shortage; Q_S > Q_D -> surplus
- 5 - Calculate the size - Subtract the smaller from the larger

Understanding these price mechanisms helps you predict what will happen when governments intervene, when technology changes production costs, or when consumer tastes shift. The next lesson will show how the whole curves shift, not just move along a single curve.

## 3.6 6. Shifts vs. Movements

The idea of a **shift** versus a **movement** is simple once we separate two things that often get mixed up: the *price* of a good and the *other factors* that affect how much of it buyers want or sellers are willing to offer.

> **Definition:** A **movement** is a change **along** the same demand (or supply) curve caused only by a change in the price of the good.

> **Definition:** A **shift** is a change of the **whole** demand (or supply) curve caused by a change in any other factor - for demand, things like income, tastes, or the price of related goods; for supply, things like technology, input prices, or the number of sellers.

Why does this matter? If we know whether a price change is moving us along a curve or moving the curve itself, we can predict what will happen to the equilibrium price and quantity. Policy makers, businesses, and consumers all use this knowledge to decide whether to raise wages, subsidize production, or change advertising.

### *Step■by■step picture*

- **Start with one curve.**

Imagine a straight■line demand curve: Q_d = 100 - 2P, where Q_d is quantity demanded and P is price.

- **Change the price only.**

If the price falls from P=20 to P=15, we plug the new price into the same equation:

Q_d = 100 - 2(15) = 70.

The curve has not moved; we have simply moved to a new point **along** the same line.

- **Change something else (e.g., income).**

Suppose consumers become richer and now want to buy more at every price. The demand equation becomes Q_d = 120 - 2P. The whole line slides rightward; every price now corresponds to a larger quantity. This is a **shift** of the demand curve.

> **In daily life:** Think of a bus route (the curve). If the bus runs faster (price change), you simply get to a different stop on the same route (movement). If the city adds a new bus line that goes through more neighborhoods (shift), the whole route changes.

### *How to tell the difference on a graph*

- **Movement:** The point slides up or down the same line. The slope (steepness) stays the same.
- **Shift:** The entire line moves left or right (for demand) or up or down (for supply). The slope stays the same, but the intercept changes.

> **Common mistake:** Assuming that any change in quantity bought or sold is a shift. Remember: only a change in a non■price factor moves the whole curve.

### *Worked Example - Demand Shift from Higher Income*

**Example 8 (Hard): Income rise shifts demand right**

*Question:* If consumer income rises, demand shifts right to Q_d = 120 - 5P. Find the new equilibrium with the original supply curve Q_s = 20 + 3P.

**Steps**

- **Write down the new demand equation** (already given).

Q_d = 120 - 5P

- **Set demand equal to supply** to find the equilibrium where the quantity buyers want equals the quantity sellers want.

120 - 5P = 20 + 3P

- **Solve for the price P**.
- Bring the P terms to one side:

120 - 20 = 5P + 3P

100 = 8P

- Divide both sides by 8:

P = frac1008 = 12.5

- **Find the equilibrium quantity** by plugging P = 12.5 into either the demand or supply equation (both give the same result). Using supply:

Q_s = 20 + 3(12.5) = 20 + 37.5 = 57.5

**Answer:** The new equilibrium price is **12.5** and the equilibrium quantity is **57.5 units**. Because the demand curve shifted right, both price and quantity are higher than before the income increase.

### *Summary of the mechanics*

- **Movement**: Only price changes -> use the same equation, plug in the new price, read the new quantity.
- **Shift**: A non■price factor changes -> first rewrite the equation to reflect the new intercept or slope, then solve for the new equilibrium as we did above.

> **Remember:** When a curve shifts, the whole relationship between price and quantity changes. When we move along a curve, the relationship stays the same; only the price we are looking at is different.

Understanding this distinction lets you answer questions like: *Will a tax on producers raise the price buyers pay?* (It shifts supply left, then we find a new equilibrium.) Or *What happens if a popular new smartphone model drops in price?* (That is a movement along the demand curve.) By keeping the two ideas separate, you can analyze any market change clearly and correctly.

## 3.7 7. Real■World Application: Policy Impact

When a government changes the rules of the market, the ideas of supply, demand, and equilibrium help us see who really feels the impact.

Two common tools are **taxes** (money the seller must pay to the government) and **subsidies** (money the government gives to the buyer).

Both shift the cost that participants face, so the price that buyers pay and the price that sellers receive are no longer the same.

> **Definition:** *Tax on sellers* - a compulsory payment per unit sold that the seller must send to the government. It raises the seller's cost of each unit.

> **Definition:** *Subsidy to buyers* - a payment per unit that the government gives to the buyer, effectively lowering the price the buyer pays out■of■pocket.

Why does this matter?

If we know the original supply and demand curves, we can predict the new equilibrium after the policy change. This tells us how much of the good will be bought and sold, and who bears the burden of the tax or enjoys the benefit of the subsidy.

### *Step■by■step for a tax on sellers*

Suppose the original market is described by

- Demand: Q_D = 20 - 2P

- Supply: Q_S = 4P

where Q is quantity (units) and P is price (dollars).

Without any tax, set Q_D = Q_S to find the equilibrium:

20 - 2P = 4P Longrightarrow 20 = 6P Longrightarrow P*= frac206approx 3.33

Q*=4P*=4(3.33)approx 13.33

Now the government imposes a **2 tax on sellers**. The seller must pay2 for every unit sold, so the price the seller receives, call it P_S, is 2 less than the price the buyer pays,P_B:

P_S = P_B - 2

The supply curve expressed in terms of the buyer's price becomes:

Q_S = 4P_S = 4(P_B-2) = 4P_B - 8

The demand curve stays the same because buyers do not face the tax directly.

**Example 8 (Hard): Tax on sellers**

*Impose a 2 tax on sellers in the original supply equation and determine the new equilibrium price paid by buyers and received by sellers.*

- **Write the new supply equation in terms of the buyer's price.**

Q_S = 4(P_B-2) = 4P_B - 8

- **Set the new supply equal to demand and solve for the buyer's price.**

20 - 2P_B = 4P_B - 8

20 + 8 = 6P_B

28 = 6P_B Longrightarrow P_B = frac286approx 4.67

- **Find the price the seller actually receives** (subtract the tax).

P_S = P_B - 2 = 4.67 - 2 = 2.67

**Find the new quantity** by plugging P_B into either equation.

Q = 20 - 2(4.67) = 20 - 9.33 = 10.67

**Answer:** After the 2 tax, buyers pay about **4.67** per unit, sellers receive about **2.67**, and the market trades **10.7 units**. The tax reduces the quantity sold compared with the original 13.3 units.

> **Note:** The tax creates a *gap* between the price buyers pay and the price sellers receive. This gap is the tax amount, and the size of the gap shows who bears more of the burden. In this example, buyers pay 1.34 more than before, while sellers receive0.66 less.

### *Step■by■step for a subsidy to buyers*

Now keep the original curves but give the government a **1 subsidy to each buyer**. The buyer's effective price after the subsidy,P_B^{\text{eff}}, is the market priceP minus the subsidy:

P_B^eff = P - 1

Because the buyer feels the good is cheaper by 1, the demand curve shifts outward (to the right). In terms of the market priceP, the new demand relation is:

Q_D = 20 - 2(P - 1) = 20 - 2P + 2 = 22 - 2P

Supply is unchanged: Q_S = 4P.

**Example 9 (Hard): Subsidy to buyers**

*Give a 1 subsidy to buyers and compute the resulting equilibrium.*

- **Write the shifted demand equation in terms of the market price.**

Q_D = 22 - 2P

- **Set the new demand equal to supply and solve for the market price.**

22 - 2P = 4P

22 = 6P

P = frac226approx 3.67

- **Determine the effective price paid by buyers** (subtract the subsidy).

P_B^eff = 3.67 - 1 = 2.67

**Find the new quantity** using either curve.

Q = 4P = 4(3.67) = 14.68

**Answer:** With a 1 buyer subsidy, the market price is about **3.67**, buyers actually pay **2.67**, sellers receive **3.67**, and the market trades **14.7 units**. The subsidy raises the quantity traded compared with the original 13.3 units.

> **In daily life:** Think of a tax on soda as the government adding a 0.10 charge per can that the store must pay. The store raises its shelf price, so you end up paying more. A subsidy is like a coupon the government gives you for each kilogram of vegetables you buy; you see a lower price at checkout, and farmers may sell more because demand rises.

> **Common mistake:** Assuming the tax or subsidy changes only one side of the market. In reality, a seller tax shifts the supply curve, while a buyer subsidy shifts the demand curve. Forgetting which curve moves leads to the wrong equilibrium.

By following these steps - rewrite the affected curve, set it equal to the unchanged curve, solve for price, then compute quantity - you can evaluate any policy that changes the cost or benefit of buying or selling. This systematic approach lets economists predict the real■world impact of taxes, subsidies, price controls, or other interventions.

## Terminology at a glance

| Term | Meaning |
|---|---|
| Law of Demand | When price rises, quantity demanded falls, assuming other factors stay constant. |
| Law of Supply | When price rises, quantity supplied rises, assuming other factors stay constant. |
| Ceteris Paribus | Latin for 'all other things being equal'; used to isolate one variable's effect. |
| Market Equilibrium | The price and quantity where quantity demanded equals quantity supplied. |
| Surplus | A situation where quantity supplied exceeds quantity demanded at a given price. |
| Shortage | A situation where quantity demanded exceeds quantity supplied at a given price. |
| Shift (of curve) | A change in the entire demand or supply curve caused by factors other than price. |
| Movement (along curve) | A change in quantity demanded or supplied caused only by a change in price. |

## Practical application

Exercise 3.1:

Exercise 3.2:

Exercise 3.3:

Exercise 3.4:

Exercise 3.5:

Exercise 3.6:

## Checklist

- [ ] Define the law of demand and the law of supply
- [ ] Draw and interpret demand and supply curves
- [ ] Identify the market equilibrium point
- [ ] Explain how price changes signal shortages or surpluses
- [ ] Calculate equilibrium price and quantity from simple data

LESSON 4

# 4. Elasticity and Its Applications

## Lesson overview

This lesson explains price elasticity of demand and supply and shows how elasticity influences total revenue and policy decisions.

## Learning outcomes

- Define price elasticity of demand and supply
- Calculate elasticity using percentage changes
- Interpret elasticity values as elastic, inelastic, or unit■elastic
- Analyze how elasticity affects total revenue
- Apply elasticity concepts to tax and subsidy policy

## 4.1 What is Price Elasticity of Demand?

The price elasticity of demand measures **how much the quantity demanded changes when the price changes**.

In other words, it tells us how "responsive" buyers are to a price move.

If a small price change causes a big change in the amount people buy, demand is said to be *elastic*.

If the same price change causes only a tiny change in the amount bought, demand is *inelastic*.

> **Definition:** *Price elasticity of demand* (often just "elasticity") is the percentage change in quantity demanded divided by the percentage change in price.

Mathematically we write

E_d \;=\; \frac{\% \Delta Q}{\% \Delta P}

where

- % Delta Q = percentage change in quantity demanded, and
- % Delta P = percentage change in price.

Both percentages are calculated as

\% \Delta X \;=\; \frac{\text{New value} - \text{Old value}}{\text{Old value}} \times 100\%

The sign of E_d is usually negative because price and quantity move in opposite directions (when price goes up, quantity demanded typically goes down). Economists often ignore the minus sign and talk about the *absolute value*.

### *Why is this useful?*

Knowing elasticity helps businesses and policymakers predict the effect of price changes.

- A firm that faces **elastic** demand will lose a lot of sales if it raises prices, so it may keep prices low to sell more units.
- A government that wants to raise revenue with a tax can look at the elasticity of the taxed good; if demand is **inelastic**, the tax will raise revenue without causing a huge drop in consumption.

### *Everyday comparison*

Think of a rubber band. Pull it a little (small change in length) and it stretches a lot (big response). That is like an elastic product: a tiny price increase makes people buy much less.

Now imagine a steel rod. Push it a little and it hardly bends. That is like an inelastic product: even a big price jump does not change how much people buy (e.g., life■saving medicine).

> **In daily life:** When the price of gasoline rises a lot but people still drive almost the same amount, gasoline demand is relatively inelastic.

### *Worked example - coffee shop latte*

**Example 1 (Easy): Latte price change**

A coffee shop raises the price of a latte from 3 to3.30 and sales drop from 200 to 180 cups. Calculate the price elasticity of demand.

- **Find the percentage change in price**

% Delta P = frac3.30 - 3.003.00 times 100% = frac0.303.00 times 100% = 10%

- **Find the percentage change in quantity demanded**

% Delta Q = frac180 - 200200 times 100% = frac-20200 times 100% = -10%

- **Apply the elasticity formula**

E_d = frac% Delta Q% Delta P = frac-10%10% = -1

**Answer:** The price elasticity of demand is -1 (absolute value = 1). This means demand is unit■elastic: the percentage drop in quantity exactly matches the percentage rise in price.

> **Note:** Because we got an elasticity of -1, total revenue (price x quantity) stays the same after the price change. If the absolute value had been larger than 1, revenue would fall; if smaller than 1, revenue would rise.

### *Interpreting the number*

- **|E_d| > 1** -> *elastic*: quantity changes proportionally more than price.
- **|E_d| = 1** -> *unit■elastic*: quantity changes by the same proportion as price.
- **|E_d| < 1** -> *inelastic*: quantity changes proportionally less than price.

The sign tells us the direction (normally negative). When we discuss "elastic" or "inelastic," we refer to the absolute value.

> **Common mistake:** Forgetting to use *percentage* changes. Using raw dollar or unit changes will give the wrong elasticity.

### *Quick checklist for calculating elasticity*

- Record the **initial** price and quantity, then the **new** price and quantity.
- Compute **percentage change in price** using the initial price as the denominator.
- Compute **percentage change in quantity** using the initial quantity as the denominator.
- Divide the percentage change in quantity by the percentage change in price.
- Interpret the absolute value and remember the sign is usually negative.

By mastering this simple formula, you can quickly assess how sensitive consumers are to price moves, a skill that will be reused throughout the rest of the course.

[Figure to be added: A bakery quickly increasing loaf output after a price rise.]

Figure 4.1. A bakery quickly increasing loaf output after a price rise.

[Figure to be added: How a 1 tax is shared between buyers and sellers depending on elasticity.]

Figure 4.2. How a 1 tax is shared between buyers and sellers depending on elasticity.

[Figure to be added: Dynamic pricing in a ride■sharing app during high demand.]

Figure 4.3. Dynamic pricing in a ride■sharing app during high demand.

[Figure to be added: Elastic vs Inelastic Demand]

Figure 4.4. Elastic vs Inelastic Demand




Figure 4.5. How Tax Burden Is Shared




Figure 4.6. Components of the Elasticity Formula




Figure 4.7. Key Elasticity Thresholds

## 4.2 Interpreting Elasticity Values

When we talk about **price elasticity of demand** we are asking: *how much does the quantity that people buy change when the price changes?*

If a tiny price change makes people buy a lot more (or a lot less), we say demand is **elastic**.

If the same price change makes people buy only a little more (or a little less), we say demand is **inelastic**.

When the change in quantity is exactly proportional to the change in price, we call demand **unit■elastic**.

Why does this matter? Knowing whether demand is elastic or inelastic tells us how consumers will react to price moves, and it helps businesses and governments predict revenue, tax burden, or the effect of a subsidy.

> **Definition:** *Elasticity* is a ratio that compares the percentage change in quantity demanded to the percentage change in price.

> E_d=frac% change in quantity demanded% change in price

The sign of E_d is usually negative because price and quantity move in opposite directions (when price goes up, quantity demanded goes down). Economists often drop the minus sign and just talk about the **absolute value** |E_d|.

### *1. Elastic demand (|E_d|>1)*

If |E_d| is larger than 1, the percentage change in quantity is **greater** than the percentage change in price.

*Plain language*: Consumers are very sensitive to price. A small price rise causes a relatively large drop in the amount they buy.

**Everyday comparison** - Think of a luxury chocolate bar that costs \5. If the shop raises the price to5.50 (a 10 % increase) and sales fall from 100 bars to 70 bars (a 30 % drop), the demand is elastic because 30%/10% = 3 > 1.

### *2. Inelastic demand (|E_d|<1)*

If |E_d| is smaller than 1, the percentage change in quantity is **smaller** than the percentage change in price.

*Plain language*: Consumers are not very sensitive to price. Even a big price change only slightly changes the amount they buy.

**Everyday comparison** - Consider gasoline. Suppose the price goes up from \3.00 to3.30 per gallon (a 10 % rise) and the average driver reduces consumption from 12 gallons to 11.5 gallons per week (a 4 % drop). Here 4%/10% = 0.4 < 1, so gasoline demand is inelastic.

### *3. Unit■elastic demand (|E_d|=1)*

When |E_d| equals exactly 1, the percentage change in quantity matches the percentage change in price.

*Plain language*: The drop (or rise) in quantity bought is proportionate to the price change.

**Everyday comparison** - Imagine a concert ticket that costs \50. If the price is increased by 20 % to60 and the number of tickets sold falls from 200 to 160 (also a 20 % drop), the demand is unit■elastic because 20%/20% = 1.

> **In daily life:** Knowing the elasticity helps you predict what will happen if a store raises the price of your favorite snack. If the snack is elastic, you might stop buying it; if it is inelastic, you'll probably keep buying it despite the higher price.

> **Remember:** Elasticity is a *ratio of percentages*, so it has **no units**. It only tells you the *direction* and *strength* of the response.

### *Worked Example*

**Example 2 (Easy): Elastic or Inelastic?**

A coffee shop sells 200 cups of coffee each day at \2 per cup. After raising the price to2.20, sales fall to 180 cups per day. Using the elasticity formula, decide whether demand for coffee at this shop is elastic or inelastic.

- **Calculate the percentage change in price.**

% change in price= frac2.20-2.002.00times 100 = frac0.202.00times 100 = 10%

- **Calculate the percentage change in quantity demanded.**

% change in quantity= frac180-200200times 100 = frac-20200times 100 = -10%

- **Compute the elasticity (use absolute values).**

|E_d| = frac|% change in quantity||% change in price|= frac10%10%=1

**Answer:** The absolute elasticity is 1, so demand for coffee at this shop is **unit■elastic**. A 10 % price rise leads to a 10 % drop in quantity demanded.

### *Interpreting the Numbers*

- **If |E_d|>1 (elastic):**
- A price increase *reduces* total revenue (price x quantity).
- Consumers can easily switch to substitutes or forego the product.
- **If |E_d|<1 (inelastic):**
- A price increase *increases* total revenue.
- Consumers have few substitutes or the product is a necessity.
- **If |E_d|=1 (unit■elastic):**
- Changes in price do **not** affect total revenue; the revenue stays the same because the percentage loss in quantity exactly offsets the higher price.

> **Common mistake:** Forgetting to use *percentage* changes rather than absolute changes. Elasticity compares *relative* changes; using raw dollar or unit numbers will give the wrong result.

Understanding these three categories lets you anticipate how a market will behave when prices move. In the next sections we will see how this knowledge feeds into revenue calculations, tax incidence, and real■world pricing decisions such as those made by ride■sharing companies.

## 4.3 Price Elasticity of Supply

> **Definition:** *Price elasticity of supply* measures how much the quantity supplied of a good changes when its price changes. It tells us whether producers can quickly adjust the amount they bring to market when they receive a higher or lower price.

Why do we care? If supply is very elastic, a small price rise will cause a large increase in output, and producers can respond fast (think of a bakery that can bake more loaves in a day). If supply is inelastic, even a big price jump will only cause a tiny change in output (like a mine that needs years to dig more ore). Knowing the elasticity helps businesses plan production and helps governments predict the impact of taxes, subsidies, or price controls.

### *The formula*

The price elasticity of supply (often written \(E_s\) or simply \( \varepsilon_s\)) is defined as the percentage change in quantity supplied divided by the percentage change in price:

\varepsilon_s \;=\; \frac{\%\ \text{change in quantity supplied}}{\%\ \text{change in price}}

In practice we use the *mid■point* (or arc) method so that the elasticity does not depend on which direction we measure the change. The steps are:

- Compute the change in quantity: \(\Delta Q = Q_2 - Q_1\).
- Compute the change in price: \(\Delta P = P_2 - P_1\).
- Find the average quantity: \(\displaystyle \bar Q = \frac{Q_1 + Q_2}{2}\).
- Find the average price: \(\displaystyle \bar P = \frac{P_1 + P_2}{2}\).
- Plug into the mid■point formula

\varepsilon_s \;=\; \frac{\Delta Q / \bar Q}{\Delta P / \bar P}

The result is a pure number (no units). If \(\varepsilon_s > 1\) supply is **elastic**; if \(\varepsilon_s < 1\) it is **inelastic**; if \(\varepsilon_s = 1\) it is **unit■elastic**.

**Example 3 (Easy): Wheat farmer's response to a price rise**

A farmer increases the price of wheat from \5 to5.50 per bushel and production rises from 1,000 to 1,150 bushels. Find the price elasticity of supply.

- **Calculate the changes.**

\(\Delta Q = 1,150 - 1,000 = 150\) bushels

\(\Delta P = 5.50 - 5.00 = 0.50\) dollars

- **Find the averages.**

\(\displaystyle \bar Q = \frac{1,000 + 1,150}{2} = 1,075\) bushels

\(\displaystyle \bar P = \frac{5.00 + 5.50}{2} = 5.25\) dollars

- **Insert into the mid■point formula.**

\varepsilon_s \;=\; \frac{150 / 1,075}{0.50 / 5.25}

Compute each fraction:

\(\frac{150}{1,075} \approx 0.1395\) (13.95 % change in quantity)

\(\frac{0.50}{5.25} \approx 0.0952\) (9.52 % change in price)

Then

\varepsilon_s \;=\; \frac{0.1395}{0.0952} \;\approx\; 1.46

**Answer:** The price elasticity of supply is about **1.46**, meaning supply is elastic; the farmer can increase output relatively easily when price goes up.

> **In daily life:** Think of a coffee shop that can brew an extra 20 cups per hour when the price per cup rises a little. Because the shop can quickly hire another barista or turn on another espresso machine, its supply is elastic. By contrast, a theater that can only add a few seats each year has inelastic supply.

### *Factors that make supply more elastic*

- Factor - How it works
- **Time horizon** - In the short run, firms are stuck with existing factories, machines, and labor contracts, so supply tends to be inelastic. Over a longer period they can build new plants or train workers, making supply more elastic.
- **Availability of inputs** - If raw materials are abundant and easy to obtain (e.g., water for a bottling plant), producers can raise output quickly, giving a higher elasticity. Scarce inputs (like rare minerals) limit responsiveness.
- **Production flexibility** - Industries that use general■purpose equipment (e.g., bakeries, clothing factories) can switch output levels fast, leading to elastic supply. Highly specialized production lines (e.g., semiconductor fabs) are rigid, so supply is inelastic.
- **Storage capacity** - Goods that can be stored (grain, oil) allow producers to respond to price changes by releasing or holding inventory, increasing elasticity. Perishable goods (fresh fruit) cannot be stored long, reducing elasticity.
- **Number of sellers** - When many firms compete, each can adjust output without affecting the market price much, making the market supply curve flatter (more elastic). A monopoly or a few large firms face a steeper (less elastic) supply curve.

> **Note:** Elasticity is *not* a fixed property of a product; it changes with the circumstances listed above. A smartphone's supply may be elastic in the long run (factories can add lines) but inelastic in the short run (current factories are at full capacity).

### *Interpreting the number*

- **Elastic supply (\(\varepsilon_s > 1\))** - Producers respond strongly to price changes. Policies that raise price (like a tax on a polluting good) may lead to a large increase in output, potentially offsetting the intended effect.
- **Inelastic supply (\(\varepsilon_s < 1\))** - Output changes little when price moves. A tax will raise revenue without causing a big drop in quantity, which is why governments often tax gasoline (short■run supply is relatively inelastic).
- **Unit■elastic supply (\(\varepsilon_s = 1\))** - Percentage change in quantity equals percentage change in price.

Understanding these patterns helps economists predict market reactions and design effective policies.

> **Remember:** The price elasticity of supply tells you *how* quantity supplied reacts *relative* to price, and the value depends on time, input availability, production flexibility, storage, and market structure. Always check these conditions before assuming a supply curve is steep or flat.

## 4.4 Elasticity and Total Revenue

When the price of a good changes, the money a seller earns is the **total revenue** (TR).

Total revenue is simply the price that each unit sells for multiplied by the number of units sold:

TR = P \times Q

where

- P = price per unit
- Q = quantity sold

Understanding how TR moves when price changes is useful for businesses and policymakers. It tells us whether raising the price will bring in more money or whether cutting the price will actually increase earnings. The key to this answer is the **price elasticity of demand** - a measure of how sensitive the quantity demanded is to a change in price.

> **Definition:** *Price elasticity of demand* (denoted E_d) is the percentage change in quantity demanded divided by the percentage change in price.

> E_d = frac%Delta Q%Delta P

If |E_d|>1, demand is **elastic**: consumers react strongly to price changes.

If |E_d|<1, demand is **inelastic**: consumers react only a little.

If |E_d|=1, demand is **unit■elastic**: the percentage change in quantity exactly offsets the percentage change in price.

### *Why elasticity matters for revenue*

Think of a rubber band. When you pull it a little (small price change), it stretches a lot if the band is loose (elastic demand). If the band is stiff (inelastic demand), pulling the same amount only stretches a little.

For a seller, the "stretch" is the change in quantity sold. When the band is loose, a small price cut can cause a large rise in sales, possibly raising total revenue. When the band is stiff, the same price cut only brings a modest increase in sales, and revenue may fall.

Mathematically, we can see the relationship by looking at the sign of E_d:

- **Elastic demand (|E_d|>1)** - a price *decrease* leads to a *larger* percentage increase in quantity, so TR goes **up**. Conversely, a price *increase* makes TR go **down**.
- **Inelastic demand (|E_d|<1)** - a price *decrease* brings a *smaller* percentage increase in quantity, so TR goes **down**. A price *increase* makes TR go **up**.
- **Unit■elastic (|E_d|=1)** - the percentage changes exactly offset, leaving TR unchanged.

> **Note:** The direction of the revenue change depends on *whether* demand is elastic or inelastic, not on the absolute size of the price change.

### *Worked example*

**Example 4 (Medium): Theater ticket price change**

A theater lowers its ticket price from \12 to10. Sales rise from 500 tickets to 560 tickets. Calculate the new total revenue and decide whether revenue increased because demand is elastic.

- **Compute the original total revenue.**

TR_old = P_old times Q_old = 12 times 500 = 6,000

- **Compute the new total revenue after the price change.**

TR_new = P_new times Q_new = 10 times 560 = 5,600

- **Compare the two revenues.**

Since TR_new < TR_old, total revenue fell when price fell.

**Answer:** The new total revenue is \5,600, which is lower than the original6,000. Because revenue dropped after a price cut, demand in this range is **inelastic** (the percentage increase in quantity (12 %) was smaller than the percentage decrease in price (16.7 %)).

### *Interpreting the result*

The example shows that even though more tickets were sold, the loss per ticket outweighed the gain in quantity, so the theater earned less money. This is exactly what we expect when demand is inelastic: consumers do not buy enough extra units to compensate for the lower price.

> **In daily life:** When you see a grocery store put a "sale" sign on a product you already buy a lot of, the store hopes the price cut will boost revenue. If the product is something you need regardless of price (e.g., basic bread), demand is usually inelastic, and the store may actually earn less during the sale.

### *Quick test for elasticity using revenue change*

If you know the direction of the revenue change after a price move, you can infer elasticity without calculating E_d:

- Revenue **up** after a price **down** -> demand is elastic.
- Revenue **down** after a price **down** -> demand is inelastic.
- Revenue unchanged -> demand is unit■elastic.

This shortcut is handy when data are limited.

### *Why businesses care*

A firm deciding whether to raise the price of a premium coffee will look at the elasticity of its customers. If customers are very sensitive (elastic), a small price hike could cause a big drop in sales, reducing revenue. If they are loyal and less price■sensitive (inelastic), the firm can raise price and increase revenue.

> **Common mistake:** Assuming that a higher price always means higher revenue. The relationship holds only when demand is inelastic; otherwise, higher price can shrink revenue.

### *Summary of the revenue■elasticity link*

- Price change - Elastic demand ( - E_d - >1) - Inelastic demand ( - E_d - <1)
- Decrease - Revenue **increases** - Revenue **decreases**
- Increase - Revenue **decreases** - Revenue **increases**

Remember that elasticity is not fixed; it can vary at different price levels. Therefore, firms often test small price adjustments and observe the resulting revenue before making larger moves.

> **Remember:** Total revenue moves in the *opposite* direction of price when demand is elastic, and in the *same* direction as price when demand is inelastic. This simple rule helps managers make quick, informed pricing decisions.

## 4.5 Policy Application: Taxes

When a government adds a tax to a good, the price that buyers pay and the price that sellers receive are no longer the same.

The **tax burden** (also called the incidence of the tax) tells us how much of the tax is paid by consumers and how much is paid by producers.

Which side carries the larger share depends on how **elastic** (responsive) each side is to price changes.

> **Definition:** *Elasticity* measures how much the quantity demanded or supplied changes when the price changes.

> • *Price elasticity of demand* (PED) = % change in quantity demanded / % change in price.

> • *Price elasticity of supply* (PES) = % change in quantity supplied / % change in price.

If demand is **elastic** (large absolute value), buyers quickly reduce the amount they buy when the price rises, so they can avoid most of the tax.

If demand is **inelastic** (small absolute value), buyers keep buying almost the same amount even if the price goes up, so they end up paying most of the tax.

The same logic applies to sellers: an elastic supply means producers can easily cut back production or switch to other products, shifting the tax onto buyers; an inelastic supply means producers cannot change output much, so they bear more of the tax.

### *Why the relative elasticities matter*

Think of a tug■of■war rope. The tax is a weight placed in the middle of the rope. If the demand side pulls hard (highly inelastic), the rope moves toward the supply side, meaning consumers bear the weight. If the supply side pulls hard (highly inelastic), the rope moves toward the demand side, meaning producers bear the weight. The side that is *less* able to move - i.e., the side with the smaller absolute elasticity - holds more of the tax.

Mathematically, the share of the tax paid by consumers ($S_c$) and producers ($S_p$) can be expressed as:

$$S_c = \frac{|\text{PES}|}{|\text{PED}| + |\text{PES}|}, \qquad$$

$$S_p = \frac{|\text{PED}|}{|\text{PED}| + |\text{PES}|}$$

where $|\text{PED}|$ and $|\text{PES}|$ are the absolute values of the elasticities (we ignore the sign because we only need the magnitude). The side with the larger denominator fraction pays less.

> **Note:** The formulas use absolute values because the sign of PED is negative (price up, quantity down) while PES is positive. Only the size matters for incidence.

### *Everyday comparison*

Imagine you buy a daily coffee. If the price goes up by 0.50, you might still buy it because you love coffee (demand is inelastic). The coffee shop, however, can't easily stop making coffee; its supply is also inelastic in the short run. If the city adds a0.20 tax per cup, most of that 0.20 will appear in the price you pay because you are not willing to give up coffee. Conversely, for a luxury car, if a tax is added, buyers will quickly decide not to purchase, so the car dealer will have to absorb most of the tax to keep sales.

### *Worked example*

**Example 5 (Medium):** A 2 per pack tax on cigarettes is imposed. Demand elasticity is -0.4 and supply elasticity is 0.6. Determine which side bears a larger share of the tax.

- **Write down the absolute elasticities.**

$|\text{PED}| = 0.4, \qquad |\text{PES}| = 0.6$

- **Calculate the consumer's share using the formula.**

$$S_c = \frac{|\text{PES}|}{|\text{PED}| + |\text{PES}|}$$

$$= \frac{0.6}{0.4 + 0.6}$$

$$= \frac{0.6}{1.0}$$

$$= 0.60$$

- **Calculate the producer's share (or note that it is the remainder).**

$S_p = 1 - S_c = 1 - 0.60 = 0.40$

**Answer:** Consumers bear 60 % of the 2 tax (about1.20 per pack) and producers bear 40 % (about 0.80 per pack). Because demand is more inelastic than supply, buyers carry the larger burden.

### *Implications for policy*

Policymakers can use this knowledge to achieve specific goals:

- **Raise revenue with minimal market distortion:** Tax goods with inelastic demand (e.g., cigarettes, gasoline). The tax will raise money while causing only a small reduction in quantity, so the market stays close to its original level.
- **Correct a negative externality without over■penalising producers:** If the supply side is very elastic (e.g., a commodity that can be imported easily), a tax will fall mostly on consumers, which may be desirable if the goal is to discourage consumption.
- **Avoid hurting small producers:** When supply is inelastic (farmers cannot quickly change output), a tax would fall heavily on them. In such cases, a subsidy or a tax exemption might be preferable.

> **In daily life:** When you see a "sugar tax" on soft drinks, most of the extra cost you pay is because people are not very responsive to price changes for sugary drinks; the tax is designed to raise money and slightly reduce consumption.

> **Common mistake:** Assuming that the side that *receives* the tax money (the government) automatically pays it. The tax is levied on transactions, but the economic burden is split between buyers and sellers according to elasticities.

### *Quick checklist for students*

- Identify the sign■less elasticities of demand and supply.
- Plug them into the incidence formulas.
- Interpret the percentages: larger consumer share -> more inelastic demand relative to supply.
- Relate the result to real■world policy goals.

Understanding tax incidence helps you see why some taxes feel "fair" to the public while others spark controversy. It also shows the power of elasticity as a tool for predicting how markets adjust to government actions.

## 4.6 Policy Application: Subsidies

When a government gives producers or consumers extra money - a **subsidy** - it changes the incentives that drive buying and selling.

Think of a subsidy as a "price discount" that the government pays for you. If you buy a cup of coffee and the government adds 10 cents to the seller, you still pay the same price, but the seller receives 10 cents more. This extra payment shifts the supply curve (if it goes to producers) or the demand curve (if it goes to consumers).

> **Definition:** *Subsidy* - a payment from the government to buyers or sellers that lowers the effective price paid by buyers or raises the effective price received by sellers.

The size of the market's response depends on **elasticity** - a measure of how much quantity changes when price changes.

- **Elastic demand** (|elasticity| > 1) means buyers are very responsive: a small price change causes a large change in quantity bought.
- **Inelastic demand** (|elasticity| < 1) means buyers are not very responsive: quantity changes only a little when price changes.
- **Elastic supply** (elasticity > 1) means producers can quickly change how much they produce when price changes.
- **Inelastic supply** (elasticity < 1) means producers cannot adjust output easily.

Why does this matter for subsidies?

If the subsidy is given to the side of the market that is **more elastic**, the quantity change will be larger, and the government will have to spend more money to achieve the same price effect. Conversely, if the subsidy hits the **more inelastic** side, the quantity change is smaller, so the government's cost is lower, but the price effect for consumers may be smaller too.

> **Note:** The incidence (who actually enjoys the price change) depends on the relative elasticities. The side that is less elastic bears a larger share of the subsidy benefit.

### *Step■by■step reasoning*

- **Identify which curve receives the subsidy.**
- A *producer subsidy* shifts the supply curve downward (or to the right).
- A *consumer subsidy* shifts the demand curve upward (or to the right).
- **Write the new equilibrium condition.**

The market clears where quantity demanded equals quantity supplied. After a subsidy, the effective price faced by one side changes by the subsidy amount \(s\).

- **Use elasticity to estimate the change in price and quantity.**

For small changes, the percentage change in quantity (\(\Delta Q/Q\)) is the elasticity times the percentage change in price (\(\Delta P/P\)).

fracDelta QQ= varepsilon times fracDelta PP

where \(\varepsilon\) is the relevant elasticity (demand or supply).

- **Solve for the new price and quantity.**

Combine the two equations (one for demand, one for supply) and solve simultaneously.

### *Worked example*

**Example 6 (Hard): Corn subsidy**

The government offers a \1 subsidy per kilogram of corn. Demand elasticity is \(-0.8\) and supply elasticity is \(1.2\). Calculate the change in equilibrium price and quantity.

- **Set up the price change for each side.**
- Because the subsidy goes to producers, the supply curve shifts down by \1.
- The effective price received by producers is \(P_s = P + 1\), while the price paid by consumers remains \(P\).
- **Express percentage price changes.**

Let the original equilibrium price be \(P_0\). The percentage change in the price received by suppliers is

fracDelta P_sP_0= frac1P_0

(since the subsidy is a fixed \1). The price paid by consumers does not change directly, but quantity will adjust.

- **Link quantity changes to price changes using elasticities.**
- For supply:

fracDelta QQ_0= varepsilon_stimesfracDelta P_sP_0=1.2 times frac1P_0

- For demand: the price consumers face falls because the market price will adjust downward. Let the new consumer price be \(P = P_0 + \Delta P\). The percentage change in consumer price is \(\frac{\Delta P}{P_0}\). Demand quantity change is

fracDelta QQ_0= varepsilon_dtimesfracDelta PP_0= -0.8 times fracDelta PP_0

- **Equate the two expressions for \(\Delta Q/Q_0\) (market clears).**

1.2 times frac1P_0= -0.8 times fracDelta PP_0

Solve for \(\Delta P\):

Delta P = -frac1.20.8= -1.5 dollars

So the consumer price falls by \1.5.

- **Find the change in quantity.**

Substitute \(\Delta P\) back into the demand formula:

fracDelta QQ_0= -0.8 times frac-1.5P_0= frac1.2P_0

The percentage increase in quantity is the same as the supply side result, \(1.2/P_0\). If the original price was, say, \5 per kg, then

fracDelta QQ_0= frac1.25=0.24

meaning quantity rises by 24 %.

**Answer:** The consumer price falls by \1.5, and the equilibrium quantity rises by 24 % (e.g., from 100 kg to 124 kg if the original quantity was 100 kg). The subsidy cost to the government is the subsidy per unit (1) times the new quantity (124 kg) = \124.

### *What the example shows*

- Because **supply is more elastic** (1.2) than demand (■0.8), the quantity response is driven mainly by producers.
- The **price paid by consumers falls** more than the subsidy amount because the inelastic demand does not expand enough to absorb the whole subsidy.
- The government's total outlay depends on the **new quantity**, which is larger when the side receiving the subsidy is elastic.

> **In daily life:** Think of a school cafeteria that receives a voucher for each sandwich sold. If the cafeteria can quickly make more sandwiches (elastic supply), the voucher leads to many more sandwiches being produced, raising the total cost of the program. If the students are not very sensitive to sandwich price (inelastic demand), they will buy almost the same number even without the voucher, so the program's impact on quantity is small.

> **Remember:** When a subsidy is given to the more elastic side of the market, quantity changes a lot and government spending rises; when it is given to the more inelastic side, quantity changes little and the price benefit to the other side is smaller. This principle helps policymakers decide where to target subsidies for the desired effect.

## 4.7 Real■World Case Study: Ride■Sharing Pricing

Imagine you are using a ride■sharing app during a busy Friday night. The app shows a higher fare than usual because many people want rides at the same time. This is called **dynamic pricing** - the price changes in response to how many riders want a ride and how many drivers are available.

Why does the app do this? It is using the idea of **price elasticity of demand**.

- **Price elasticity of demand** measures how much the quantity demanded (here, the number of rides) changes when the price changes.
- If demand is **elastic**, a small price increase causes a large drop in rides.
- If demand is **inelastic**, the same price increase only causes a small drop in rides.

The app wants to keep its drivers earning enough while not losing too many riders. Understanding elasticity helps it set a price that balances those goals.

> **Definition:** *Price elasticity of demand* (PED) = % change in quantity demanded / % change in price.

> **Note:** A negative sign is usually omitted; we report the absolute value because we know demand falls when price rises.

### *Step■by■step: How elasticity works for ride■sharing*

- **Calculate the percentage change in price.**

The app may raise the *base fare* (the starting price before any surge multiplier).

- **Calculate the percentage change in the number of rides.**

This is the *quantity demanded* - how many people actually request a ride.

- **Divide the two percentages** to obtain the elasticity.

If the result is greater than 1, demand is elastic; if it is less than 1, demand is inelastic.

> **In daily life:** Think of a coffee shop that raises the price of a latte by 10 %. If the shop sees a 30 % drop in latte sales, the demand for lattes is elastic (30 % / 10 % = 3).

### *What does elasticity tell the app about revenue?*

Total revenue (TR) = price x quantity.

- When demand is **elastic**, raising the price *reduces* total revenue because the loss in quantity outweighs the higher price.
- When demand is **inelastic**, raising the price *increases* total revenue because the quantity falls only a little.

The app can therefore predict whether a surge price will bring more money overall or will scare riders away.

### *Worked Example*

**Example 7 (Hard):** *Base■fare hike and ridership drop*

The ride■sharing app raises its base fare by 15 %. Ridership falls from 10,000 rides per week to 8,500 rides per week. Compute the price elasticity of demand and discuss whether the price change will increase or decrease total weekly revenue.

- **Find the percentage change in price.**

The price increase is 15 %, so

% Delta P = frac15100=0.15 (or 15%)

- **Find the percentage change in quantity demanded.**

Quantity falls from 10,000 to 8,500, a drop of 1,500 rides.

% Delta Q = frac{8,500-10,000}{10,000}= frac{-1,500}{10,000}= -0.15 (or -15%)

- **Compute elasticity using the definition.**

PED= frac|% Delta Q|% Delta P= frac0.150.15=1.0

The absolute value is 1, indicating **unit■elastic** demand.

**Answer:** The price elasticity of demand is 1.0 (unit■elastic). Because demand is exactly unit■elastic, the increase in price will leave total weekly revenue unchanged: the 15 % higher fare is exactly offset by a 15 % lower number of rides.

### *Implications for drivers and riders*

- **Drivers:** When the app's surge multiplier pushes the price into a range where demand becomes elastic (elasticity > 1), fewer riders request trips. Drivers may end up waiting longer for passengers, reducing their hourly earnings even though each ride pays more.
- **Riders:** If the price increase moves demand into the elastic region, riders quickly switch to alternatives (public transport, walking, or other apps). This can lower overall usage of the platform.

> **Common mistake:** Assuming that a higher price always means higher revenue. Without checking elasticity, the app may actually lose money.

### *Using elasticity to set a "sweet spot"*

Suppose the app wants to test a smaller price increase, say 5 %. It can estimate the expected change in rides using the previously calculated elasticity (≈1).

- Expected % change in rides = elasticity x % change in price = 1 x 5 % = 5 % drop.
- New expected rides = 10,000 x (1 - 0.05) = 9,500 rides.
- New price = original price x 1.05.

Total revenue before the change = P times 10,000.

Total revenue after the change = (1.05P) times 9,500 = 9,975P.

Because 9,975P < 10,000P, even a modest 5 % hike would **decrease** revenue when demand is unit■elastic. The app would need to keep the price increase small enough that demand becomes inelastic (elasticity < 1) before it can raise revenue.

### *Policy angle: How taxes or subsidies would work*

If a city imposes a tax of 0.50 per ride, the effective price to riders rises. Using the same elasticity, the city can predict the drop in rides and the resulting tax revenue. Conversely, a subsidy (e.g., a discount coupon) lowers the price; if demand is elastic, the subsidy may dramatically increase rides, boosting overall traffic and possibly reducing congestion if more people share rides.

> **Remember:** Elasticity is a *ratio* that tells you the direction and strength of response. It does not give the exact number of rides; you still need the original quantities and prices to calculate changes.

### *Quick checklist for the app*

- **Step 1:** Estimate current elasticity from recent price■quantity data.
- **Step 2:** Decide on a target revenue change (increase or maintain).
- **Step 3:** Choose a price adjustment that keeps the resulting elasticity on the inelastic side if revenue growth is desired.
- **Step 4:** Monitor actual ridership after the change; update the elasticity estimate regularly because rider behavior can shift (e.g., after a new competitor enters the market).

By treating each price move as a small experiment and applying the elasticity formula, the ride■sharing platform can fine■tune its dynamic pricing, keep drivers earning fairly, and avoid pushing riders away. This systematic approach turns what might feel like "random surge pricing" into a predictable, data■driven strategy.

### Terminology at a glance

| Term | Meaning |
|---|---|
| Price elasticity of demand (PED) | The percentage change in quantity demanded divided by the percentage change in price. |
| Elastic demand | When the absolute value of PED is greater than 1; quantity changes a lot relative to price. |
| Inelastic demand | When the absolute value of PED is less than 1; quantity changes little relative to price. |
| Unit■elastic demand | When the absolute value of PED equals 1; quantity changes proportionally with price. |
| Price elasticity of supply (PES) | The percentage change in quantity supplied divided by the percentage change in price. |
| Total revenue | Revenue equals price times quantity sold (TR = P x Q). |
| Tax incidence | The division of a tax's burden between buyers and sellers, determined by relative elasticities. |
| Subsidy | A government payment that lowers the effective price for buyers or raises the price received by sellers. |

### Practical application

Exercise 4.1:

Exercise 4.2:

Exercise 4.3:

Exercise 4.4:

Exercise 4.5:

Exercise 4.6:

### Checklist

[ ] Define price elasticity of demand and supply

[ ] Calculate elasticity using percentage changes

[ ] Interpret elasticity values as elastic, inelastic, or unit■elastic

[ ] Analyze how elasticity affects total revenue

[ ] Apply elasticity concepts to tax and subsidy policy

LESSON 5

# 5. Production and Costs

### Lesson overview

This lesson explains the factors of production, distinguishes short■run and long■run costs, and introduces economies of scale. Students learn how firms decide how much to produce and why costs change over time.

### Learning outcomes

- Identify the four factors of production and give real■world examples
- Explain the difference between short■run and long■run cost concepts
- Calculate total, average, and marginal costs in the short run
- Describe economies of scale and recognize them in everyday businesses
- Analyse a simple scenario to decide whether a firm is operating at an efficient scale

### 5.1 1. Factors of Production

Factors of production are the building blocks that any economy uses to create goods and services.

In the simplest terms, they are the **inputs** - the things we put into the production process - that allow a bakery to bake bread, a factory to assemble phones, or a teacher to prepare a lesson. Understanding these inputs helps us see why prices change, why some businesses grow faster than others, and how resources are allocated in society.

> **Definition:** *Factors of production* are the four essential inputs needed to produce anything: **land**, **labor**, **capital**, and **entrepreneurship**.

### *1. Land*

The word *land* does not only mean a plot of soil. In economics it stands for **all natural resources** that are not created by humans. This includes farmland, forests, minerals, water, and even sunlight. These resources are limited in supply; you cannot create more sunlight by working harder.

*Why it matters*: If a company needs iron to make cars, the amount of iron available will affect how many cars can be produced and at what cost.

> **In daily life:** Think of a coffee shop that needs a location (the building and the space around it) and a steady supply of water. The building and water are part of "land" because they come from nature or are owned as natural assets.

### *2. Labor*

*Labor* is the human effort used in production. It includes physical work (like a construction worker lifting bricks) and mental work (like a software developer writing code). Labor is measured in **hours worked** and its quality depends on education, training, and health.

*Why it matters*: More skilled workers can produce more output in the same amount of time, which can lower the cost per unit.

> **Note:** Labor is not just the number of people; it is the *effective* work they contribute. Two workers with different skill levels do not produce the same amount of output.

### *3. Capital*

*Capital* refers to man■made tools, machines, buildings, and equipment that help workers be more productive. A tractor, a computer, a factory building - all are forms of capital. Capital is different from money; it is the **physical assets** that are used repeatedly in production.

*Why it matters*: Capital allows a small workshop to become a large factory. For example, a hand■sewing needle can make one stitch per minute, while a sewing machine (capital) can make dozens of stitches in the same time.

> **Common mistake:** Confusing money with capital. Money is a *medium of exchange*; capital is the *productive equipment* bought with that money.

### *4. Entrepreneurship*

*Entrepreneurship* is the ability and willingness to combine land, labor, and capital to create new products or services. An entrepreneur takes **risk** - the chance that the venture might fail - and seeks **profit**, the reward for successful risk■taking. Entrepreneurs also innovate, finding better ways to use the other three factors.

*Why it matters*: Without entrepreneurs, resources would sit idle. They are the spark that turns raw inputs into finished goods, like a chef turning ingredients (land) and kitchen tools (capital) into a meal.

> **Remember:** Entrepreneurship is about *organizing* the other factors and *bearing risk* for potential reward.

### *Putting the pieces together*

When a firm decides to produce, it must acquire each factor:

Total Input = Land + Labor + Capital + Entrepreneurship

The firm then mixes these inputs in a *production function*, often written as Q = f(L, K, N, E), where Q is the quantity of output, L is labor, K is capital, N represents natural resources (land), and E stands for entrepreneurship. This equation simply says that the amount of output depends on how much of each input is used and how efficiently they are combined.

Understanding the four factors helps us answer practical questions:

- How much will a new factory cost to build? (Capital and land)
- Will hiring more workers speed up production? (Labor)
- Can a new technology replace some natural resources? (Capital vs. land)
- Who decides which combination to use? (Entrepreneurship)

By recognizing each factor and its role, students can better analyze real■world economic situations, from a farmer deciding whether to buy a new tractor to a tech startup planning its first product launch.

[Figure to be added: Workers and equipment turning raw ingredients into loaves of bread.]

Figure 5.1. Workers and equipment turning raw ingredients into loaves of bread.

[Figure to be added: Machines that run continuously (fixed) and stations where workers add materials (variable).]

Figure 5.2. Machines that run continuously (fixed) and stations where workers add materials (variable).

[Figure to be added: A small workshop expanding into a large plant to lower average costs.]

Figure 5.3. A small workshop expanding into a large plant to lower average costs.

[Figure to be added: Factors of Production]

Figure 5.4. Factors of Production

[Figure to be added: Fixed vs Variable Costs]

Figure 5.5. Fixed vs Variable Costs

[Figure to be added: Short■Run vs Long■Run Decision Process]

Figure 5.6. Short■Run vs Long■Run Decision Process

[Figure to be added: Key Cost Formulas]

Figure 5.7. Key Cost Formulas

## 5.2 2. Short■Run vs Long■Run

**Definition:** The **short run** is a time period in which at least one **factor of production** (such as land, a factory building, or major machinery) cannot be changed. The **long run** is a later period when **all** factors can be varied; the firm can rent new space, buy new machines, or even relocate.

**Note:** In the short run the firm faces a **fixed factor** and therefore a **fixed cost**. In the long run there are no fixed costs; every cost can be adjusted.

> **In daily life:** Think of a bakery that owns a brick■oven. While the oven is being used, the baker cannot instantly get a bigger oven; the oven size is **fixed** for a while. If the bakery decides later to open a second shop, it can rent a new space and buy another oven - now all inputs are **variable**.

> **Remember:** The distinction is about *time* and *flexibility*, not about the size of the business.

### *Why the distinction matters*

Economists need the short■run/long■run split because costs behave differently when some inputs are stuck at a certain level.

- In the short run, the firm can only change **variable inputs** (like labor or raw materials) while the fixed input stays the same.
- In the long run, the firm can change **all inputs**, so it can choose the most efficient combination of land, capital, and labor for any level of output.

Understanding which period we are in tells us which cost curves to use, how to predict the firm's response to price changes, and how to plan investment.

### *Fixed vs. Variable Factors*

- Term - Meaning - Example
- **Fixed factor** - An input whose quantity cannot be altered in the short run. - The size of a factory building.
- **Variable factor** - An input that can be increased or decreased quickly. - Number of workers hired each week.
- **Fixed cost** - Cost tied to a fixed factor; it does not change with output. - Monthly rent for the factory.
- **Variable cost** - Cost that changes as output changes. - Cost of flour used to bake each loaf.

When the firm moves to the long run, the previously fixed factor becomes variable, and its associated cost becomes variable as well.

### *Worked Example*

**Example 1 (Basic):** A small workshop rents a workshop space for 2,000 per month. The rent is a fixed cost because the space cannot be enlarged instantly. The workshop also buys raw wood, costing5 per unit of furniture produced. If the workshop makes 300 pieces in a month, what are its total fixed cost, total variable cost, and total cost for that month?

- **Identify the fixed cost.**

The rent does not depend on output, so

Fixed Cost= 2,000.

- **Calculate the variable cost.**

Variable cost equals the cost per unit times the number of units:

Variable Cost=5 /unittimes 300 units=1,500.

- **Add fixed and variable costs to get total cost.**

Total Cost= Fixed Cost+Variable Cost= 2,000+1,500= 3,500.

**Answer:** The workshop's fixed cost is **2,000**, its variable cost is **1,500**, and total cost for the month is **3,500**.

This example shows the short■run situation: the rent is fixed, while the cost of wood varies with output.

### *Moving to the Long Run*

Suppose the same workshop decides to move to a larger facility where it can produce more furniture. In the long run, the rent becomes a **choice**: the firm can select a bigger space with higher rent or a smaller one with lower rent. The cost of the space is now a **variable cost** because the firm can adjust it.

Mathematically, the **total cost (TC)** in the long run is expressed only with variable components:

TC = sum_i=1^n cost of input_i times quantity of input_i,

where each input \(i\) (land, capital, labor, materials) can be changed.

### *Short■Run vs. Long■Run Cost Curves*

- **Short■run total cost (SRTC)** includes a fixed part:

SRTC = FC + VC,

where \(FC\) is fixed cost and \(VC\) is variable cost.

- **Long■run total cost (LRTC)** has no fixed part:

LRTC = VC_all.

Because the short■run curve is "pinned down" by the fixed cost, it lies **above** the long■run curve for any given output level (except at the point where the fixed factor exactly matches the long■run optimum).

### *Why the curves differ*

Imagine you own a lemonade stand. In the short run you have one table (fixed). You can hire more helpers (variable) to serve more customers, but you cannot add another table until next summer. Your cost of the table stays the same no matter how many cups you sell, so the short■run cost curve starts higher.

When summer arrives and you can buy a second table, you can spread the fixed cost of the first table over more sales, and you may even find a cheaper supplier for lemons because you buy in bulk. The long■run cost curve reflects these new choices and therefore lies lower.

### *Key Take■aways*

- The **short run** has at least one fixed factor -> fixed cost present.
- The **long run** has all factors variable -> no fixed cost.
- Cost calculations differ: add a fixed component in the short run, omit it in the long run.
- Understanding the period helps firms decide whether to adjust labor, invest in new capital, or change production scale.

By keeping the distinction clear, you can correctly apply cost formulas, interpret cost curves, and make sound production decisions.

## 5.3 3. Short■Run Cost Curves

Total cost (TC) is the amount of money a firm spends to produce a certain quantity of output.

It is made up of two parts:

- **Fixed cost (FC)** - money that must be paid even if the firm produces nothing (rent, insurance, salaries of permanent staff).
- **Variable cost (VC)** - money that changes with the amount produced (raw materials, hourly wages, electricity for machines).

So

TC = FC + VC

> **Definition: Average total cost (ATC)** is the total cost per unit of output.

> ATC = fracTCQ where Q is the quantity produced.

> **Definition: Average variable cost (AVC)** is the variable cost per unit of output.

> AVC = fracVCQ

> **Definition: Marginal cost (MC)** is the extra cost of producing one more unit.

> MC = fracDelta TCDelta Q = fracDelta VCDelta Q

> (Because fixed cost does not change when output changes, the change in total cost equals the change in variable cost.)

Why do we care?

ATC tells us how much each unit "costs on average," which helps a firm decide whether the price it can charge covers its costs.

MC tells us the cost of the *next* unit; comparing MC to the market price shows whether it is profitable to expand production a little more.

### *Everyday comparison*

Imagine you run a lemonade stand. You rent a table for 10 a day - that is your fixed cost. Every cup of lemonade needs0.50 of lemons, sugar, and cups - that is your variable cost per cup.

*If you sell 0 cups, you still pay 10 (FC). If you sell 20 cups, your VC is0.50 x 20 = 10, so TC =10 + 10 =20.*

Your ATC at 20 cups is 20 / 20 =1 per cup. The MC of the 21st cup is just the extra 0.50 of ingredients, because the rent does not change.

### *Worked examples*

### *Example 2 (Easy): Bakery cost basics*

A bakery has fixed rent of 200 per week and variable ingredient costs of2 per loaf. Find TC, ATC, and MC for producing 100 loaves.

- **Calculate variable cost.**

VC = (cost per loaf) times Q = 2 times 100 = 200

- **Add fixed cost to get total cost.**

TC = FC + VC = 200 + 200 = 400

- **Find average total cost.**

ATC = fracTCQ = frac400100 = 4

- **Determine marginal cost.**

Because each additional loaf requires exactly 2 of ingredients and rent does not change,

MC = 2

**Answer:** For 100 loaves, TC = 400, ATC =4 per loaf, and MC = 2 per additional loaf.

#### *Example 3 (Medium): Computing AVC and MC*

A factory's variable cost rises from 0 to500 as output goes from 0 to 50 units, then to 1,200 at 100 units. Compute AVC and MC at 50 and 100 units.

- **At 50 units, find AVC.**

AVC_50 = frac{VC_50}{Q_50} = frac50050 = 10

- **At 100 units, find AVC.**

AVC_100 = frac{VC_100}{Q_100} = frac1200100 = 12

- **Compute MC from 0 to 50 units.**

Change in VC = 500 - 0 = 500, change in Q = 50 - 0 = 50

MC_0to50 = fracDelta VCDelta Q = frac50050 = 10

- **Compute MC from 50 to 100 units.**

Change in VC = 1200 - 500 = 700, change in Q = 100 - 50 = 50

MC_50to100 = frac70050 = 14

**Answer:**

- At 50 units, AVC = 10 per unit and MC =10 per unit.
- At 100 units, AVC = 12 per unit and MC =14 per unit.

> **Note:** MC equals the change in VC because fixed cost never changes with output. If you ever see a formula that adds Delta FC, it is a mistake.

> **Common mistake:** Forgetting that ATC includes both fixed and variable components. Some students compute ATC as just VC/Q, which actually gives AVC. Remember **ATC = AVC + AFC**, where AFC = FC/Q.

> **Remember:** When MC is below ATC, producing one more unit pulls the average cost down; when MC is above ATC, the average cost rises. This relationship explains the typical U■shaped ATC curve in the short run.

#### *Putting it together*

- **Start with FC** - a constant amount, independent of output.
- **Add VC** - grows as you make more.
- **Total cost** is the sum.
- **Divide TC by output** to get ATC; divide VC by output to get AVC.
- **Find the change in VC** when output rises by one unit to obtain MC.

Understanding these three cost measures lets you read the short■run cost curves that textbooks draw:

- The **AVC curve** usually falls at first (spreading out cheap inputs) then rises (crowding of resources).
- The **ATC curve** lies above AVC because it also spreads the fixed cost over each unit.
- The **MC curve** intersects both AVC and ATC at their minimum points - a useful fact for profit■maximizing decisions.

By mastering the definitions, the simple arithmetic, and the intuition behind each cost, you will be ready to interpret the graphical representations that follow in later sections.

## 5.4 4. Long■Run Cost and Returns to Scale

All inputs can be changed in the long run.

In the short run at least one factor (for example a factory building) is fixed, so the firm can only adjust labor or raw material.

In the **long run** the firm is free to vary **every** factor of production - labor, machines, land, and even the size of the plant itself. This flexibility lets the firm choose the most efficient scale of operation.

> **Definition:** *Long■run average cost (LRAC)* is the lowest possible average cost of producing any given level of output when the firm can adjust all inputs.

Why do we care about LRAC?

If a firm knows the cheapest way to produce each output level, it can decide whether to expand, stay the same, or shrink. The shape of the LRAC curve tells us about **returns to scale**, which describe how output responds when **all** inputs are increased proportionally.

### *Returns to scale*

- **Increasing returns to scale**: doubling every input more than doubles output.
- **Constant returns to scale**: doubling every input exactly doubles output.
- **Decreasing returns to scale**: doubling every input less than doubles output.

These concepts are linked to the LRAC curve:

- When the firm experiences increasing returns, the LRAC falls as output rises (the curve slopes downward).
- With constant returns, the LRAC is flat (horizontal).
- With decreasing returns, the LRAC rises (upward■sloping).

Thus, the LRAC curve is a visual summary of how average costs change with scale, and the slope of the curve reveals the type of returns to scale the firm enjoys.

### *Worked example - checking returns to scale*

**Example 4 (Medium): Tech startup scaling**

A tech startup can double both labor and equipment. Output rises from 1,000 to 2,200 units. Determine if the firm experiences increasing, constant, or decreasing returns to scale.

- **State the proportional change in inputs.**

The firm doubles each input, so the input factor change is a factor of 2.

- **Calculate the proportional change in output.**

Output change factor= frac2,2001,000=2.2

- **Compare the two factors.**
- If output factor > 2 -> increasing returns.
- If output factor = 2 -> constant returns.
- If output factor < 2 -> decreasing returns.

Here 2.2 > 2, so the output grew more than the inputs.

**Answer:** The startup exhibits **increasing returns to scale** because output increased by a factor of 2.2 when all inputs were doubled.

### *From the example to the LRAC curve*

When a firm enjoys increasing returns, each additional unit of output costs less than the previous one, so the LRAC curve falls. Imagine a bakery that can buy a larger oven and hire more bakers. If the larger oven lets each baker bake 20 % more loaves, the cost per loaf drops, and the LRAC moves downwards.

Conversely, if the same bakery kept expanding but each new baker added only a tiny extra amount of loaves, the average cost would rise and the LRAC would slope upward, indicating decreasing returns.

> **In daily life:** Think of buying a bulk pack of pencils. When you buy a small box, the price per pencil is higher. Buying a huge box lowers the price per pencil because you spread fixed costs (shipping, packaging) over many more pencils. That is the same idea as a falling LRAC due to increasing returns.

### *How to draw the LRAC curve*

- **Choose several output levels** (e.g., 100, 200, 400 units).
- **For each level, find the cheapest combination of inputs** that can produce that output.
- **Compute average cost** = total cost / output.
- **Plot output on the horizontal axis and average cost on the vertical axis.**
- **Connect the points smoothly**; the resulting curve shows the lowest cost possible at each scale.

If the points fall as you move right, the firm has increasing returns at those sizes. If they level off, the firm is at constant returns. If they rise, the firm faces decreasing returns.

> **Note:** The LRAC is not a single "average cost" line for a given plant; it is the **envelope** of many short■run average cost (SRAC) curves, each representing a different fixed plant size. The firm picks the SRAC that lies lowest for the desired output, and the collection of those lowest points forms the LRAC.

### *Why returns to scale matter for decision■making*

When a firm plans to enter a new market or launch a product, it must estimate the cost of producing the required quantity. If the LRAC is falling at that quantity, the firm can expect lower per■unit costs as it expands - making large■scale production attractive. If the LRAC is rising, the firm may decide to stay small or look for ways to improve technology to shift the LRAC downward.

> **Remember:** In the long run, *all* inputs are variable, so the LRAC tells you the cheapest way to produce any output, and its slope tells you whether you get **increasing**, **constant**, or **decreasing** returns to scale. Use this insight to choose the optimal size of your operation.

## 5.5 5. Economies of Scale

> **Definition:** *Economies of scale* are the cost advantages that a firm obtains when it increases the size of its production.

> When a company produces more units, the average cost of each unit (average total cost, **ATC**) can fall because resources are used more efficiently.

### *Why economies of scale matter*

If a firm can lower its ATC, it can either raise profit while keeping prices the same, or lower prices and attract more customers. Understanding the sources of these cost savings helps managers decide whether to expand, merge, or stay small.

### *Internal vs. external economies*

*Internal* economies arise **inside** the firm, from its own decisions and resources.

*External* economies arise **outside** the firm, from conditions in the whole industry or region.

### *1. Technical economies*

These come from using larger or more advanced machines, better production layouts, or automation. A bigger plant can spread the cost of expensive equipment over many more units.

> **In daily life:** Think of a bakery that buys a big industrial oven. The oven costs a lot, but if the bakery bakes 10,000 loaves a month instead of 1,000, the oven's cost per loaf becomes very small.

### *2. Managerial economies*

When a firm grows, it can afford specialized managers (e.g., a dedicated purchasing manager). Specialization reduces mistakes and speeds up decisions, lowering overall costs.

> **Note:** A small shop owner may handle buying, sales, and accounting alone. A larger firm can hire a professional accountant, whose expertise avoids costly errors.

### *3. Financial economies*

Large firms can borrow money at lower interest rates because banks view them as less risky. They may also issue bonds or sell shares more cheaply.

> **Common mistake:** Assuming that any firm can get cheap loans simply by producing more. The lower rate comes from the firm's size and credit history, not just output volume.

### *4. Marketing economies*

Advertising, branding, and distribution become cheaper per unit when spread over many sales. For example, a national TV ad reaches millions of potential buyers, but the cost per viewer drops as sales increase.

> **Remember:** Marketing economies are internal only when the firm itself creates the campaign. If the whole industry benefits from a shared trade show, that is an external economy.

### *5. External economies*

These arise when the whole industry or region becomes more productive. Examples include:

- **Supplier clustering:** When many firms locate near each other, suppliers set up nearby, reducing transportation costs for everyone.
- **Labor market pooling:** A region with many similar firms attracts skilled workers, making it easier for each firm to hire.
- **Knowledge spillovers:** Firms learn from each other's innovations, raising overall productivity.

> **In daily life:** Silicon Valley's many tech companies benefit from a shared pool of software engineers and venture■capital firms. Each individual company enjoys lower hiring and financing costs, even though they did not create the pool themselves.

### *Worked example - internal technical economies*

**Example 5 (Hard): Car manufacturer reduces ATC**

A car maker produces 10,000 cars per year with an average total cost of \25,000 per car. After expanding to 15,000 cars per year, ATC falls to22,000.

Calculate the percentage reduction in ATC and explain which type of economy of scale is likely responsible.

- **Find the change in ATC.**

Delta ATC=25,000-22,000=3,000 dollars

- **Compute the percentage reduction relative to the original ATC.**

\text{Percent reduction}= \frac{\Delta \text{ATC}}{25{,}000}\times 100

=\frac{3{,}000}{25{,}000}\times 100=12\%

- **Identify the likely source of the economy.**

The firm increased output by 50 % (from 10,000 to 15,000). The large drop in ATC is typical of **technical economies of scale**: the manufacturer can spread the cost of expensive assembly lines, robotics, and tooling over more cars, lowering the cost per vehicle.

**Answer:** The average total cost fell by **12 %**. The reduction is most plausibly due to **technical economies of scale**, because larger production lets the firm use its costly machinery more efficiently.

### *Putting it together*

When a firm grows, it may enjoy several internal economies at once - better machines (technical), specialized managers (managerial), cheaper financing (financial), and cheaper advertising (marketing). All of these push the ATC curve downward in the short run.

External economies, on the other hand, are not under the firm's direct control but can still lower costs. They often appear as a shift of the whole industry's long■run average cost curve to the left.

Understanding which type of economy is at work helps managers answer questions such as:

- Should we invest in a bigger factory (technical) or hire a dedicated supply■chain analyst (managerial)?
- Is our region attracting suppliers that will lower input prices for everyone (external)?
- Will a larger scale let us negotiate lower interest rates (financial)?

By recognizing the source, firms can plan the most efficient path to lower costs and stronger competitive positions.

## 5.6 6. Diseconomies of Scale

When a firm keeps getting bigger, it does **not** always become cheaper to run.

After a certain size, the average cost of each unit can start to go up.

This rise in cost is called **diseconomies of scale**.

> **Definition:** *Diseconomies of scale* are the increase in per■unit cost that occurs when a firm becomes so large that its internal inefficiencies outweigh any advantages of size.

### *Why does this happen?*

- **Coordination problems** - Imagine a small kitchen where one chef prepares all dishes. When the restaurant expands to 20 chefs, they must share recipes, ingredients, and timing. If the head chef cannot give clear instructions to everyone, some dishes are prepared late or incorrectly, wasting time and ingredients. In a firm, more workers and more departments mean more communication is needed. Mis■understandings or delays raise the cost of each product.
- **Bureaucracy** - As the organization grows, it often creates many layers of management (supervisors, managers, directors). Each layer adds paperwork, approvals, and meetings. These activities do not produce the product itself, but they consume time and resources. The extra "overhead" pushes up the average cost.

Both effects mean that the firm's **total cost (TC)** grows faster than the increase in **output (Q)**, so the **average total cost (ATC = TC / Q)** rises.

> **In daily life:** Think of a family road trip. With two people, you can pack the car quickly and decide the route together. With ten people, you need to assign seats, agree on music, and maybe stop for extra bathroom breaks. The trip takes longer and feels more expensive per person.

### *Simple formula*

If diseconomies are present, the slope of the ATC curve is positive:

$$\frac{\Delta \text{ATC}}{\Delta Q} > 0$$

where Delta means "change in". A positive slope tells us that each extra unit costs more than the previous one.

### *Worked Example*

**Example 6 (Hard): Rising ATC for a clothing retailer**

A clothing retailer's average total cost (ATC) rises from $30 to$35 when output increases from 5,000 to 7,000 shirts. Compute the change in ATC and discuss a possible diseconomy.

- **Find the change in ATC.**

Subtract the initial ATC from the final ATC.

$$\Delta \text{ATC} = \$35 - \$30 = \$5$$

- **Find the change in output (Q).**

Subtract the initial quantity from the final quantity.

$$\Delta Q = 7{,}000 - 5{,}000 = 2{,}000 \text{ shirts}$$

- **Calculate the rate of ATC increase per shirt.**

Divide the change in ATC by the change in output.

$$\frac{\Delta \text{ATC}}{\Delta Q} = \frac{\$5}{2{,}000 \text{ shirts}} = \$0.0025 \text{ per shirt}$$

**Answer:** The ATC rises by $5, or $0.0025 for each additional shirt produced between 5,000 and 7,000 units. This upward movement suggests a diseconomy of scale. A plausible cause is **coordination difficulty**: as the retailer orders larger batches, the supply‑chain team must manage more suppliers, more inventory locations, and more paperwork, leading to higher per‑shirt costs.

> **Note:** A common mistake is to think that any increase in ATC automatically means the firm is "inefficient." In reality, ATC can rise temporarily while the firm reorganizes to eliminate the coordination problem.

### *How to spot diseconomies in real data*

- Plot ATC against output.
- Look for a segment where the curve slopes upward.
- Ask: *What internal changes occurred when the firm entered this range?*
- New management layers?
- More distant factories?
- Complex scheduling systems?

If the answer points to added complexity, you have identified a diseconomy.

> **Remember:** Diseconomies are not permanent. Firms can often redesign their structure (e.g., flatter hierarchies, better information systems) to bring the ATC curve back down.

### *Quick checklist for students*

- **Identify** whether the ATC curve is rising or falling.
- **Link** a rise to a specific coordination or bureaucratic issue.
- **Explain** how that issue adds cost without adding output.
- **Suggest** a possible remedy (e.g., better communication tools, reducing management layers).

By understanding why costs can increase when a firm becomes too large, you can better analyze real‑world businesses and advise on ways to keep production efficient.

## 5.7 7. Applying Cost Concepts to Decision‑Making

When a firm decides how much to produce, it compares two important ideas: **marginal cost (MC)** and **marginal revenue (MR)**.

> **Definition:** *Marginal cost* is the extra cost of making one more unit of output.

> **Definition:** *Marginal revenue* is the extra revenue earned from selling one more unit.

If the firm makes a unit whose MC is lower than the MR it earns more money than it spends on that unit, so profit rises. If MC is higher than MR, producing that extra unit would lower profit. The profit■maximising rule is therefore:

Produce up to the point where MC = MR

If the market price is fixed (as in perfect competition), MR equals the price, so the firm simply produces where its MC curve meets the market price.

### *Why the rule works*

Think of a lemonade stand. Each cup costs you a little more to make as you run out of cheap lemons and have to buy pricier ones (that's MC). Each cup also brings you the same price from a customer (that's MR). You keep adding cups until the cost of the next cup would be higher than the money you get for it. At that point you stop - any more would lose money.

### *Step■by■step use of MC = MR*

- **Find the market price** (or MR if the firm can set price).
- **List the MC for each additional unit**.
- **Compare MC to price**: produce the last unit whose MC is still <= price.

Below we apply these steps in a worked example.

**Example 7 (Very Hard): profit■maximising output**

A firm faces a market price of \12 per unit. Its MC schedule is5, \8,11, \14 for the first four units. Determine the profit■maximising output and calculate total profit if fixed costs are10.

- **Identify price (MR)**:

P = MR = 12

- **Compare MC to price for each unit**:
- Unit 1: MC =5 \le \12 -> produce.
- Unit 2: MC =8 \le \12 -> produce.
- Unit 3: MC =11 \le \12 -> produce.
- Unit 4: MC =14 > \12 -> stop before this unit.

The last unit with MC le MR is the third unit, so **output = 3 units**.

- **Calculate total revenue (TR) and total cost (TC)**:

TR = P times Q = 12 times 3 =36

Variable cost (VC) is the sum of the first three MC values:

VC = 5 + 8 + 11 = 24

Add fixed cost (FC) of \10:

TC = VC + FC = 24 +10 = 34

Profit = TR - TC:

pi = 36 -34 = 2

**Answer:** The firm should produce 3 units and earns a profit of **\2**.

### *How economies of scale affect market entry*

*Economies of scale* mean that as a firm produces more, its **average total cost (ATC)** falls because fixed costs are spread over more units and because larger production can use cheaper inputs. When ATC is lower than competitors' at the same output, a firm has a **cost advantage** and is more likely to enter a market profitably.

> **In daily life:** Imagine two bakeries. One has a big oven that can bake 1,000 loaves at once, lowering the cost per loaf. The smaller bakery must bake in batches of 100, paying more per loaf. The larger bakery enjoys economies of scale.

If a firm expects to achieve lower ATC, it can set a price that still covers its costs while undercutting rivals, making entry attractive. Conversely, if a firm faces **diseconomies of scale** (ATC rises with output), it may avoid entering because costs become too high.

### *Step■by■step entry decision using ATC*

- **Estimate ATC at the likely output level** for your firm.
- **Compare ATC to the market price** (or to rivals' ATC).
- **If ATC < price and lower than rivals**, entry is likely profitable; otherwise, reconsider.

The next example follows these steps.

**Example 8 (Very Hard): cost advantage and entry**

Two firms consider entering a market. Firm A can achieve an ATC of \9 at 1,000 units, while Firm B's ATC is11 at the same output. Discuss which firm has a cost advantage and how economies of scale influence their entry decision.

- **Write down each firm's ATC**:

ATC_A = 9,qquad ATC_B =11

- **Compare ATC values**:

Since \9 <11, **Firm A's average cost is lower**, giving it a cost advantage.

- **Interpret the advantage with respect to economies of scale**:
- Firm A's lower ATC suggests it either has larger scale or more efficient production, so it can spread fixed costs better and possibly buy inputs cheaper.
- Firm B, with higher ATC, may face diseconomies of scale or simply less efficient technology.

If the market price is, say, \12, both firms could earn a profit, but Firm A's profit margin (12■\9 =3 per unit) is larger than Firm B's (\12■11 = \1 per unit). Therefore **Firm A is more likely to enter** and may even deter Firm B because the latter would have a thin margin.

**Answer:** Firm A enjoys a cost advantage because its ATC (\9) is lower than Firm B's (11). Economies of scale that lower ATC make entry attractive for Firm A, while Firm B's higher ATC makes entry riskier.

> **Remember:** The profit■maximising rule (produce where MC = MR) tells a firm exactly how many units to make, while economies of scale tell the firm whether it can **afford** to produce enough units to be competitive in the first place. Both concepts together guide real■world decisions about output and market entry.

## Terminology at a glance

| Term | Meaning |
|---|---|
| Factors of production | The four essential inputs - land, labor, capital, and entrepreneurship - needed to produce any good or service. |
| Short run | A time period in which at least one factor of production is fixed, creating fixed costs. |
| Long run | A time period when all factors can be varied, so there are no fixed costs. |
| Fixed cost (FC) | Costs that must be paid even when output is zero, such as rent or salaries of permanent staff. |
| Variable cost (VC) | Costs that change with the level of output, like raw materials or hourly wages. |
| Marginal cost (MC) | The extra cost of producing one more unit of output. |
| Economies of scale | Cost advantages that arise when a firm expands production, causing average costs to fall. |
| Diseconomies of scale | When a firm becomes so large that average costs start to rise due to inefficiencies. |

## Practical application

Exercise 5.1:

Exercise 5.2:

Exercise 5.3:

Exercise 5.4:

Exercise 5.5:

Exercise 5.6:

## Checklist

[ ] Identify the four factors of production and give real■world examples

[ ] Explain the difference between short■run and long■run cost concepts

[ ] Calculate total, average, and marginal costs in the short run

[ ] Describe economies of scale and recognize them in everyday businesses

[ ] Analyse a simple scenario to decide whether a firm is operating at an efficient scale

LESSON 6

# 6. Market Structures Overview

## Lesson overview

This lesson introduces the four main market structures - perfect competition, monopoly, oligopoly, and monopolistic competition - and explains how they differ in terms of firms, pricing power, and efficiency. Students will learn to identify each structure in real■world examples and understand why the differences matter for consumers and policy makers.

## Learning outcomes

- Describe the key characteristics of each market structure
- Compare how firms set prices and output in the four structures
- Identify real■world examples of each market structure
- Explain the impact of each structure on consumer welfare and economic efficiency

## 6.1 What Is a Market Structure?

A market structure is simply the way a market is organized.

Think of a market as a playground where buyers and sellers meet.

The "structure" tells us **how many kids are on the playground, what games they are playing, and how easy it is for a new kid to join the game**. In economics the "kids" are firms (companies), the "games" are the products they sell, and the "rules for joining" are the barriers to entry.

> **Definition:** *Market structure* describes the pattern of competition in a market, focusing on three main features: the number of firms, how similar (or different) their products are, and how easy or hard it is for new firms to enter.

Why do we care about market structure?

Because it shapes the choices that consumers have, the prices they pay, and how efficiently resources are used. If we know the structure, we can predict whether a market will push prices down, keep them high, or stay somewhere in between.

### *1. Number of firms*

The first clue is **how many firms are selling the same kind of product**.

- **Many firms** -> each one is small compared with the whole market.
- **Few firms** -> each one holds a larger share of the market.

When there are many firms, no single firm can set the price alone; they must watch what others do. When there are only a few, each firm's decision matters a lot to the whole market.

### *2. Product similarity (or "product differentiation")*

Products can be **identical** or **different**.

- **Identical products** (also called *homogeneous*) mean that a buyer does not care which firm they buy from; the only difference might be price.
- **Differentiated products** have features, branding, or quality that make buyers prefer one firm over another, even if the price is similar.

For example, a bag of plain white rice from two supermarkets is almost identical - buyers care mainly about price. A smartphone, however, comes in many designs, operating systems, and brand reputations, so buyers may choose based on features, not just price.

### *3. Barriers to entry*

These are obstacles that make it hard for a new firm to start selling in the market.

Barriers can be **legal** (licenses, patents), **economic** (high start■up costs, need for large economies of scale), or **strategic** (strong brand loyalty, control of key resources).

If barriers are low, new firms can join easily, increasing competition. If barriers are high, the market stays dominated by the existing firms.

> **Note:** *Barriers to entry* are not the same as "competition". A market can have many firms *and* still have high barriers if, for example, the industry requires huge investments that only a few can afford.

### *Putting the three pieces together*

Economists combine these three characteristics to classify markets into four classic structures: perfect competition, monopoly, oligopoly, and monopolistic competition. Each structure represents a different mix of firm numbers, product similarity, and entry barriers.

- **Perfect competition**: many firms, identical products, no barriers.
- **Monopoly**: one firm, unique product, high barriers.
- **Oligopoly**: few firms, may have similar or differentiated products, moderate to high barriers.
- **Monopolistic competition**: many firms, differentiated products, low to moderate barriers.

Understanding which category a real■world market falls into helps us answer questions like: *Will prices fall if a new competitor arrives?* or *Is the government likely to intervene to protect consumers?*

> **In daily life:** Think of a local farmers' market where dozens of vendors sell the same kind of apples at similar prices - that's close to perfect competition. Compare it with your city's water supply, which is provided by a single utility company that controls the whole network - that's a monopoly.

> **Remember:** The three pillars - *number of firms*, *product similarity*, and *entry barriers* - are the building blocks for every market structure. By checking each pillar, you can quickly sketch the competitive landscape of any industry.

[Figure to be added: Kids on a playground represent firms and the games they play show different market structures.]

Figure 6.1. Kids on a playground represent firms and the games they play show different market structures.

[Figure to be added: A bakery and nearby bakeries illustrate interdependence in an oligopoly.]

Figure 6.2. A bakery and nearby bakeries illustrate interdependence in an oligopoly.

[Figure to be added: Many small farms selling identical crops demonstrate perfect competition.]

Figure 6.3. Many small farms selling identical crops demonstrate perfect competition.

[Figure to be added: Perfect Competition vs Monopoly]

Figure 6.4. Perfect Competition vs Monopoly

[Figure to be added: Oligopoly vs Monopolistic Competition]

Figure 6.5. Oligopoly vs Monopolistic Competition

[Figure to be added: Three Key Features of a Market Structure]

Figure 6.6. Three Key Features of a Market Structure




Figure 6.7. Two Types of Economic Efficiency




Figure 6.8. Efficiency and Consumer Welfare

## 6.2 Perfect Competition

In a perfectly competitive market there are **many sellers** who all offer exactly the same product. Because each firm is tiny compared with the whole market, no single seller can influence the overall price. This situation creates what economists call a *price taker*: the firm must accept whatever price the market sets and cannot charge more or less without losing all its customers.

> **Definition:** *Perfect competition* is a market structure where many firms sell identical goods, entry and exit are free, and each firm is a price taker.

Why does this matter? When firms cannot set prices, the market price reflects the true cost of producing the good for society. If a firm tried to raise its price even a little, buyers would simply buy from another seller who is offering the same product at the lower market price. This competition forces firms to be efficient - producing at the lowest possible cost - because any wasteful spending would make them less profitable than their rivals.

### *Many sellers*

Imagine a farmer's market where dozens of farmers bring the same type of apple. No single farmer controls the overall price of apples; the price is determined by the total amount of apples available and how much buyers are willing to pay. Each farmer's individual decision about how many apples to bring does not move the market price.

> **In daily life:** Think of a popular brand of bottled water sold in supermarkets. Hundreds of stores stock the same brand, and none can charge more than the shelf price without losing sales.

### *Identical products*

When products are *identical* (also called *homogeneous*), buyers do not care which firm they purchase from. The only thing that matters is the price. If one firm tried to add a fancy label or a slightly different shape but kept the price the same, buyers would still be indifferent because the core product - the water itself - has not changed.

### *Free entry and exit*

*Free entry* means that anyone can start a new firm without facing huge legal or financial barriers. *Free exit* means a firm can leave the market whenever it wants, for example if it cannot cover its costs. These freedoms keep the market competitive. If profits become high, new firms are attracted, increasing supply and pushing the price down. If profits become low, some firms quit, reducing supply and pushing the price up. This dynamic pushes the market toward an equilibrium where firms earn just enough to stay in business.

> **Note:** The ability to enter and leave easily prevents long■term super■profits. Any excess profit attracts new entrants, which erodes the profit.

### *Firms are price takers*

Because each firm's output is tiny relative to the whole market, it cannot affect the market price by changing its own quantity. The firm's *demand curve* - the graph that shows how much it can sell at each price - is therefore a horizontal line at the market price. This is different from a monopoly, where the firm faces a downward■sloping demand curve and can set a higher price by restricting output.

Mathematically, the condition for a price■taking firm is

P = MR = MC

where P is the market price, MR is marginal revenue (the extra revenue from selling one more unit), and MC is marginal cost (the extra cost of producing one more unit). In perfect competition, MR equals the price because each additional unit sold brings in exactly the market price.

> **Remember:** In perfect competition, profit maximisation occurs where the firm's marginal cost equals the market price.

### *Why the model is useful*

Even though no real market meets every assumption perfectly, the model gives a benchmark for efficiency. It shows what happens when competition is intense and firms have no market power. Policymakers can compare actual markets to this benchmark to decide whether regulation is needed.

> **Common mistake:** Thinking that "many sellers" automatically means "perfect competition." The other conditions - identical products and free entry/exit - are just as essential. A market with many sellers but differentiated products (like restaurants) is not perfectly competitive.

## 6.3 Monopoly

A monopoly is a market situation where **one firm is the only seller of a product that has no close substitutes**. Because there is no competition, the firm can decide how much to produce and what price to charge.

> **Definition:** A monopoly exists when a single seller supplies a unique product and faces very high barriers that stop other firms from entering the market.

### *Why does a single seller matter?*

In most markets, many firms compete. Competition forces each firm to take the market price as given - they are *price takers*. In a monopoly, the opposite happens: the firm is a *price maker*. It chooses a price that maximizes its profit, subject to the demand of consumers.

The profit■maximising rule is the same as in other market structures:

Choose Q such that MR = MC

where

- MR = marginal revenue (the extra revenue from selling one more unit)
- MC = marginal cost (the extra cost of producing one more unit)

Because the monopoly faces the whole market demand curve, its marginal revenue falls faster than price. This means the monopoly will usually set a price above marginal cost, unlike perfect competition where P = MC.

### *High barriers to entry*

A barrier to entry is anything that makes it difficult or impossible for a new firm to start selling the same product. Common barriers include:

- **Legal barriers** - patents, copyrights, or government licences that give exclusive rights.
- **Control of essential resources** - owning a rare mineral or a unique technology.
- **Economies of scale** - the incumbent can produce at a lower average cost only when it is very large, so a newcomer cannot compete on price.

When these barriers are strong, potential rivals stay out, and the monopoly remains the only seller.

### *Price■setting power*

Because the monopoly does not have to worry about competitors undercutting it, it can raise the price above the competitive level. Imagine the demand curve is P = 20 - 0.5Q. If the monopoly's marginal cost is constant at MC = 4, we first find marginal revenue:

MR = 20 - Q

Set MR = MC:

20 - Q = 4 => Q*=16

Plug Q* back into the demand equation to get the price:

P*=20 - 0.5(16)=12

So the monopoly sells 16 units at a price of 12, while the competitive price would be P = MC = 4. The higher price means higher profit for the firm but lower consumer surplus.

> **In daily life:** Think of a local water utility that is the only provider of tap water in a town. The company cannot be undercut by another water supplier because building a parallel pipe system would be enormously expensive (a barrier). It therefore sets the price it charges households.

### *What does this mean for consumers?*

When a monopoly charges a higher price and produces less than the socially optimal quantity, there is a **deadweight loss** - a loss of total welfare that is not captured by anyone. Graphically, the area between the demand curve and the marginal cost curve, from the monopoly quantity to the competitive quantity, represents this loss.

> **Note:** A monopoly is not always "bad." Some monopolies arise from natural circumstances (e.g., a single railway line) where having one provider is more efficient than duplicating expensive infrastructure. In such cases, regulation may be used to limit price■setting power and protect consumers.

> **Common mistake:** Assuming that because a monopoly can set any price, it will always charge the highest possible price. In reality, the monopoly must consider how price affects the quantity demanded; setting the price too high would cause sales to fall dramatically, reducing total profit.

#### *Key take■aways*

> **Remember:** A monopoly's power comes from being the sole seller of a unique product and from barriers that keep others out. This gives the firm the ability to set price above marginal cost, leading to higher profits for the firm but usually lower overall efficiency and consumer welfare.

## 6.4 Oligopoly

In an **oligopoly** a market is dominated by only a few large firms.

Because there are so few players, each firm's decisions - especially about price - affect the others. This creates a situation called **interdependence**: a firm must think about how its rivals will react before it changes anything.

> **Definition:** An oligopoly is a market structure with a small number of firms whose actions are mutually dependent.

#### *Why interdependence matters*

Imagine you own a bakery that sells chocolate chip cookies. If you are the only bakery in town (a monopoly), you can set the price as you like. If there are hundreds of bakeries (perfect competition), your price is forced to match the market. In an oligopoly, suppose there are only three bakeries in the city. If you raise your price, customers might simply go to one of the other two bakeries, so you could lose sales. If you lower your price, the other bakeries might also cut theirs, starting a price war that hurts everyone's profit. Therefore, each bakery watches the others closely and often chooses a price that keeps the market stable.

#### *Strategic behavior*

Firms in an oligopoly use **strategic behavior**: they plan their moves while anticipating the reactions of rivals. A classic way to think about this is the **game■theoretic** model called the *prisoner's dilemma*. In the simplest version, each firm has two choices: keep a high price (cooperate) or cut the price (defect). The payoff matrix might look like this:

\begin{array}{c|c|c}

& \text{Rival High} & \text{Rival Low} \\

\hline

\text{High} & (\pi_H,\pi_H) & (\pi_L,\pi_{LL}) \\

\text{Low} & (\pi_{LL},\pi_L) & (\pi_{LL},\pi_{LL}) \\

\end{array}

Here pi_H is the profit when both keep high prices, pi_L is the profit for the firm that lowers price while the rival stays high, and pi_LL is the profit when both lower price. Usually pi_H > pi_LL > pi_L, meaning both firms are better off cooperating, but each has an incentive to defect.

#### *Collusion*

Because cooperation yields higher profits, firms sometimes try to **collude** - that is, they make an agreement (explicit or tacit) to act together, usually by fixing prices or dividing the market. Explicit collusion, such as a formal price■fixing agreement, is illegal in most countries because it harms consumers. Tacit collusion is harder to prove; firms simply signal their intentions by, for example, matching each other's price changes.

> **Note:** Collusion is not the same as competition. In a competitive market firms set prices independently; in collusion they coordinate to keep prices high.

#### *Real■world clues of oligopoly*

- **Airlines** on a particular route often have only a few carriers. You may notice that ticket prices tend to move together after one airline changes them.

- **Smartphone operating systems** (iOS and Android) dominate the market, leaving little room for a third major system.
- **Automobile manufacturers** in many countries are limited to a handful of large firms that closely watch each other's model releases and pricing.

### *How firms protect themselves*

- **Price leadership** - One firm (often the largest) sets a price, and the others follow.
- **Non■price competition** - Firms compete by improving quality, advertising, or adding features instead of lowering price.
- **Barriers to entry** - High start■up costs, patents, or control of essential resources keep new competitors out, reinforcing the oligopoly.

> **In daily life:** When you see two big supermarkets raising their prices at the same time, they are likely responding to each other's moves, not to a change in your local grocery store.

### *Common pitfalls*

> **Common mistake:** Assuming that because there are only a few firms, each one can act like a monopoly. In reality, the need to consider rivals' reactions limits how much power any single firm has.

### *Bottom line*

An oligopoly sits between perfect competition (many small firms) and monopoly (one firm). Its defining feature is **mutual interdependence**, which leads firms to behave strategically, sometimes colluding, and often focusing on non■price ways to win customers. Understanding this helps explain why some markets have stable, high prices even though there is more than one seller.

## 6.5 Monopolistic Competition

Monopolistic competition describes a market where **many firms** sell products that are similar but not identical.

Each firm faces a **downward■sloping demand curve**, meaning that if it raises its price, it will lose some customers, and if it lowers its price, it will attract more. Because the products are differentiated - by brand, quality, design, or extra features - each seller has a little bit of **pricing power**: they can set a price above the pure■competition level without losing all customers.

> **Definition:** *Monopolistic competition* is a market structure with many sellers, product differentiation, some control over price, and relatively low barriers to entry.

Why does this situation arise? Imagine a town with dozens of coffee shops. All of them sell coffee, but each shop decorates its space differently, offers unique drink flavors, or provides a cozy atmosphere. Customers care about these differences, so they are willing to pay a bit more for their favorite shop's "special" latte. The shops cannot ignore each other - if one raises its price too much, coffee■drinkers will switch to another shop that offers a similar product for less. At the same time, because opening a small coffee shop does not require a huge investment, new competitors can enter the market fairly easily.

### *How firms decide how much to produce*

In monopolistic competition, a firm chooses the quantity where its **marginal cost (MC)** equals its **marginal revenue (MR)**. Marginal cost is the extra cost of producing one more unit, while marginal revenue is the extra revenue earned from selling that unit. The profit■maximising rule is:

MC = MR

Because the demand curve is downward sloping, MR lies below the demand curve. The firm sets its price by looking up to the demand curve at the quantity where MC = MR. This price is usually higher than the **average total cost (ATC)** in the short run, allowing the firm to earn a **positive economic profit**.

### *Entry and long■run adjustment*

When firms earn profit, new sellers are attracted by the opportunity. Because entry barriers are low, new firms can open quickly. The entry of competitors shifts each existing firm's demand curve leftward (they now have fewer loyal customers). This process continues until economic profit is driven to zero. In the long run, the firm's price equals its average total cost:

P = ATC

At this point, the firm makes **normal profit** - just enough to stay in business, but no extra profit to lure more entrants.

> **Note:** In the long■run equilibrium of monopolistic competition, firms produce where **price = average total cost**, not where **price = marginal cost**. This creates a small "deadweight loss" compared to perfect competition.

### *Efficiency and consumer welfare*

Because firms charge a price above marginal cost, the market does not achieve **allocative efficiency** (the condition P = MC). Some consumers who value the product more than its marginal cost but less than the price will not buy, leaving a small loss in total welfare.

However, product differentiation provides **variety** and can increase consumer satisfaction. The trade■off is between a little inefficiency and a richer selection of goods.

> **In daily life:** Think of the many brands of toothpaste. Each brand adds a different flavor, whitening claim, or packaging style. You might pay a bit more for a brand you trust, even though the basic cleaning function is similar across all brands.

### *Common pitfalls*

> **Common mistake:** Assuming that because many firms exist, the market must be perfectly competitive. The key difference is **product differentiation**; without it, firms would be price takers, not price setters.

### *Summing up*

Monopolistic competition captures many real■world markets - restaurants, clothing stores, smartphone apps - where firms compete on more than just price. The structure explains why we see a wide array of choices, why firms can charge a premium for brand or style, and why new competitors can appear relatively easily. Understanding this helps us see how markets balance efficiency with diversity, and why policy makers sometimes intervene (e.g., advertising regulations) to ensure competition remains healthy.

## 6.6 Comparing Efficiency and Consumer Welfare

A market's **efficiency** tells us how well it turns resources into the goods and services people want. Economists use two kinds of efficiency:

- **Allocative efficiency** - the point where the value consumers place on a product (their willingness to pay) exactly equals the cost of producing the next unit (the marginal cost). In symbols, it is where

P = MC

with P the price and MC the marginal cost.

- **Productive efficiency** - the situation where a firm produces at the lowest possible cost per unit. This happens when the firm operates where average total cost (ATC) is at its minimum.

When a market reaches both kinds of efficiency, society's total welfare - the sum of consumer and producer surplus - is as high as possible. Let's see how each market structure measures up.

**Perfect competition** is the benchmark. Because many firms sell identical products and no single firm can influence price, each firm is a "price taker." The market price is set where the overall demand curve meets the industry supply curve, which is also where P = MC. Therefore, firms produce exactly the quantity that consumers value most, achieving allocative efficiency. Since firms also operate where ATC is minimized (they cannot earn economic profit in the long run), productive efficiency is also attained.

> **Note:** In perfect competition, the price consumers pay equals the cost of producing the last unit, so there is no waste of resources.

In a **monopoly**, a single seller controls the whole market. The monopolist chooses the quantity where its marginal revenue (MR) equals marginal cost, then sets the highest price consumers are willing to pay for that quantity. Because MR is always below the price, the monopoly price ends up **above** marginal cost:

P > MC

This gap means the firm produces less than the socially optimal amount, creating a dead■weight loss - a loss of total welfare that no one receives. The monopolist also does not have to produce at the lowest possible cost; it may operate at a higher average total cost because there is no competitive pressure to cut waste.

> **Definition:** *Dead■weight loss* is the reduction in total surplus that occurs when market output is below the efficient level.

An **oligopoly** consists of a few large firms. Their behavior depends on how they interact. If they collude (explicitly or tacitly) to act like a monopoly, the outcome is similar: price above marginal cost, under■production, and dead■weight loss. If they compete fiercely (e.g., price wars), the market can move closer to the competitive outcome, but strategic interdependence often keeps prices above marginal cost. Productive efficiency is also uncertain; large firms may enjoy economies of scale that lower average costs, but without competition they lack the incentive to eliminate all inefficiencies.

> **In daily life:** Think of a few smartphone makers. When they agree on a standard price, you pay more than the cost of making the phone, and fewer phones are sold than if many small makers competed.

**Monopolistic competition** features many firms selling slightly differentiated products (different flavors, brands, or styles). Each firm has some price■setting power, so it charges a price above marginal cost, like a monopoly, leading to some allocative inefficiency. However, because there are many firms, entry of new competitors erodes excess profits, pushing price toward marginal cost in the long run. The result is *partial* allocative efficiency: the gap P - MC is smaller than in a monopoly but not zero. Productive efficiency is also not fully achieved; firms produce where price equals average total cost, which is above the minimum ATC.

> **Common mistake:** Assuming that "many firms" automatically means "perfectly efficient." Differentiation gives each firm market power, so some inefficiency remains.

**Consumer welfare** - the benefit consumers receive from buying a product at a price lower than the maximum they are willing to pay - is highest when prices are low and output is high. Perfect competition gives the greatest consumer surplus because prices equal marginal cost and the quantity is maximal. Monopoly and colluding oligopolies give the smallest consumer surplus: higher prices and fewer units sold. Monopolistic competition and competitive oligopolies fall in between.

> **Remember:** Higher consumer welfare does not always mean higher producer profit. In perfectly competitive markets, firms earn zero economic profit in the long run, yet consumers enjoy the most surplus.

In summary, the farther a market is from the perfect■competition ideal, the larger the gap between price and marginal cost, the greater the dead■weight loss, and the lower the overall welfare for consumers. Understanding these efficiency concepts helps policymakers decide when to intervene (e.g., antitrust laws) to move markets closer to the efficient benchmark.

## 6.7 Real■World Examples and Policy Implications

A market structure is the way firms are organised in a particular industry. Knowing which structure a real■world market resembles helps us see why prices are high or low, why some firms are very powerful, and what governments can do about it.

**Agriculture - Perfect Competition**

Many small farms sell virtually the same product: wheat, corn, or milk. Because each farm's output is almost identical, no single farmer can set the price. The market price is decided by the total amount that all farms together are willing to sell (the **supply**) and the amount buyers want to buy (the **demand**). If a farmer tries to charge more than the market price, buyers simply buy from another farmer.

> **In daily life:** Think of a farmer's market where dozens of stalls sell the same kind of apple. You pick the cheapest stall because the apples are indistinguishable.

Governments usually do not need to intervene much in perfectly competitive markets, but they may provide **price supports** or **subsidies** to protect farmers from sudden price drops caused by bad weather. A price support is a guaranteed minimum price paid by the government; a subsidy is a cash payment that lowers the farmer's cost.

**Utilities - Monopoly**

Electricity, water, and natural gas are often provided by a single company in a region. Building two separate sets of power lines or water pipes would be wasteful, so the industry is a **natural monopoly** - one firm can supply the whole market at lower cost than several firms could. Because there is no competition, the monopoly could raise prices and reduce service quality if left unchecked.

> **Note:** A **natural monopoly** occurs when a single firm can produce the whole market output at a lower average cost than multiple firms.

To protect consumers, governments either **regulate** the monopoly (set price caps and service standards) or **public■own** the utility. Price caps are often expressed as a formula:

P_max = MC + frac(R - MC)e

where MC is marginal cost, R is a fair return on capital, and e is the price elasticity of demand. This ensures the firm covers its costs but does not earn excessive profit.

**Smartphone Market - Oligopoly**

The global smartphone market is dominated by a few large firms (Apple, Samsung, Huawei, etc.). This is an **oligopoly**: a market with only a handful of sellers, each holding a big share. Because each firm's decisions affect the others, they watch each other's prices, new features, and advertising very closely.

> **Definition:** An **oligopoly** is a market where a small number of firms have most of the market power, and they must consider rivals' reactions when making choices.

In everyday life, think of two rival pizza chains on the same street. If one raises its price, the other may keep theirs low to attract more customers. In the smartphone world, a new iPhone launch often prompts Samsung to release a competing flagship at a similar price, because both firms know that a big price change could shift many buyers.

Governments may apply **antitrust laws** to prevent collusion (secret agreements to fix prices) and to stop mergers that would give one firm too much power. For example, the U.S. Department of Justice blocked a proposed merger between two large telecom companies because it would have reduced competition.

**Fast■Food Chains - Monopolistic Competition**

Restaurants like McDonald's, Burger King, and Wendy's sell similar but not identical products. Each chain differentiates itself through branding, menu items, or restaurant ambience. This is **monopolistic competition**: many firms sell slightly different products and have some control over price, but because there are many alternatives, no single chain can dominate the market.

> **In daily life:** Imagine a street with several coffee shops. One offers free Wi■Fi, another has a cozy interior, and a third sells the cheapest latte. Your choice depends on what matters most to you.

Because firms compete on quality, price, and advertising, consumers usually enjoy a wide variety of choices and relatively low prices. Governments rarely need to regulate these markets, but they may enforce **health and safety standards** (e.g., nutrition labeling) to protect consumers.

**Why policy matters**

Each market structure creates different challenges for policymakers:

- **Perfect competition** - focus on stability (subsidies, safety nets).
- **Monopoly** - need price regulation or public ownership to avoid abuse of power.
- **Oligopoly** - enforce antitrust rules to keep competition alive and prevent collusion.
- **Monopolistic competition** - ensure truthful advertising and product safety.

> **Remember:** Understanding the market structure helps decide whether the government should step in, and if so, what tools (price caps, subsidies, antitrust actions, standards) are most appropriate.

## Terminology at a glance

| Term | Meaning |
|---|---|
| Market structure | The pattern of competition in a market, defined by the number of firms, product differentiation, and ease of entry. |
| Perfect competition | A market with many sellers of identical products, free entry, and firms that are price takers. |
| Monopoly | A market with a single seller of a unique product and high barriers that prevent competition. |
| Oligopoly | A market dominated by a few large firms whose decisions are interdependent. |
| Monopolistic competition | A market with many firms selling differentiated products, giving each some price■setting power. |
| Allocative efficiency | When price equals marginal cost (P=MC), meaning resources go to their highest valued uses. |
| Productive efficiency | When a firm produces at the lowest possible average total cost. |
| Barriers to entry | Obstacles that make it difficult for new firms to enter a market, such as high startup costs or legal restrictions. |

## Practical application

Exercise 6.1:

Exercise 6.2:

Exercise 6.3:

Exercise 6.4:

Exercise 6.5:

Exercise 6.6:

## Checklist

- [ ] Describe the key characteristics of each market structure
- [ ] Compare how firms set prices and output in the four structures
- [ ] Identify real■world examples of each market structure
- [ ] Explain the impact of each structure on consumer welfare and economic efficiency

LESSON 7

# 7. Macroeconomic Indicators

## Lesson overview

This lesson explains the main macroeconomic indicators - GDP, inflation, and unemployment - and shows how each is measured. Students will learn to calculate simple versions of these indicators and understand what they reveal about an economy.

## Learning outcomes

- Define GDP, inflation, and unemployment
- Explain how each indicator is measured
- Calculate nominal and real GDP using basic data
- Compute the inflation rate with a price index
- Determine the unemployment rate from labor■force figures

## 7.1 What is GDP?

Imagine you want to know how big a country's economy is, just like you might want to know how much money you earned in a week from all your part■time jobs. Economists use a single number for this purpose: **Gross Domestic Product**, or **GDP** for short.

> **Definition: GDP** is the total market value of all *final* goods and services produced within a country's borders during a specific period of time (usually one year or one quarter).

Let's break that sentence down piece by piece.

- **Total market value** means we add up the prices that buyers actually paid for the items. If a loaf of bread costs 2 and a car costs20,000, we count 2 +20,000, not the raw physical amount of bread or metal.
- **Final goods and services** are the products that are ready for the consumer, not the intermediate parts used to make something else. For example, the steel used to build a car is *intermediate*; the finished car is *final*. Counting only final goods avoids double■counting the same value.
- **Produced within a country's borders** tells us we look at everything made inside the nation, no matter who owns the company. A factory in the United States owned by a Japanese firm still counts toward U.S. GDP, while a U.S. company's factory in Mexico does not.
- **Given period** means we measure over a set time, such as a calendar year or a three■month quarter, so we can compare one year to the next.

Why do we need GDP? It gives policymakers, businesses, and ordinary people a single, comparable figure that tells how much economic activity is happening. If GDP is growing, the economy is expanding; if it shrinks, the economy is contracting. This helps governments decide whether to raise or lower taxes, whether to spend more on public projects, or whether to change interest rates.

Think of GDP like the score on a video■game leaderboard that shows how many points each player has earned. The higher the score, the better the player is doing compared to others. In daily life, we use GDP scores to rank countries, just as we might rank friends by the number of steps they walked in a week.

To calculate GDP we add up the value of everything produced. One common way is the **expenditure approach**, which says that total spending on final goods and services equals total output. The formula looks like this:

GDP = C + I + G + (X - M)

where

- C = consumption spending by households (food, clothes, entertainment)
- I = investment spending by businesses (machinery, new factories) and residential construction
- G = government spending on goods and services (roads, schools, salaries)
- X = exports (goods made domestically and sold abroad)
- M = imports (goods made abroad and bought domestically)

The term (X - M) is called **net exports**; it adds the value of what we sell to other countries and subtracts what we buy from them.

> **In daily life:** When you buy a new smartphone, your purchase is part of the C component. When a construction company builds a new office building, that counts as I. When the government repairs a highway, that is G. If a car made in your country is shipped to another country, its price adds to X; if you buy a foreign■made laptop, its price is subtracted as M.

It is important to remember that GDP does **not** measure everything that matters. It ignores unpaid work (like caring for a child), environmental damage, and the distribution of income. A country could have a high GDP but still have many people living in poverty.

> **Common mistake:** Adding the value of intermediate goods (like the steel used to make a car) together with the finished car. This would count the same value twice and make GDP too large.

Finally, keep in mind that GDP is a *snapshot* of economic activity, not a judgment of well■being. It tells us how much is being produced and sold, which is useful for many decisions, but it does not answer questions about happiness, health, or fairness.

> **Remember:** GDP = the market value of all final goods and services produced inside a country during a set time period. It is a tool for comparing the size and growth of economies, not a complete measure of a nation's overall welfare.

[Figure to be added: Shows a collection of goods and services representing a country's total production.]

Figure 7.1. Shows a collection of goods and services representing a country's total production.

[Figure to be added: A typical shopping basket used to calculate the CPI.]

Figure 7.2. A typical shopping basket used to calculate the CPI.

[Figure to be added: Shows who is counted in the labor force and who is not.]

Figure 7.3. Shows who is counted in the labor force and who is not.

[Figure to be added: How to calculate Nominal GDP]

Figure 7.4. How to calculate Nominal GDP

[Figure to be added: Nominal GDP vs Real GDP]

Figure 7.5. Nominal GDP vs Real GDP

[Figure to be added: Labor Force Components]

Figure 7.6. Labor Force Components

[Figure to be added: Key Macro Indicators]

Figure 7.7. Key Macro Indicators

## 7.2 Calculating Nominal GDP

When we talk about **nominal GDP**, we are simply adding up the value of everything a country produces **in the prices that are actually paid this year**.

Think of it like a grocery receipt: you list each item, multiply the number of units you bought by the price you paid for each, and then add all those amounts together. The total on the receipt tells you how much money you spent today. Nominal GDP works the same way, but instead of a grocery basket it uses the whole economy's "basket" of goods and services.

> **Definition:** *Nominal Gross Domestic Product (GDP)* is the sum of the market values of all final goods and services produced in a country during a given year, measured using the prices that prevail in that same year.

Why do we calculate it this way?

Using current■year prices lets us see the **size of economic activity** as it is experienced right now. Policymakers, businesses, and news reporters often quote nominal GDP because it reflects the actual amount of money flowing through the economy today. Later we will adjust for price changes to get *real* GDP, but the first step is always the nominal figure.

### *Step■by■step method*

- **List every product or service** that is produced in the economy.

Only *final* goods are counted - items that are ready for consumption, not intermediate parts that will be used to make something else.

- **Record the quantity** of each product that was produced during the year.
- **Find the price** that buyers actually paid for each product in that same year.
- **Multiply** the quantity by its current price to get the *value* of that product.
- **Add** all those values together. The sum is the nominal GDP.

Mathematically, if the economy produces \(n\) different goods, the formula is

\text{Nominal GDP} = \sum_{i=1}^{n} (Q_i \times P_i)

where \(Q_i\) is the quantity of good \(i\) and \(P_i\) is its price this year.

> **In daily life:** Imagine you run a tiny lemonade stand that also sells cookies. To know how much you earned this summer, you multiply the number of lemonades sold by the price per lemonade, do the same for cookies, and then add the two totals. That total is your "nominal GDP" for the stand.

### *Example 1 (Easy): Small economy of apples and bread*

**Question:** A small economy produces 100 apples at \2 each and 50 loaves of bread at3 each. What is its nominal GDP?

**Steps**

- **Identify the goods and their quantities.**
- Apples: \(Q_{\text{apple}} = 100\)
- Bread: \(Q_{\text{bread}} = 50\)
- **Record the current-year prices.**
- Apples: \(P_{\text{apple}} = \2\)
- Bread: \(P_{\text{bread}} = \3\)
- **Calculate the value of each good.**

Value of apples = 100 times 2 = 200

Value of bread = 50 times 3 = 150

- **Add the values together.**

Nominal GDP = 200 + 150 = 350

**Answer:** The nominal GDP of this tiny economy is **\350**.

### *Adding more goods*

Suppose the economy also produces 30 bicycles sold at \120 each. We simply extend the same steps:

- Quantity of bicycles \(Q_{\text{bike}} = 30\)
- Price per bicycle \(P_{\text{bike}} = \120\)
- Value of bicycles 30 times 120 = 3,600
- New nominal GDP 350 + 3,600 = 3,950

You can see how each additional product is just another term in the sum.

> **Common mistake:** Forgetting to use *current* prices. If you accidentally use last year's prices, the result is not nominal GDP; it is a mixed measure that can mislead you about the true size of today's economy.

### *Why we multiply, not add quantities directly*

Quantities alone (e.g., "100 apples + 50 loaves") have no monetary meaning because apples and loaves are different items. Multiplying each quantity by its price converts every product into a common unit - dollars - so they can be summed meaningfully.

> **Remember:** The multiplication step turns physical units (apples, loaves) into monetary units (dollars). Only after this conversion can we add them together.

### *Quick checklist for calculating nominal GDP*

- **List** every final good or service.
- **Record** the year's quantity $Q_i$.
- **Record** the year's price $P_i$.
- **Compute** $Q_i \times P_i$ for each good.
- **Sum** all the products.

When you follow these five tiny steps, you will always arrive at the correct nominal GDP, no matter how large or complex the economy.

> **Note:** Nominal GDP does not adjust for inflation. If prices rise from one year to the next, nominal GDP will increase even if the actual amount of goods and services stays the same. That is why economists later calculate *real* GDP, but the nominal figure is the essential first building block.

## 7.3 Real GDP and Price Indices

Real GDP is the value of all goods and services produced in a country **measured with the prices of a fixed year**.

Why do we need this? Imagine you run a lemonade stand. In year 1 you sell 10 glasses at 2 each, so you earn20. In year 2 you sell the same 10 glasses, but the price has risen to 3. Your earnings are now30.

If we only look at the dollar amount, it looks like the stand is doing better, but the *quantity* of lemonade sold has not changed. The higher number is just because prices went up. To see whether the economy really produced more, we must strip out the effect of price changes. That stripped■out measure is **real GDP**.

> **Definition:** *Real GDP* = nominal GDP adjusted for changes in the overall price level, using the prices of a chosen base year.

### *How do we adjust nominal GDP?*

First we need a way to compare the price level of the current year with the price level of the base year. A simple tool is a **price index**.

A price index is a number that shows how much the overall price level has changed relative to the base year. If the index is 120, prices are 20 % higher than in the base year; if it is 95, prices are 5 % lower.

> **Note:** The base year is the year whose prices we treat as "unchanged" (index = 100). All other years are expressed as a percentage of that base.

### *Step■by■step construction of a simple price index*

- **Choose a base year** - pick a year that is recent and has reliable data.
- **Select a basket of goods** - pick the goods and services that represent the economy (e.g., apples, bread, cars).
- **Record the quantities** of each good produced in the current year.
- **Calculate the cost of the basket** using current■year prices (this gives *current■year expenditure*).
- **Calculate the cost of the same basket** using base■year prices (this gives *base■year expenditure*).
- **Form the index** by dividing current■year expenditure by base■year expenditure and multiplying by 100.

Mathematically:

$$\text{Price Index} = \frac{\displaystyle\sum_{i} P_{i}^{\text{current}} Q_{i}}{\displaystyle\sum_{i} P_{i}^{\text{base}} Q_{i}} \times 100$$

where $P_i^{current}$ is the current price of good i, $P_i^{base}$ is its base■year price, and $Q_i$ is the quantity produced in the current year.

### *Converting Nominal GDP to Real GDP*

Once we have the price index, we can remove the effect of price changes:

$$\text{Real GDP} = \frac{\text{Nominal GDP}}{\text{Price Index}/100}$$

Dividing by the index (expressed as a decimal) scales the nominal figure back to base■year dollars.

**Example 2 (Medium): Real GDP with a simple price index**

Suppose in Year 2 the economy produces 100 apples and 50 loaves of bread. The current■year prices are 2 for apples and3 for bread. The base■year prices (the year we will use as a reference) are 1.5 for apples and2.5 for bread.

- **Compute nominal GDP** - multiply each quantity by its current price and add them.

Nominal GDP=100times2 + 50times3 = 200 + 150 = 350

- **Compute the cost of the same basket at base■year prices** - this is the denominator of the price index.

Base■year expenditure=100times1.5 + 50times2.5 = 150 + 125 = 275

- **Form the price index** - divide nominal expenditure by base■year expenditure and multiply by 100.

Price Index= frac350275times100 approx 127.27

- **Convert nominal GDP to real GDP** - divide nominal GDP by the index (as a decimal).

Real GDP= frac350127.27/100= frac3501.2727approx 275

**Answer:** Real GDP equals 275, which is the value of Year 2 output measured in base■year dollars.

> **In daily life:** Think of a "real" salary as the amount of groceries you could buy with your pay, not the number of dollars on your paycheck. If prices double but your paycheck also doubles, your "real" buying power is unchanged.

### *Why the price index matters*

- **Comparability:** Real GDP lets us compare output across years without the distortion of inflation or deflation.
- **Policy decisions:** Governments use real growth rates to decide whether the economy is expanding or contracting.
- **International comparison:** When comparing countries, we first adjust each country's GDP to a common price base (often using purchasing■power■parity methods).

> **Common mistake:** Forgetting to keep the *quantities* the same when forming the index. The index must use the *current* quantities but *base■year* prices; swapping them gives a wrong adjustment.

### *Quick checklist for calculating real GDP*

- Choose a base year and keep its prices fixed at 100.
- Use the same quantities for both the numerator (current prices) and denominator (base prices).
- Compute the price index, then divide nominal GDP by the index (as a decimal).

When you follow these steps, you obtain a measure of economic activity that reflects true changes in production, not just changes in the price level. This is the foundation for later topics such as growth rates and productivity analysis.

## 7.4 Measuring Inflation with the CPI

The **Consumer Price Index (CPI)** is a way to measure how the overall price of everyday goods and services changes over time.

Think of the CPI as a "shopping basket" that contains items most families buy - food, clothing, rent, transportation, and so on. Each month the government checks how much that basket would cost today compared with a base year. If the basket becomes more expensive, the CPI goes up, indicating inflation; if it becomes cheaper, the CPI falls, indicating deflation.

> **Definition:** *Consumer Price Index (CPI)* - the percentage change in the cost of a fixed basket of goods and services bought by typical households.

Why do we need the CPI?

Prices do not stay the same. When you go to the grocery store each week, you may notice that the price of milk or bread has risen. The CPI captures these changes in a single number, allowing economists, policymakers, and you as a consumer to see whether the cost of living is going up or down. It also helps adjust wages, pensions, and tax brackets so that they keep pace with price changes.

> **In daily life:** If your monthly rent is 800 and the CPI shows a 3 % inflation rate, a landlord might raise the rent by roughly24 to keep the real value of the rent unchanged.

### *How the CPI is turned into an inflation rate*

The CPI itself is an index number, not a percentage. To express how fast prices are moving, we calculate the **inflation rate** between two periods (usually one year apart). The formula is:

\text{Inflation Rate (\%)} = \frac{\text{CPI}_{\text{new}} - \text{CPI}_{\text{old}}}{\text{CPI}_{\text{old}}} \times 100\%

Here:

- CPI_new is the index for the later period (e.g., this year).
- CPI_old is the index for the earlier period (e.g., last year).
- The fraction tells us the proportionate change; multiplying by 100% turns it into a percentage.

> **Note:** The CPI is *relative* to a base year that is set to 100. If the CPI is 120, prices are 20 % higher than in the base year.

### *Worked example*

**Example 3 (Medium): Calculating the annual inflation rate from CPI values**

*Question:* Last year the CPI was 120 and this year it is 126. What is the inflation rate for the year?

- **Identify the two CPI numbers.**
- CPI_old = 120 (last year)
- CPI_new = 126 (this year)
- **Plug the numbers into the inflation■rate formula.**

\text{Inflation Rate} = \frac{126 - 120}{120} \times 100\%

- **Do the arithmetic step by step.**
- Subtract: 126 - 120 = 6
- Divide: frac6120 = 0.05
- Multiply by 100%: 0.05 times 100% = 5%

**Answer:** The inflation rate for the year is **5 %**, meaning the overall price level rose by five percent compared with last year.

### *Interpreting the result*

A 5 % inflation rate tells us that, on average, the items in the CPI basket cost five percent more this year than they did last year. If your salary stayed the same, your purchasing power would have fallen because you can now buy less with the same amount of money.

> **Common mistake:** Forgetting to divide by the *old* CPI. Using the new CPI in the denominator would give a smaller percentage and misrepresent the true rate of price change.

### *Why the CPI uses a fixed basket*

The CPI holds the composition of the basket constant over the measurement period. This is important because it isolates **price changes** from **quantity changes**. If the basket were allowed to change each year (for example, if people started buying more smartphones), the index would mix price effects with changes in what people buy, making it harder to see pure inflation.

> **Remember:** The CPI measures *price* changes, not *quantity* changes. It answers the question, "How much more does the same set of goods cost today?"

### *Quick check: applying the formula to a different scenario*

Suppose the CPI was 150 two years ago and 158 today. What is the inflation rate over the two■year span?

- Identify: CPI_old = 150, CPI_new = 158
- Insert: Inflation Rate = frac158 - 150150 times 100%
- Compute: 158 - 150 = 8, frac8150 approx 0.0533, 0.0533 times 100% approx 5.33%

**Answer:** Over the two years, the inflation rate is **about 5.33 %**.

### *Connecting CPI to other macro indicators*

When you later study **real GDP**, you will use a price index (often the CPI or a similar index) to strip out inflation and see how the *quantity* of goods and services produced has changed. The CPI therefore serves as a bridge between nominal (money■term) values and real (inflation■adjusted) values.

> **Note:** Real GDP = Nominal GDP / (CPI / 100). This shows why a clear understanding of the CPI and its inflation rate is essential for many other calculations in economics.

By mastering the CPI and the simple inflation■rate formula, you gain a practical tool for interpreting news headlines ("inflation hit 4 % this month") and for evaluating how price changes affect your own budget. Keep the steps - identify the two CPI numbers, subtract, divide by the old CPI, multiply by 100% - and you'll be able to calculate inflation for any pair of periods.

## 7.5 Unemployment Rate Basics

Unemployment is a key concept that tells us how many people who want a job cannot find one.

To measure it we first need to know who is counted as part of the **labor force**.

**Labor force** - all people who are either working (employed) or looking for work (unemployed) and are old enough to work.

People who are not looking for a job - such as full■time students, retirees, or stay■at■home parents - are **outside** the labor force and do not affect the unemployment rate.

**Employed** - anyone who did any work for pay or profit during the reference week, or who was temporarily absent from a job (e.g., on vacation or sick leave).

**Unemployed** - anyone who is not working **and** has actively looked for a job in the past four weeks, or is waiting to be called back to a job they already have.

Once we have these three numbers, we can compute the **unemployment rate**:

\text{Unemployment Rate} = \frac{\text{Number of Unemployed}}{\text{Labor Force}} \times 100\%

> **Definition:** *Unemployment rate* is the percentage of the labor force that is jobless but actively seeking work.

Why do we use a percentage?

A percentage lets us compare unemployment across countries or over time even when the total number of workers changes. It is a **relative** measure, not an absolute count.

> **In daily life:** Imagine a school club with 20 members. If 2 members are absent because they are sick and cannot attend meetings, the "absence rate" would be 2/20 times 100% = 10%. The same idea works for a whole economy: we look at the share of people who want to work but cannot.

### *Worked example - calculating the unemployment rate*

**Example 4 (Easy): Labor force calculation**

A small town has 200 people who are either working or looking for work. 180 of them have jobs, and 20 are looking for a job. What is the unemployment rate?

- **Identify the labor force.**

The labor force includes everyone who is employed **or** unemployed.

Labor Force=200

- **Identify the number of unemployed.**

The problem tells us there are 20 people looking for work.

Unemployed=20

- **Apply the unemployment■rate formula.**

Unemployment Rate= frac20200times 100%

**Answer:** The unemployment rate is **10 %**. This means that out of every 100 people who are either working or looking for work, 10 cannot find a job.

### *Understanding each component*

- **Why count only the labor force?**

People who are not seeking work are not "available" for jobs, so including them would make the rate look lower than the true level of job scarcity.

- **What does "actively looking" mean?**

It means the person has taken concrete steps such as sending résumés, filling out online applications, or contacting employers in the last four weeks. Simply wanting a job is not enough.

- **What if someone is temporarily laid off?**

If they expect to be recalled soon and are still looking for work, they are counted as unemployed.

> **Common mistake:** Adding people who are not looking for work (students, retirees) to the labor force. This makes the denominator too large and the unemployment rate artificially low.

### *Another quick calculation*

**Example 5 (Easy): Checking consistency**

In a city, 1,500 people are employed and 300 are unemployed. Find the labor force and the unemployment rate.

- **Add employed and unemployed to get the labor force.**

Labor Force=1,500+300=1,800

- **Insert the numbers into the unemployment■rate formula.**

Unemployment Rate= frac3001,800times 100%

**Answer:** The labor force is **1,800 people**, and the unemployment rate is **16.7 %** (rounded to one decimal place).

### *Interpreting the rate*

- **Low unemployment rate** (e.g., 3■4 %) usually signals a strong economy where most people who want jobs can find them.
- **High unemployment rate** (e.g., 10 % or more) indicates that many people are unable to secure work, which may reflect economic downturns, mismatches between skills and job requirements, or other structural problems.

> **Remember:** The unemployment rate is a snapshot for a specific period (usually a month). It can change quickly as people find jobs or stop looking.

### *Putting it together with other indicators*

When we study macroeconomics, we look at the unemployment rate alongside **GDP** and **inflation**. Together they give a picture of:

- **How much the economy is producing** (GDP).
- **How prices are moving** (inflation).
- **How many people are without work** (unemployment).

If GDP is growing, inflation is moderate, and unemployment is low, the economy is generally considered to be in a healthy state. Conversely, a falling GDP, rising inflation, and high unemployment suggest a recession.

By mastering the definitions and the simple formula above, you can now read any unemployment statistic and understand what it really means for people's lives and for the overall economy.

## 7.6 Putting It All Together

Imagine you have three pieces of information about an economy: how much it produced in current dollars (nominal GDP), how prices have changed (the Consumer Price Index, CPI), and how many people are working or looking for work (the labor■force numbers).

Putting these pieces together lets you see **whether the country is really getting richer**, **how fast prices are rising**, and **how many people are without a job**.

> **Definition:** *Nominal GDP* is the total market value of all final goods and services produced in a year, measured using the prices that are current in that same year.

> **Definition:** *Real GDP* adjusts nominal GDP for changes in the overall price level, so it reflects the actual quantity of goods and services produced.

> **Note:** The CPI is a price index that tracks the cost of a typical basket of consumer goods. It is used to turn nominal values into real values and to compute inflation.

> **Remember:** The unemployment rate looks only at people who are **in the labor force** (those who are either employed or actively looking for a job).

### *Step 1 - Turn nominal GDP into real GDP*

First, we need a *base■year* price index. The problem tells us to use the CPI of the previous year (150) as the base. The formula is

Real GDP= fracNominal GDPCPI this year times CPI base year

Why does this work? If prices have risen, the same basket of goods now costs more. Dividing by the current CPI removes the effect of higher prices, and multiplying by the base■year CPI puts the value back into the price level of the base year.

### *Step 2 - Find the inflation rate from the CPI*

Inflation tells us how fast overall prices are rising. We compare the CPI of this year with the CPI of last year:

Inflation rate= fracCPI this year-CPI last yearCPI last yeartimes 100%

If the CPI goes up by 10 %, we say inflation is 10 %.

### *Step 3 - Compute the unemployment rate*

The unemployment rate measures the share of the labor force that does **not** have a job but is looking for one.

Unemployment rate= fracLabor force-EmployedLabor forcetimes 100%

Only the labor■force numbers matter; people who are not looking for work (students, retirees) are excluded.

### *Worked Example*

**Example 5 (Hard): Real■world data set**

An economy has:

- Nominal GDP = 1,200 million
- CPI last year = 150
- CPI this year = 165
- Labor force = 500 people
- Employed = 460 people

Calculate real GDP (using base■year price index 150), inflation rate, and unemployment rate.

- **Convert nominal GDP to real GDP**
- Use the formula for real GDP.

Real GDP= frac{1,200}165times150

- **Compute the inflation rate**
- Compare this year's CPI to last year's CPI.

Inflation rate= frac165-150150times100%

- **Find the unemployment rate**
- Subtract the number employed from the labor force, then divide by the labor force.

Unemployment rate= frac500-460500times100%

**Answer:**

- Real GDP = frac{1,200}165times150 = 1,090.9 million (rounded to 1,091 million).
- Inflation rate = frac15150times100% = 10%.
- Unemployment rate = frac40500times100% = 8%.

So, after removing the effect of a 10 % price increase, the economy's output is about 1.09 billion in base■year dollars. Prices have risen by 10 % over the year, and 8 % of the labor force is unemployed.

### *Putting the three indicators together*

Now that we have the numbers, ask yourself what they mean for everyday life:

- **Real GDP** tells us whether people are actually producing more goods and services, not just paying higher prices. In our example, real GDP is slightly lower than nominal GDP, showing that part of the 1.2 billion figure is just price growth, not extra production.
- **Inflation** of 10 % means that the same basket of groceries, clothes, and services now costs ten percent more than a year ago. If wages do not rise at the same pace, households feel poorer.
- **Unemployment** of 8 % indicates that out of every 100 people who want to work, eight cannot find a job. This can signal under■utilised resources and may prompt the government to consider policies that stimulate hiring.

> **In daily life:** Think of a video■game where you have three gauges: one for "how many points you scored" (real GDP), one for "how fast the game is getting harder" (inflation), and one for "how many teammates are missing" (unemployment). All three together tell you whether you're winning, struggling, or need a new strategy.

### *Why we do all three*

If you looked at only one indicator, you could be misled. A high nominal GDP might look impressive, but if inflation is also high, the real increase in production could be modest. Likewise, a low unemployment rate is good, but if inflation is soaring, wages may not keep up, eroding purchasing power.

> **Common mistake:** Assuming that a rise in nominal GDP automatically means people are better off. Without adjusting for inflation, the increase could be purely price■driven.

By calculating real GDP, inflation, and unemployment together, economists get a balanced picture of economic health. Policymakers use this picture to decide whether to tighten or loosen monetary policy, invest in job programs, or adjust taxes.

### *Quick checklist for the next time you see data*

- Identify **nominal GDP** and the **current CPI**.
- Use the base■year CPI to compute **real GDP**.

- Compare this year's CPI with last year's to get the **inflation rate**.
- Use the **labor■force** and **employed** numbers to find the **unemployment rate**.

When you follow these steps, you turn raw numbers into a story about how an economy is really performing.

### Terminology at a glance

| Term | Meaning |
|---|---|
| GDP | Gross Domestic Product, the total market value of all final goods and services produced within a country in a given period. |
| Nominal GDP | GDP measured using the prices that prevail in the year the output is produced. |
| Real GDP | GDP adjusted for changes in the overall price level, using prices from a fixed base year. |
| Consumer Price Index (CPI) | A price index that tracks the cost of a fixed basket of goods and services bought by typical households. |
| Labor Force | All people who are either employed or actively looking for work and are of working age. |
| Unemployment Rate | The percentage of the labor force that is unemployed, calculated as unemployed divided by labor force. |
| Base Year | A reference year whose prices are used to calculate real GDP and price indices. |
| Inflation | A sustained increase in the overall price level, reflected by a rising CPI. |

### Practical application

Exercise 7.1:

Exercise 7.2:

Exercise 7.3:

Exercise 7.4:

Exercise 7.5:

Exercise 7.6:

### Checklist

[ ] Define GDP, inflation, and unemployment

[ ] Explain how each indicator is measured

[ ] Calculate nominal and real GDP using basic data

[ ] Compute the inflation rate with a price index

[ ] Determine the unemployment rate from labor■force figures

LESSON 8

# 8. Fiscal Policy and Government Role

### Lesson overview

This lesson explains how taxation, government spending, and budget deficits work and how they affect the overall economy. Students will see the link between fiscal decisions and macro■economic outcomes like growth and inflation.

### Learning outcomes

- Define taxation, government spending, and budget deficit
- Explain how each component influences aggregate demand
- Calculate simple tax revenue and deficit amounts
- Analyse the short■run impact of a fiscal stimulus
- Evaluate the trade■offs of expansionary vs. contractionary fiscal policy

### 8.1 What Is Fiscal Policy?

Fiscal policy is simply the way a government tries to steer the whole economy by changing two big levers: **taxes** and **spending**.

When the government collects money through taxes, it takes money out of households and businesses. When it spends money - building roads, paying teachers, or buying equipment - it puts money back into the economy. By adjusting how much it takes and how much it gives, the government can try to make the economy grow faster, slow it down, or keep it stable.

> **Definition:** *Fiscal policy* = the use of government taxation and spending to influence overall economic activity.

Why does this matter? Imagine a classroom where the teacher wants the class to finish a big project on time. If some students are too busy with other work, the teacher might give them extra help (spending) or ask them to finish a smaller part of the project (taxing their time). The goal is to keep the whole class moving toward the deadline without anyone falling too far behind or racing ahead too fast.

### *How Taxes Work*

A **tax** is a compulsory payment that individuals or firms must give to the government. Taxes can be taken as a percentage of income (income tax), a share of sales (sales tax), or a charge on property (property tax). When taxes go up, households have less disposable income - money they can spend on things like food, clothes, or entertainment. When taxes go down, they have more disposable income, so they can buy more.

> **Note:** Higher taxes usually *reduce* consumer spending, while lower taxes usually *increase* it.

### *How Government Spending Works*

**Government spending** is the money the state uses to buy goods and services or to pay wages. This includes building highways, funding schools, or providing health care. When the government spends more, it directly creates jobs and purchases that add to the total amount of goods and services produced in the economy.

> **In daily life:** Think of a city deciding to repair a bridge. The construction crew gets paid, suppliers sell cement, and nearby restaurants see more customers. All of these activities add to the city's economic activity.

### *The Balance Between the Two*

The combination of taxes and spending determines the **budget balance**. If the government spends more than it collects, it runs a **budget deficit**; if it collects more than it spends, it has a **budget surplus**. Both situations affect the amount of money circulating in the economy.

The basic relationship can be shown with a simple equation:

\text{Budget Balance} = \text{Tax Revenue} - \text{Government Spending}

When the balance is negative (a deficit), the government must borrow money, usually by selling bonds. When the balance is positive (a surplus), the government can pay down debt or save the extra funds.

### *Why Use Fiscal Policy?*

Governments use fiscal policy to achieve three main goals:

- **Stabilize the economy** - smooth out big ups and downs (boom■bust cycles).
- **Promote growth** - encourage more production and jobs over time.
- **Redistribute income** - help reduce inequality by taking more from those who can afford it and giving more to those who need it.

For example, during a recession (when many people lose jobs and spend less), a government might lower taxes and increase spending on public projects. This puts more money into people's hands, encouraging them to buy goods and services, which helps businesses stay open and hire workers.

> **Remember:** Fiscal policy works *slowly* because changing tax laws or approving new spending projects takes time, and the effects on the economy may appear months later.

### *A Simple Analogy*

Picture a bathtub with water flowing in from the tap (government spending) and draining out through the plughole (taxes). If the water level is too low, the tub looks empty (the economy is sluggish). Turning the tap up (more spending) or plugging the hole tighter (lower taxes) raises the water level. If the water is spilling over (the economy is overheating), you can turn the tap down (cut spending) or open the plughole more (raise taxes) to bring the level back to a comfortable height.

By understanding this basic idea - how taxes pull money out and spending pushes money in - students can see why governments constantly adjust fiscal policy, much like a caretaker adjusting the water flow to keep the bathtub at just the right level.

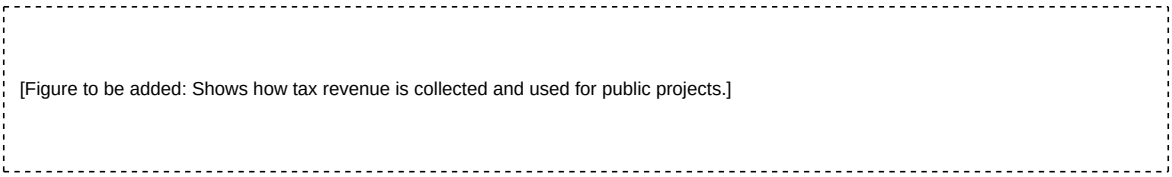



Figure 8.1. Shows how tax revenue is collected and used for public projects.

[Figure to be added: Illustrates government spending on infrastructure.]

Figure 8.2. Illustrates government spending on infrastructure.

[Figure to be added: Shows how a tax cut leaves more money for household spending.]

Figure 8.3. Shows how a tax cut leaves more money for household spending.

[Figure to be added: Main Tools of Fiscal Policy]

Figure 8.4. Main Tools of Fiscal Policy

[Figure to be added: Expansionary vs. Contractionary Fiscal Policy]

Figure 8.5. Expansionary vs. Contractionary Fiscal Policy

[Figure to be added: How Expansionary Fiscal Policy Works]

Figure 8.6. How Expansionary Fiscal Policy Works

## 8.2 Taxation Basics

Taxes are the main way governments collect money from households and businesses.

When you pay a tax, a part of the money you would otherwise keep for yourself is taken by the government. This taken■away money becomes **tax revenue**, which the government can then use to pay for things like schools, roads, and hospitals.

> **Definition:** *Tax revenue* - the total amount of money a government receives from taxes in a given period.

Why does this matter?

If the government has more tax revenue, it can spend more on public services, which can improve the overall well■being of the community. On the other hand, if taxes are too high, households have less **disposable income** - the money left after taxes that can be spent on everyday items or saved.

> **Note:** *Disposable income* = total income - taxes paid.

Think of taxes like a shared pizza. Imagine a group of friends ordering a large pizza that costs \20. If each friend contributes an equal share, the pizza is paid for and everyone gets a slice. The contribution each friend makes is like a tax: it is taken from each person's pocket so that the whole group can enjoy a benefit (the pizza, or public services). The amount each friend keeps after paying is their disposable income.

### *How Taxes Are Calculated*

Most taxes are expressed as a **rate** - a percentage of the amount being taxed. For example, a 5 % sales tax means you pay an extra 5 cents for every dollar you spend on a taxable good.

The basic formula for tax revenue from a single transaction is

Tax revenue = Tax base times Tax rate

where the **tax base** is the amount on which the tax is applied (e.g., the price of a good).

### *Worked Example 1 (Easy): Sales tax on a purchase*

**Example 1 (Easy): City sales tax**

A city imposes a 5 % sales tax on a \200 purchase. How much tax revenue does the city collect?

- Identify the tax base: the purchase price is \200.
- Identify the tax rate: 5 % = 0.05 (as a decimal).
- Multiply the tax base by the tax rate:

Tax revenue = 200 times 0.05

**Answer:** The city collects \10 in tax revenue.

### *Effect on Disposable Income*

When a household buys the same \200 item, the total amount it must pay becomes the original price plus the tax. The extra cost reduces the money left for other uses.

The formula for **disposable income after tax** is

Disposable income = Income - Taxes paid

If a family earns \1,000 in a month and pays100 in various taxes, their disposable income is

Disposable income = 1,000 - 100 = 900

They can now spend or save the remaining \900.

### *Worked Example 2 (Easy): Impact on a household*

**Example 2 (Easy): Monthly budget**

A family earns \2,500 per month and pays250 in taxes (including income tax and sales tax). What is their disposable income?

- Write down total income: \2,500.
- Write down total taxes paid: \250.
- Subtract taxes from income:

Disposable income = 2,500 - 250

**Answer:** The family's disposable income is \2,250 per month.

### *Different Types of Taxes*

- **Income tax** - a percentage of the money you earn from work or investments.
- **Sales tax** - a percentage added to the price of goods and services you buy.
- **Property tax** - a percentage of the assessed value of land or buildings you own.

Each type follows the same basic idea: *tax revenue = tax base x tax rate*. The only difference is what the tax base is.

> **In daily life:** When you buy a coffee for \3 and the local sales tax is 8 %, you actually pay3 x 1.08 = \3.24. The extra0.24 goes to the city's tax revenue.

### *Why Governments Choose Different Tax Rates*

Governments balance two goals:

- **Raise enough revenue** to fund public services.
- **Leave enough disposable income** so households can keep spending, which helps the economy grow.

If the tax rate is set too high, disposable income falls sharply, and people may buy less. This can slow down economic activity. If the tax rate is too low, the government may not have enough money to provide essential services.

> **Common mistake:** Thinking that higher taxes always mean better public services. In reality, if taxes are so high that people cannot afford to buy goods, the overall economy may shrink, reducing the total tax base and possibly lowering total revenue.

### *Quick Check*

Suppose a city wants to raise exactly \5,000 from a 2 % sales tax on a popular local festival. How much total spending (tax base) must occur at the festival?

- Let the unknown total spending be X.
- Use the tax revenue formula:

5,000 = X times 0.02

- Solve for X by dividing both sides by 0.02:

X = frac{5,000}0.02

**Answer:** The festival must generate \250,000 in total sales for the city to collect5,000 in tax revenue.

Understanding taxation basics gives you the tools to see how every purchase you make contributes to the larger picture of government finance, and how the amount you keep after taxes shapes your own spending choices. Keep these simple formulas in mind, and you'll be able to follow any tax discussion in later lessons.

## 8.3 Government Spending

Government spending is the money that a government uses to buy goods, pay for services, and give transfers to households or firms.

When the government buys a new school building, hires teachers, or pays unemployment benefits, it is putting money directly into the economy. This injection raises **aggregate demand** - the total amount of goods and services that people want to buy at a given overall price level.

> **Definition:** *Aggregate demand* is the sum of consumption, investment, government spending, and net exports. It shows the total spending on a country's output.

Why does government spending matter?

Every dollar the government spends becomes income for someone else: a construction worker, a teacher, a supplier of cement, or a family receiving a benefit. Those recipients then spend part of that income on other things - food, clothing, entertainment - which creates a chain of additional purchases. This chain is called the **multiplier effect**. The more times the money changes hands, the larger the boost to overall demand.

**In daily life:** Think of a school fundraiser. If the school buys a new computer for 1,000, the computer store receives that1,000 as revenue. The store pays its staff and buys parts from another company. Those workers and suppliers then spend part of their earnings on groceries, rent, etc. The original 1,000 has now helped many people, not just the school.

### *How government spending enters the aggregate■demand equation*

The aggregate■demand formula can be written as

AD = C + I + G + (X - M)

where

- C = consumption by households
- I = investment by firms
- G = government spending
- X = exports, M = imports, so (X-M) is net exports.

When G rises, holding everything else constant, AD rises by the same amount plus the multiplier effect.

> **Note:** The multiplier is larger when people are likely to spend the extra income rather than save it. In a recession, many households have unused capacity to spend, so the multiplier tends to be higher.

### *Worked example - Easy*

**Example 2 (Easy): Building a road**

The government decides to build a new road that costs **3 million**. How does this spending affect aggregate demand?

- **Identify the change in government spending.**

The increase in G is 3 million dollars.

Delta G = 3 million

- **Add the change directly to aggregate demand.**

Since AD = C + I + G + (X-M), a rise in G adds the same amount to AD (before multiplier).

Delta AD_initial = Delta G = 3 million

- **Recognize the multiplier effect (optional for this easy level).**

If the marginal propensity to consume (MPC) is 0.8, the multiplier k is 1/(1-MPC)=1/(1-0.8)=5. The total impact would be

Delta AD_total = k times Delta G = 5 times 3 million =15 million

**Answer:** The road project adds 3 million directly to aggregate demand; with a typical multiplier of 5, the total boost could be about15 million.

### *Types of government spending*

- **Direct purchases of goods and services** - building highways, buying military equipment, funding research.

- **Transfer payments** - unemployment benefits, pensions, cash grants. These do not buy a product directly, but they give households money to spend, which still raises G in the aggregate■demand formula.

Even though transfer payments are not buying a physical good, they are counted as government spending because they increase household income and therefore consumption C.

> **Common mistake:** Thinking that transfer payments do not affect aggregate demand because the government is not buying anything. In reality, the money given to households becomes part of their income and is usually spent, so it still raises total demand.

### *The short■run impact on the economy*

When the economy is below full employment (i.e., there are idle workers and factories), increasing G can:

- Raise output (more goods and services are produced).
- Reduce unemployment (more workers are needed for the new projects).
- Slightly increase the price level if the economy is close to capacity.

If the economy is already at full capacity, extra spending mainly pushes up prices rather than output, leading to inflation.

### *Worked example - Moderate*

**Example 3 (Moderate): Unemployment benefits**

A government raises monthly unemployment benefits by 200 for 10,000 households. Estimate the immediate increase in aggregate demand, assuming an MPC of 0.75.

- **Calculate the total extra transfer payment.**

Delta G = 200 times 10,000 = 2 million

- **Find the multiplier.**

k = frac11 - MPC = frac11 - 0.75 = 4

- **Compute the total impact on aggregate demand.**

Delta AD = k times Delta G = 4 times 2 million =8 million

**Answer:** The benefit increase raises aggregate demand by about 8 million in the short run.

### *Why governments track spending carefully*

Because each dollar of spending can have a multiplied effect, policymakers must decide where to spend for the greatest benefit. Infrastructure projects often have high multipliers because they use many domestic suppliers and create long■term productive capacity. Transfer payments are useful for quickly supporting households, but the multiplier may be smaller if recipients save a portion of the extra income.

> **Remember:** Government spending directly adds to aggregate demand, and the final impact depends on the multiplier, which reflects how much of the new income is re■spent.

In summary, government spending - whether on roads, schools, or cash transfers - injects money into the economy, raises aggregate demand, and can stimulate production and employment, especially when the economy has unused resources. Understanding the size of the multiplier helps us predict how big the overall effect will be.

## 8.4 Budget Deficit and Surplus

When a government looks at its money box, it compares two numbers: how much it **takes in** and how much it **spends out**.

The money it takes in comes mainly from taxes, fees, and sometimes from selling assets.

The money it spends goes to things like schools, roads, health care, and paying salaries of public workers.

If the amount taken in (called **revenue**) is **smaller** than the amount spent, the government has to borrow the difference. This situation is called a **budget deficit**.

If the revenue is **larger** than the spending, the government ends up with extra money. That is a **budget surplus**.

> **Definition:** A **budget deficit** occurs when **government spending > government revenue**.

> **Definition:** A **budget surplus** occurs when **government revenue > government spending**.

Both concepts are calculated with a very simple formula:

Budget Balance = Revenue - Spending

- If the result is **positive**, the balance is a **surplus**.
- If the result is **negative**, the balance is a **deficit** (the absolute value tells you how big the deficit is).

Why do we care about this?

Knowing whether a budget is in deficit or surplus tells us whether the government needs to borrow money (by issuing bonds) or can pay down existing debt. It also signals the state of the economy: large, persistent deficits may lead to higher public debt, while large surpluses might mean the government is taking too much money out of the economy.

### *Everyday comparison*

Think of a household. Imagine you earn 3,000 a month from your job (your **revenue**) and you spend3,500 on rent, food, and other bills (your **spending**). You are 500 short, so you use a credit card or a loan to cover the gap. That500 shortfall is like a **budget deficit** for the household.

If instead you earned 3,500 and only spent3,000, you would have 500 left over. You could save it or invest it - this is like a **budget surplus** for the household. The same arithmetic works for a whole country, just on a much larger scale.

### *Worked example*

**Example 3 (Medium): Calculating a budget deficit**

Tax revenue this year is 120 billion, spending is150 billion. What is the budget balance, and does the government have a deficit or a surplus?

- **Write down the revenue and spending values.**

Revenue = 120 billion, Spending =150 billion.

- **Apply the budget balance formula.**

Budget Balance = Revenue - Spending

- **Insert the numbers and compute.**

Budget Balance = 120 billion - 150 billion = -30 billion

**Answer:** The budget balance is ■**30 billion**, which means the government runs a **budget deficit of 30 billion**.

> **In daily life:** Just as a family uses a credit card when expenses exceed income, a government issues bonds when it has a deficit.

### *Interpreting the sign*

The minus sign in the result tells us the direction of the imbalance. A **positive** number means the government ends the year with extra money (surplus). A **negative** number means it owes money (deficit). The absolute value tells us the size of the gap.

### *Quick check*

- Revenue = 200 billion, Spending =180 billion -> 200 - 180 = 20 -> **20 billion surplus**.
- Revenue = 85 billion, Spending =100 billion -> 85 - 100 = -15 -> **15 billion deficit**.

> **Note:** The terms "deficit" and "surplus" refer to the **annual** balance. A government can have a deficit one year and a surplus the next; the overall debt level depends on the cumulative result over many years.

### *Why the government might choose a deficit*

Sometimes a government deliberately runs a deficit to **stimulate the economy**. By spending more than it collects, it injects extra money into the market, encouraging businesses to invest and households to spend. This is called **expansionary fiscal policy** (covered later). Conversely, a surplus can be used to **pay down debt** or **cool down an overheating economy**.

> **Common mistake:** Assuming a deficit is always bad. In reality, a modest deficit during a recession can help the economy recover, while a large surplus in a booming economy can prevent inflation.

### *Summary steps for any budget calculation*

- Identify total **revenue** (all money the government expects to receive).
- Identify total **spending** (all money the government plans to pay out).
- Subtract spending from revenue using the formula Budget Balance = Revenue - Spending.
- Look at the sign:
- Positive -> **surplus** (extra money).
- Negative -> **deficit** (shortfall, need to borrow).

> **Remember:** The budget balance is a simple subtraction, but its implications affect borrowing, debt, and overall economic health. Understanding it is the first step to grasping how fiscal policy works.

## 8.5 Impact on Aggregate Demand

Fiscal policy works through the **aggregate■demand (AD) curve**, which shows the total amount of goods and services that households, firms, the government and foreigners want to buy at each overall price level.

When the government changes taxes or its own spending, it directly changes one of the components of AD, so the whole curve shifts left (down) or right (up).

> **Definition:** *Aggregate demand* is the total planned spending in an economy at a given overall price level, usually written as

> AD = C + I + G + (X-M)

> where **C** is consumption, **I** investment, **G** government spending, **X** exports and **M** imports.

### *Why a tax change moves AD*

A tax cut puts more money into the hands of households and firms. That extra money is called **disposable income** - the income left after taxes. People usually spend part of any increase in disposable income. The fraction they spend is the **marginal propensity to consume (MPC)**. If the MPC is 0.6, households spend 60 % of every extra dollar they receive and save the rest.

Because consumption **C** is a component of AD, a rise in **C** makes the whole AD curve shift to the right. The shift is larger than the initial tax cut because the spending that results from the extra income can itself generate further income for others (the multiplier effect).

> **In daily life:** Think of a school cafeteria that decides to lower the price of lunch. More students can afford lunch, so they buy more meals. The cafeteria's total sales go up, and the whole school's lunchtime activity becomes busier - that's a right■ward shift in demand.

### *Why a change in government spending moves AD*

When the government decides to build a new road, it spends money on construction firms, pays workers, buys materials, etc. This spending is the **G** component of AD. An increase in **G** adds directly to total demand, shifting the AD curve rightward by exactly the amount of the new spending (before multiplier effects). A cut in **G** does the opposite, shifting AD left.

> **Note:** The size of the shift depends on the *multiplier*, which we will see in the example below.

### *Worked Example - Tax Cut and Consumption*

**Example 4 (Medium): Tax cut raises consumption**

A 10 billion tax cut raises disposable income, increasing consumption by6 billion (assuming a marginal propensity to consume of 0.6). How far does the AD curve shift?

- **Find the change in consumption.**

The MPC tells us the fraction of the tax cut that is spent:

Delta C = MPC times Delta Tax cut = 0.6 times 10 billion = 6 billion

- **Add the change to aggregate demand.**

Since consumption is part of AD, the whole AD curve moves right by the same amount (ignoring further multiplier rounds for this simple step).

Delta AD = Delta C = 6 billion

- **State the result in words.**

The 10 billion tax cut shifts the AD curve rightward by6 billion, meaning total planned spending in the economy is now 6 billion higher at every price level.

**Answer:** The aggregate■demand curve shifts right by 6 billion.

### *Worked Example - Government Spending Increase*

**Example 5 (Medium): New highway project**

The government announces a 4 billion highway construction program. Assuming an MPC of 0.6, what is the total impact on AD after one round of multiplier effects?

- **Initial change in AD from government spending.**

Direct government spending adds straight to AD:

Delta AD_0 = Delta G = 4 billion

- **First round of induced consumption.**

The 4 billion paid to workers and firms becomes their disposable income. They spend a fraction MPC of it:

Delta C_1 = MPC times Delta AD_0 = 0.6 times 4 billion = 2.4 billion

- **Add the induced consumption to AD.**

Total AD after the first round:

Delta AD_1 = Delta AD_0 + Delta C_1 = 4 billion + 2.4 billion = 6.4 billion

**Answer:** After the first multiplier round, the AD curve shifts right by 6.4 billion.

#### *Putting It All Together*

Both tax changes and government■spending changes move the AD curve, but they do so in slightly different ways:

- **Tax cuts** first change disposable income, then consumption rises by *MPC x tax change*. The shift equals that increase in consumption (plus later multiplier rounds if we continue).
- **Government spending** adds directly to AD, and the money paid out becomes new income for others, generating additional consumption through the same MPC multiplier.

> **Remember:** A rightward shift in AD means higher total spending at every price level, which tends to raise real output and employment in the short run. A leftward shift does the opposite.

Understanding these mechanics helps us see why policymakers use fiscal tools to smooth business cycles: during a recession they may cut taxes or raise spending to push AD rightward; during an overheating economy they may raise taxes or cut spending to pull AD leftward.

## 8.6 Expansionary Fiscal Policy

When the government wants the economy to grow faster, it can use **expansionary fiscal policy**.

In plain words, this means the government either spends more money on things like roads, schools, and hospitals **or** it lowers taxes so that households and businesses have more money to spend. The extra spending or the extra cash in people's pockets raises the total amount of goods and services that people buy - this is called **aggregate demand**.

> **Definition:** *Expansionary fiscal policy* is a government action that increases overall demand by raising public spending, cutting taxes, or doing both at the same time.

Why does this work?

People tend to spend a part of any extra money they receive. Economists call the fraction of additional income that is spent the **marginal propensity to consume** (MPC). If the MPC is 0.75, for every extra dollar a household gets, it will spend 75 cents and save 25 cents. When the government adds money to the economy, the first round of spending is the direct amount it injects. Then the recipients of that money spend a portion of it, and those recipients spend a portion of what they receive, and so on. This chain of spending is called the **multiplier effect**.

The size of the multiplier depends on the MPC:

Multiplier = frac11-MPC

If MPC = 0.75, the multiplier is

frac11-0.75=4

So a 1 billion increase in government spending can ultimately raise total demand by4 billion.

> **Note:** The multiplier works the same way for tax cuts, but we first add the tax cut to the "initial change in spending" because the tax cut becomes extra income for households.

#### *Worked Example 5 (Hard): Infrastructure spending and tax cut*

**Question:** The government adds 20 billion to infrastructure projects and cuts taxes by5 billion. With an MPC of 0.75, what is the total increase in aggregate demand?

- **Add the two policy actions** to get the initial change in spending (the "first round").

Delta G_initial = 20 billion + 5 billion = 25 billion

- **Apply the MPC** to the initial change because only the fraction that people will actually spend matters.

Delta AD = MPC times Delta G_initial = 0.75 times 25 billion

- **Calculate the product** to find the total increase in demand.

Delta AD = 18.75 billion

**Answer:** The expansionary fiscal package raises aggregate demand by **18.75 billion**.

### *How the policy helps the economy*

- **Higher demand -> more production** - Firms see that more people want to buy their goods, so they increase output.
- **More jobs -> higher incomes** - To meet the higher output, firms hire additional workers or ask current workers to work extra hours.
- **Higher incomes -> more spending** - The new wages feed back into the multiplier, reinforcing the initial boost.

> **In daily life:** Think of a school fundraiser. If the school decides to spend 100 on a new playground (direct spending) and also gives each student a10 voucher to buy snacks (tax cut), the playground money is used right away by contractors. The vouchers become extra cash for students, who then buy snacks, and the snack sellers may hire extra staff. The total activity in the school community becomes larger than the original 200 the school put in.

### *Limits and cautions*

- **Timing matters.** If the economy is already near full capacity, extra demand can cause prices to rise rather than output, leading to inflation.
- **Crowding out.** If the government borrows heavily to finance the spending, interest rates may rise, making it more expensive for private firms to invest.
- **Budget balance.** Persistent deficits (spending more than revenue) increase national debt, which future generations must repay.

> **Common mistake:** Assuming that every dollar of tax cut becomes 1 of new spending. In reality, only the portion equal to the MPC is spent; the rest is saved or used to pay off debt.

### *Quick calculation checklist for expansionary policy*

- **Identify the policy components** - total new government spending (Delta G) and total tax cuts (Delta T).
- **Sum them** to get the initial change in spending: Delta G_initial = Delta G + Delta T.
- **Multiply by the MPC** to obtain the change in aggregate demand: Delta AD = MPC times Delta G_initial.
- **Interpret the result** - a positive Delta AD means the policy is expected to boost output and employment, assuming there is slack in the economy.

### *Worked Example 6 (Medium): Small tax cut only*

**Question:** The government cuts taxes by 2 billion and does not change its spending. If MPC = 0.6, what is the increase in aggregate demand?

- **Initial change** comes only from the tax cut.

Delta G_initial = 0 + 2 billion = 2 billion

- **Apply MPC** to find the spending that results from the tax cut.

Delta AD = 0.6 times 2 billion

- **Compute the product.**

Delta AD = 1.2 billion

**Answer:** The tax cut raises aggregate demand by **1.2 billion**.

### *Worked Example 7 (Easy): Pure spending increase*

**Question:** The government decides to spend an extra 3 billion on public schools. With MPC = 0.8, what is the total impact on demand?

- **Initial change** is the spending amount itself.

Delta G_initial = 3 billion

- **Multiply by MPC** to capture the portion that will be re■spent.

Delta AD = 0.8 times 3 billion

- **Calculate.**

Delta AD = 2.4 billion

**Answer:** Aggregate demand rises by **2.4 billion**.

### *Bottom line*

Expansionary fiscal policy is a tool for "jump■starting" an economy that is sluggish or in recession. By increasing spending or cutting taxes, the government injects extra income that, through the multiplier, leads to a larger rise in total demand than the original amount. The effectiveness depends on the MPC, the size of the policy change, and the current state of the economy. Use the simple steps above to estimate the impact before deciding whether the policy is appropriate.

## 8.7 Contractionary Fiscal Policy

Contractionary fiscal policy is the government's way of "turning down the heat" when the economy is running too hot.

In plain words, it means the government either **raises taxes** or **cuts its own spending** so that households and businesses have less money to spend. With less spending, the total demand for goods and services - called **aggregate demand** - falls, which helps slow down inflation and keep the economy from overheating.

Why does this work?

When people have to pay more tax, the amount of income they keep (their **disposable income**) goes down. With less disposable income, they buy fewer things. The same logic applies when the government stops buying goods or services; there is simply less money flowing through the economy. Both actions reduce the **overall spending** that drives prices upward.

> **Definition:** *Contractionary fiscal policy* - government actions that lower aggregate demand by increasing taxes or decreasing government spending.

Think of it like a bathtub with the tap running full blast (the economy). If the water level gets too high and starts spilling over (inflation), you can either turn the tap down (raise taxes) or open a drain (cut spending) to bring the level back to a safe height.

### *How the tax side works: the multiplier effect*

When the government raises taxes by a certain amount, the immediate loss in spending is not just that amount. Because people tend to spend only a part of any extra money they receive, the **marginal propensity to consume** (MPC) tells us what fraction of each dollar is spent. The rest is saved. The total change in consumption is the tax change multiplied by the MPC.

The formula for the change in consumption (Delta C) is

Delta C = MPC times Delta T

where Delta T is the change in taxes (a positive number when taxes go up).

**Example 6 (Hard): Tax increase to fight inflation**

The government decides to raise taxes by 8 billion. The economy's MPC is 0.8. How much will consumption fall, and what is the impact on aggregate demand?

- Identify the tax increase: Delta T = 8 billion.
- Use the consumption change formula:

Delta C = 0.8 times 8 billion

- Calculate the product:

Delta C = 6.4 billion

**Answer:** Consumption falls by 6.4 billion, which directly reduces aggregate demand by the same amount.

> **Note:** The reduction in aggregate demand is exactly the fall in consumption because other components (investment, net exports) are unchanged in this simple illustration.

### *How the spending side works*

Cutting government spending works in a similar way, but the initial change is already a reduction in demand, so we do not need to multiply by the MPC. The whole amount of the cut leaves the economy immediately.

If the government cuts 5 billion of spending, aggregate demand drops by5 billion right away.

**Example 7 (Medium): Government spending cut**

Suppose the government reduces its infrastructure program by 5 billion. What is the immediate effect on aggregate demand?

- Identify the spending cut: Delta G = -5 billion (negative because it is a reduction).
- Recognize that the change in aggregate demand (Delta AD) equals the change in government spending when other components stay the same:

Delta AD = Delta G

- Substitute the value:

Delta AD = -5 billion

**Answer:** Aggregate demand falls by 5 billion.

> **In daily life:** Cutting a school's budget for new computers means fewer purchases of computers, which lowers demand for those goods in the market.

### *Putting it together: the overall impact on the economy*

When both taxes rise and spending falls, the total change in aggregate demand (Delta AD_total) is the sum of the two effects:

Delta AD_total = -Delta C_tax + Delta G_spending

(The consumption change is negative because higher taxes reduce spending, while the spending change is also negative when the government cuts its outlays.)

If the government raises taxes by 8 billion (MPC = 0.8) and cuts spending by5 billion, the combined effect is:

- Compute the consumption drop: 6.4 billion (from Example 6).
- Add the spending cut: 5 billion.
- Total reduction:

Delta AD_total = -6.4 billion - 5 billion = -11.4 billion

Thus, aggregate demand falls by 11.4 billion, pulling price pressures downward.

> **Remember:** Contractionary policy is a tool to cool inflation, but if applied too strongly it can push the economy into recession. Policymakers must balance the size of tax hikes and spending cuts with the current state of the economy.

### *Why timing matters*

The effects of tax changes and spending cuts do not appear instantly. Households may take a few weeks to adjust their budgets, and businesses may delay hiring or investment decisions. Therefore, economists talk about **lags** - the delay between a policy decision and its impact on the real economy.

A common mistake is to think that a small tax increase will have a negligible effect. In reality, even modest changes can be amplified through the multiplier process, especially when the MPC is high.

> **Common mistake:** Assuming that because a tax increase is "only" a few percent, it won't affect inflation. The multiplier shows that the total impact can be several times the initial change.

### *Quick checklist for students*

- **Identify** whether the policy is tax■based or spending■based.
- **Use** the consumption formula (Delta C = MPC times Delta T) for tax changes.
- **Add** the direct spending change for government cuts.
- **Sum** the two effects to find the total shift in aggregate demand.
- **Consider** the size of the MPC and the timing lag when evaluating real■world outcomes.

By following these steps, you can see clearly how contractionary fiscal policy works to "turn down the heat" in an economy that is growing too fast.

## Terminology at a glance

| Term | Meaning |
|---|---|
| Fiscal policy | Use of government taxation and spending to influence overall economic activity |
| Tax revenue | Total money the government collects from taxes |
| Disposable income | Income left after taxes that households can spend or save |
| Aggregate demand | Total planned spending in an economy (C+I+G+X■M) |
| Budget deficit | When government spending exceeds its revenue |
| Budget surplus | When government revenue exceeds its spending |
| Expansionary fiscal policy | Government actions that increase spending or cut taxes to boost demand |
| Contractionary fiscal policy | Government actions that cut spending or raise taxes to reduce demand |

## Practical application

Exercise 8.1:

Exercise 8.2:

Exercise 8.3:

Exercise 8.4:

Exercise 8.5:

Exercise 8.6:

### Checklist

- [ ] Define taxation, government spending, and budget deficit
- [ ] Explain how each component influences aggregate demand
- [ ] Calculate simple tax revenue and deficit amounts
- [ ] Analyse the short■run impact of a fiscal stimulus
- [ ] Evaluate the trade■offs of expansionary vs. contractionary fiscal policy

LESSON 9

# 9. Monetary Policy and Central Banking

### Lesson overview

This lesson explains how central banks manage the money supply, set interest rates, and control inflation. Students learn the tools of monetary policy and why they matter for the economy.

### Learning outcomes

- Explain the main functions of a central bank
- Describe how open market operations affect the money supply
- Illustrate the link between interest rates and inflation
- Calculate the effect of a change in the reserve requirement on bank lending
- Analyze a simple scenario of monetary policy to predict its impact on prices

### 9.1 What Is a Central Bank?

The idea of a **central bank** is simple: it is the main bank of a whole country (or a group of countries) that looks after the nation's money system.

Think of it as the "head librarian" of a huge library of money. Just as the head librarian decides which books stay on the shelves, how many copies are made, and makes sure the library runs smoothly, a central bank decides how much money should exist, how it moves around, and keeps the whole financial system safe.

> **Definition:** A central bank is a public institution that controls a country's money supply, issues the national currency, and supervises commercial banks.

Why do we need such an institution?

If every bank or individual could print money whenever they wanted, the value of money would fall quickly - just like if everyone could print copies of a popular video game, the game would lose its rarity and worth. By giving one trusted body the exclusive right to create money, the economy gets a stable "unit of account" that people can rely on when they buy groceries, pay rent, or save for the future.

#### *Independence - why it matters*

Independence means the central bank can make decisions without direct pressure from the government or politicians. Imagine a school principal who must decide whether to give extra homework. If the principal had to ask the mayor each time, the decision might be swayed by politics rather than what's best for learning. Similarly, an independent central bank can focus on long■term economic health - like keeping inflation low - rather than short■term political goals such as winning an election.

> **Note:** Independence does not mean the central bank is completely free from oversight; it is still accountable to the public and often reports to a parliament or congress.

#### *Core responsibilities*

- **Issuing currency** - The central bank prints banknotes and mints coins, and it decides how many should be in circulation. This is called the *money supply*.

The equation that captures the basic relationship is

M times V = P times Y

where M is the money supply, V is how quickly money changes hands (velocity), P is the overall price level, and Y is real output (the total goods and services produced). By adjusting M, the central bank influences P and Y.

- **Setting policy interest rates** - The central bank determines the cost of borrowing money for other banks. When the policy rate is low, banks can lend cheaply, encouraging businesses and consumers to spend. When it is high, borrowing becomes expensive, which can cool down an overheating economy.
- **Supervising and regulating banks** - Commercial banks (the ones you use for checking accounts) must follow rules to stay safe. The central bank checks that they keep enough reserves, manage risks, and treat customers fairly. This supervision helps prevent bank failures that could harm ordinary people's savings.
- **Managing foreign reserves and exchange rates** - The central bank holds foreign currencies and gold. By buying or selling its own currency against others, it can influence the exchange rate, which affects import prices and export competitiveness.

> **In daily life:** When you see the price of a loaf of bread rise slowly over several years, that is often the result of the central bank's decisions about how much money to create and what interest rates to set.

### *A quick everyday comparison*

Think of a central bank like the thermostat in a house. The thermostat monitors the temperature (the economy's "heat" measured by inflation and growth) and turns the heating or cooling on or off (adjusts money supply and rates) to keep the house comfortable. If the thermostat were constantly changed by guests who just wanted the room warmer for a party, the house would become too hot or too cold. The thermostat's independence - being set by the homeowner, not the guests - keeps the temperature steady.

> **Common mistake:** Believing that a central bank "prints money" to pay for government spending. In reality, the bank creates money mainly to support the banking system and control inflation; direct financing of government deficits is usually prohibited.

### *Bottom line*

A central bank is the cornerstone of a country's financial system. Its independence lets it act on long■term goals rather than short■term politics. By issuing currency, setting interest rates, supervising banks, and managing reserves, it strives to keep prices stable, ensure that banks are safe, and support sustainable economic growth.

> **Remember:** The central bank's ultimate job is to maintain confidence in the nation's money, because confidence is what makes money useful in everyday life.

[Figure to be added: The main building where a country's central bank operates.]

Figure 9.1. The main building where a country's central bank operates.

[Figure to be added: How buying and selling government bonds changes bank reserves.]

Figure 9.2. How buying and selling government bonds changes bank reserves.

[Figure to be added: Banks keep a fraction of deposits as reserves.]

Figure 9.3. Banks keep a fraction of deposits as reserves.

[Figure to be added: What a Central Bank Does]

Figure 9.4. What a Central Bank Does




Figure 9.5. Main Monetary■Policy Tools




Figure 9.6. How Open■Market Operations Work

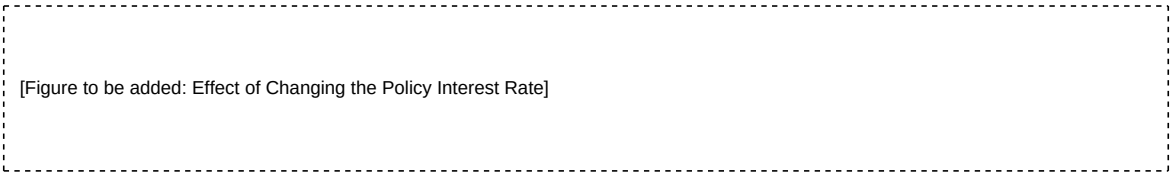



Figure 9.7. Effect of Changing the Policy Interest Rate

## 9.2 Tools of Monetary Policy

When a country wants to keep its economy stable, the central bank has a "toolbox" it can reach into.

The three main tools are **open■market operations (OMO)**, **reserve requirements**, and the **policy interest rate**.

Each tool changes how much money is available for banks to lend, which in turn influences spending, investment, and prices.

> **Definition:** *Monetary policy* is the set of actions a central bank uses to control the amount of money and credit in the economy.

### *1. Open■Market Operations (OMO)*

**Idea in plain words** - The central bank buys or sells government securities (like bonds) in the market.

*Why it works*: When the bank **buys** securities, it pays cash to the sellers, which adds reserves to the banking system. More reserves mean banks can lend more. When it **sells** securities, the buyers pay cash, taking reserves out of the system, so banks have less to lend.

**Everyday comparison** - Think of a bathtub with a faucet and a drain. The water level is the amount of money banks can use. Opening the faucet (buying bonds) lets more water in; opening the drain (selling bonds) lets water out.

**Key term** - *Reserve*: the cash that banks must keep on hand, either in their vaults or as a deposit at the central bank.

**Worked example**

**Example 1 (Basic): Open■Market Purchase and Bank Reserves**

The central bank decides to buy \20 million of government bonds from commercial banks. How does this affect the total reserves in the banking system?

- **Identify the transaction** - The central bank pays cash to the banks, so reserves increase.

Change in reserves = +20 million

- **Add the change to existing reserves** - Suppose banks already hold \80 million in reserves.

New total reserves = 80 million +20 million = 100 million

- **State the result** - The banking system now has \100 million in reserves, giving banks more capacity to make loans.

**Answer:** After the open■market purchase, total bank reserves rise to **\100 million**.

> **In daily life:** Buying bonds is like the central bank pouring water into the economy's "bathtub," making it easier for banks to lend.

### *2. Reserve Requirements*

**Idea in plain words** - The central bank tells each commercial bank what fraction of its deposits it must keep as reserves.

*Why it works*: If the required fraction is lowered, banks can keep a smaller part of deposits as reserves and lend out the rest. Raising the fraction forces banks to hold more money and lend less.

**Everyday comparison** - Imagine you have a piggy bank where you must keep 10 % of any money you receive. If the rule changes to 5 %, you can spend twice as much of each new amount.

**Key term** - *Reserve■requirement ratio*: the percentage of deposits that banks must hold as reserves.

**Worked example**

**Example 2 (Intermediate): Effect of Changing the Reserve Ratio**

A bank has \50 million in customer deposits. The reserve■requirement ratio is cut from 10 % to 5 %. How much additional money can the bank lend out?

- **Calculate original required reserves** -

Original reserves = 0.10 times 50 million =5 million

- **Calculate new required reserves** -

New reserves = 0.05 times 50 million =2.5 million

- **Find the freed■up reserves** - The bank now has

Extra reserves = 5 million -2.5 million = 2.5 million

These extra reserves can be loaned out.

**Answer:** The bank can lend an additional **\2.5 million** after the reserve■requirement cut.

> **Note:** Changing the reserve ratio is a blunt tool; it affects all banks at once, unlike OMO which can be targeted.

### *3. Policy Interest Rate (and the Discount Rate)*

**Idea in plain words** - The central bank sets a benchmark interest rate that influences the rates banks charge each other and their customers.

*Why it works*: When the policy rate is low, borrowing becomes cheap, encouraging businesses and households to take loans and spend more. When the rate is high, borrowing costs rise, slowing down spending and easing inflation pressure.

**Everyday comparison** - Think of the policy rate as the "price of money" in a grocery store. If the price goes down, shoppers buy more; if it goes up, they buy less.

**Key term** - *Discount rate*: the interest rate the central bank charges commercial banks for short■term loans directly from the central bank. It is usually a little higher than the policy rate to discourage banks from borrowing unless needed.

**Worked example**

**Example 3 (Applied): Impact of a Policy■Rate Cut on Loan Cost**

The central bank lowers its policy rate from 4 % to 2 %. A small business wants a \100,000 loan for one year. Assuming the bank adds a 1 % margin to the policy rate, what is the annual interest the business will pay before and after the cut?

- **Calculate the original interest rate** -

Original rate = 4% + 1% = 5%

- **Compute interest before the cut** -

Interest_before = 0.05 times 100,000 =5,000

- **Calculate the new interest rate** -

New rate = 2% + 1% = 3%

- **Compute interest after the cut** -

Interest_after = 0.03 times 100,000 =3,000

**Answer:** The annual interest drops from **\5,000** to **3,000** when the policy rate is cut to 2 %.

> **Remember:** The policy rate is the central bank's primary lever because it quickly changes the cost of borrowing throughout the whole economy.

### *Putting the three tools together*

All three tools ultimately change **the amount of money that banks can lend**:

- Tool - How it changes lending - Typical speed of effect
- Open■market operations - Adds or removes reserves directly - Very fast (days)
- Reserve requirements - Alters the fraction of deposits that must be held - Medium (weeks)
- Policy interest rate - Shifts the cost of borrowing for banks and consumers - Fast to medium (days■months)

By choosing the right combination, the central bank can steer the economy toward its goals of stable prices and sustainable growth.

> **Common mistake:** Assuming the tools work in isolation. In practice, the central bank coordinates them; for example, an OMO may be used to keep the policy rate at its target level after a change in the discount rate.

## 9.3 Open Market Operations (OMO)

Open market operations (OMO) are the most frequently used tool that a central bank has to change the amount of money circulating in the economy.

In plain words, an OMO is simply the buying or selling of government securities - usually short■term bonds - by the central bank.

When the central bank **buys** securities, it pays for them by adding reserves to the banks that sold the bonds.

When it **sells** securities, the buying banks must use some of their reserves to pay, so reserves fall.

Because banks need reserves to make loans, a change in reserves directly changes how much money banks can create. The chain of cause and effect looks like this:

- Central bank buys securities -> reserves of commercial banks rise.
- More reserves -> banks can lend more.
- More lending -> the total money supply expands.

The opposite happens when the central bank sells securities.

> **Definition:** *Open market operations* are transactions in which a central bank buys or sells government securities in order to change the level of reserves that commercial banks hold.

### *Why does this work?*

Commercial banks keep only a fraction of deposits as reserves; the rest can be turned into loans.

The fraction they must keep is called the **reserve requirement** (or reserve ratio).

If the reserve ratio is 10 %, a bank must keep 0.10 of every1 deposited as reserves and can lend out the remaining 0.90.

When the central bank adds 10 million of reserves to the banking system, each bank now has more "spending power" to meet its 10 % requirement and still have excess reserves to lend. The extra lending creates new deposits, which in turn create more reserves, and so on. This repeated process is captured by the **money multiplier**.

> **Note:** The money multiplier tells us how much the total money supply will change for a given change in reserves. It is calculated as Multiplier = frac1Reserve Ratio.

### *Everyday comparison*

Think of the banking system as a series of water buckets connected by hoses. The central bank's open market operation is like adding water to the first bucket (the reserves). Because each bucket only needs to keep a small amount of water (the reserve requirement) and can pass the rest along, the total amount of water in all buckets grows quickly. Selling securities is like draining water from the first bucket, which eventually reduces the water level everywhere.

### *Worked example*

**Example 2 (Easy):** The central bank buys 10 million of bonds, increasing reserves by10 million. How does this affect the total money supply if the reserve ratio is 10 %?

- **Find the money multiplier.**

Multiplier = frac10.10 = 10

- **Multiply the change in reserves by the multiplier** to get the change in the money supply.

Delta Money Supply = 10 times 10 million =100 million

- **Interpret the result.** The banking system can now create an additional 100 million of deposits through lending.

**Answer:** The total money supply increases by 100 million.

### *What happens when the central bank sells securities?*

Suppose the central bank decides to sell 5 million of bonds. Banks that buy these bonds must pay with reserves, so reserves fall by5 million. Using the same reserve ratio of 10 %, the money multiplier is still 10, so the potential contraction of the money supply is

Delta Money Supply = 10 times (-5 million) = -50 million.

Thus, selling securities can shrink the money supply, helping to cool down an overheating economy.

> **In daily life:** When you hear news that the central bank "is buying bonds," think of it as the bank putting extra cash into the hands of other banks, which then makes it easier for you to get a loan for a car or a house.

### *Limits and practical points*

- **Liquidity:** Government securities are very liquid (easy to buy and sell), so the central bank can act quickly.
- **Market impact:** Large purchases or sales can move bond prices and yields, which also affect borrowing costs for the government and businesses.
- **Transparency:** Central banks usually announce their OMO plans in advance to avoid surprise shocks to the financial system.

> **Common mistake:** Assuming that a one■time purchase of 1 million automatically creates10 million of money forever. The increase lasts only as long as banks keep lending; if they hold excess reserves, the multiplier effect is smaller.

### *Quick checklist for OMO*

- **Identify the action:** buy or sell securities?
- **Determine the change in reserves:** equal to the amount of securities transacted.
- **Know the reserve ratio:** this gives the multiplier frac1r.
- **Calculate the change in money supply:** Delta M = Multiplier times Delta Reserves.

> **Remember:** Open market operations change the money supply by first changing bank reserves; the size of the effect depends on the reserve ratio through the money multiplier.

By mastering OMO, you understand the central bank's most direct lever for steering the economy - whether the goal is to stimulate growth by expanding the money supply or to curb inflation by pulling money out of circulation.

## 9.4 Reserve Requirements

Reserve Requirements

When a central bank says that banks must keep a certain *reserve requirement*, it means each bank has to hold a fixed percentage of the money that customers have deposited. This money stays in the bank's vault or in an account at the central bank and cannot be loaned out.

> **Definition:** *Reserve requirement* - the legally mandated fraction of deposits that a bank must keep as reserves and cannot use for lending.

Why does this matter? Banks create most of the money we use by lending out a part of the deposits they receive. If a bank receives \1 of new deposits and the reserve requirement is 10 %, it must keep0.10 as reserves and can lend out the remaining \0.90. The borrower spends that0.90, the money ends up in another bank, which again keeps 10 % and lends out 90 %, and so on. This repeated process is called the *money multiplier* and it determines how much total money the banking system can create from an initial deposit.

Think of the reserve requirement like a rule for a classroom library. Suppose the teacher says each student must keep 20 % of the books they borrow on the shelf at all times. If a student borrows 5 books, they must leave 1 book on the shelf and can take home 4. Those 4 books might be lent to other classmates later, but the rule always forces a portion to stay on the shelf, limiting how many books can be out at once.

### *How the reserve requirement limits lending*

- **Calculate the required reserves.** Multiply the total deposits by the reserve■requirement percentage.
- **Find the excess reserves.** Subtract the required reserves from the bank's total reserves. This is the amount the bank can actually lend.
- **Apply the money multiplier.** The multiplier is the reciprocal of the reserve requirement ( 1 / r ). It tells how many dollars of total money can be created for each dollar of excess reserves.

> **Note:** A lower reserve requirement -> larger multiplier -> banks can lend more; a higher requirement does the opposite.

### *Worked Example*

**Example 3 (Medium): Lowering the reserve requirement**

A reserve requirement is lowered from 12 % to 8 %. If a bank has \50 million in deposits, how much additional money can it create through lending?

- **Find the original required reserves.**

Required reserves_old = 0.12 times 50 million =6 million

- **Find the new required reserves after the change.**

Required reserves_new = 0.08 times 50 million =4 million

- **Determine the increase in excess reserves.**

The bank's total reserves stay the same (assume it already held \6 million).

Additional excess reserves = 6 million -4 million = 2 million

- **Calculate the new money multiplier.**

Multiplier_new = frac10.08 = 12.5

- **Find the extra money the banking system can create.**

Multiply the additional excess reserves by the new multiplier.

Additional money created = 2 million times 12.5 =25 million

**Answer:** By lowering the reserve requirement from 12 % to 8 %, the bank can create an extra \25 million in loans and deposits.

### *Putting the pieces together*

The reserve requirement works together with other tools (like open■market operations) to control the overall amount of credit in the economy. When the central bank wants to **stimulate** growth, it may lower the requirement, freeing up reserves for banks to lend more. When it wants to **cool down** inflation, it can raise the requirement, forcing banks to keep a larger slice of deposits and thus shrink the money supply.

> **In daily life:** If you notice your bank offering more loans after a news report that the central bank cut reserve requirements, that's the mechanism in action.

### *Common pitfalls*

> **Common mistake:** Assuming that a bank can lend out *all* its deposits once the reserve requirement is lowered. The bank still needs enough actual cash (or central■bank balances) to meet the new reserve level; if it lacks those funds, it cannot lend the full theoretical amount.

### *Quick recap*

- **Reserve requirement** is the fraction of deposits banks must hold and not lend.
- Required reserves = *reserve rate* x *total deposits*.
- Excess reserves = *actual reserves* - *required reserves*.
- Money multiplier = 1 / reserve rate.
- Lowering the reserve rate increases the multiplier, allowing more lending; raising it does the opposite.

Understanding reserve requirements gives you a clear picture of one of the simplest levers central banks use to steer the economy. By adjusting how much banks must keep on hand, the central bank can expand or contract the amount of money flowing through households and businesses, influencing spending, investment, and ultimately, inflation.

## 9.5 Policy Interest Rate and the Discount Rate

The policy interest rate is the "price of money" that the central bank sets for banks that borrow directly from it.

When the central bank changes this target rate, commercial banks usually move the rates they charge their customers in the same direction.

> **Definition:** *Policy interest rate* - the rate the central bank aims to keep for short■term loans to commercial banks (often called the "target rate" or "official rate").

> **Note:** The central bank does not set every loan or deposit rate in the economy, but its target acts like a lighthouse: banks look at it to decide how much to charge or pay.

### *Why the policy rate matters*

Think of the policy rate as the thermostat in a house. If you turn the thermostat up, the whole room gets warmer; if you turn it down, the room cools. Similarly, when the central bank raises its target rate, borrowing becomes more expensive for banks, and they pass part of that cost on to households and firms. When it lowers the rate, borrowing becomes cheaper, encouraging spending and investment.

### *The discount rate*

Banks can also borrow directly from the central bank at the *discount rate*. This is usually a little higher than the policy rate, acting as a penalty for banks that need emergency funds. The discount rate sets a ceiling on how high commercial banks' own rates can go, because they would rather borrow from the central bank than charge customers even more.

> **In daily life:** If you need a short■term loan from a friend (the central bank) at 3 % interest, you will not ask another friend (a commercial bank) to lend you money at 5 % if you can get it cheaper elsewhere.

### *How the policy rate moves commercial rates*

Commercial banks add a *spread* to the policy rate to cover their costs, risk, and profit. The spread is usually expressed in percentage points. For example, if the policy rate is 2 % and a bank's typical spread is 2 %, the loan rate it offers to customers will be about 4 %.

The formula is simple:

$$\text{Commercial loan rate} = \text{Policy rate} + \text{Spread}$$

### *Worked example*

**Example 4 (Medium): Policy rate rise and loan rate calculation**

The central bank raises its policy rate by 0.5 %. A commercial bank's loan rate was 4.0 %; it usually adds a 2 % spread. What is the new loan rate?

- **Identify the original policy rate.**

The original loan rate = policy rate + spread, so

Policy rate_old = 4.0% - 2% = 2.0%.

- **Add the central■bank increase to the old policy rate.**

Policy rate_new = 2.0% + 0.5% = 2.5%.

- **Add the bank's unchanged spread to get the new loan rate.**

Loan rate_new = 2.5% + 2% = 4.5%.

**Answer:** The commercial bank's new loan rate is **4.5 %**.

### *The link to borrowing costs for the public*

When the loan rate rises, the cost of a mortgage, a car loan, or a business line of credit goes up. Higher borrowing costs tend to reduce the amount people and firms want to spend, which can slow inflation. Conversely, lower rates make credit cheap, encouraging spending and potentially raising inflation.

> **Remember:** The central bank uses the policy rate as a lever to steer the whole economy - higher rates cool down spending, lower rates warm it up.

### *Quick checklist for students*

- The **policy rate** is set by the central bank and influences all short■term bank borrowing.
- The **discount rate** is the rate banks pay for emergency loans from the central bank; it sits slightly above the policy rate.
- Commercial banks add a **spread** to the policy rate to determine the rates they charge customers.
- A change in the policy rate -> immediate, proportional change in most consumer loan and deposit rates.

Understanding this chain helps you see why news about "the central bank raising rates" quickly shows up in headlines about mortgage payments, car loans, and even the price of a cup of coffee.

## 9.6 Link Between Interest Rates and Inflation

When the central bank changes the interest rate, it is trying to steer how much people spend and how much prices rise.

Think of the interest rate as the "price of borrowing money."

If the price is high, fewer people and businesses want to borrow, so they buy less.

If the price is low, borrowing becomes cheap, so they buy more.

Because spending drives demand for goods and services, changing demand changes the pressure on prices - that is, inflation.

> **Definition:** *Inflation* is the general rise in the price level of goods and services over time, measured as a percentage change per year.

> **Note:** The *policy interest rate* (also called the *benchmark rate*) is the rate set by the central bank that influences all other interest rates in the economy, such as those on loans and mortgages.

### *Why higher rates tend to lower inflation*

- **Cost of borrowing rises** - When the central bank lifts the policy rate, banks raise the interest they charge on loans.

- **Spending falls** - Households think twice before taking a loan for a new car or a house, and firms delay investment projects because financing is more expensive.
- **Demand drops** - With less money being spent, firms see weaker sales and are less able to raise prices.
- **Price growth slows** - The lower demand puts downward pressure on inflation, so the overall price level rises more slowly.

### *Why lower rates tend to raise inflation*

- **Borrowing becomes cheap** - A cut in the policy rate makes loans cheaper.
- **Spending rises** - Consumers are more willing to finance purchases, and firms are eager to invest in new equipment.
- **Demand increases** - More money chases the same amount of goods, creating upward pressure on prices.
- **Price growth accelerates** - The higher demand pushes the inflation rate upward.

### *Everyday comparison*

Imagine a busy highway. The speed limit is like the policy interest rate.

- When the limit is high (say 80 km/h), drivers tend to drive slower because they feel it is risky to go fast; traffic moves more slowly, and fewer cars reach the city quickly.
- When the limit is low (say 40 km/h), drivers feel comfortable accelerating; more cars arrive at the city faster, crowding the streets.

In the economy, "cars" are dollars being spent, and the "city" is the market for goods. The speed limit controls how quickly money moves through the system, which in turn affects how tightly packed (inflated) the market becomes.

### *Simple relationship*

A very common way to express the link is:

\Delta \pi = \alpha \, (i - i^*)

where

- Delta pi = change in inflation (percentage points)
- i = current policy interest rate
- i^* = "neutral" rate that would keep inflation steady (often close to the long■run average)
- alpha = sensitivity factor (usually positive).

If i is above i^*, the term (i-i^*)* is positive, making Delta pi positive **or negative** depending on the sign of alpha. In most textbook models, a higher i leads to a negative Delta pi (inflation falls).

### *Worked example*

**Example 5 (Hard):** *Inflation is running at 6 % while the policy rate is 3 %. Using the simple Phillips■curve idea, estimate the inflation rate after the central bank raises the policy rate to 5 %.*

- **Identify the neutral rate** - In the simple Phillips■curve framework we assume the neutral rate i^* is the rate at which inflation would stay unchanged.

i^* = 3%

(Because the current inflation of 6 % is considered "steady" at the existing rate.)

- **Choose a sensitivity factor** - A typical textbook value is alpha = -0.5, meaning each 1 percentage■point increase in the policy rate reduces inflation by half a percentage point.
- **Apply the formula** - Plug the new policy rate i = 5% into the relationship.

\Delta \pi = \alpha \, (i - i^*) = (-0.5)\,(5\% - 3\%) = (-0.5)\,(2\%) = -1\%

This tells us inflation will fall by 1 percentage point.

**Answer:** After the rate rise, the estimated inflation rate is 6% - 1% = 5%.

The calculation shows how a modest increase in the policy rate can pull inflation down, illustrating the central bank's tool in action.

> **In daily life:** When you hear news that "the Fed has raised rates," think of it as the government making borrowing more expensive so people spend less, which helps keep grocery prices from climbing too fast.

> **Common mistake:** Assuming that a rate change instantly changes inflation. In reality, there is a lag - spending adjusts first, and price changes follow weeks or months later.

### *Putting the pieces together*

- **Higher rates -> higher borrowing cost -> less spending -> lower demand -> slower price growth -> lower inflation.**
- **Lower rates -> cheaper borrowing -> more spending -> higher demand -> faster price growth -> higher inflation.**

Understanding this chain helps you see why central banks watch inflation closely and adjust rates accordingly. The next lesson will show how the central bank decides *how much* to change the rate, using tools like open■market operations and reserve requirements.

## 9.7 Putting It All Together: A Policy Scenario

Imagine the economy is running hot: prices are rising fast (high inflation) while businesses are not growing much (low growth).

The central bank has several levers it can pull at the same time - buying bonds, changing reserve requirements, and adjusting the policy interest rate.

By looking at each lever and then adding their effects together, we can see how the bank tries to cool inflation without hurting the already weak growth.

> **Definition:** *Money supply* - the total amount of cash and bank deposits that people can spend in the economy.

> **Note:** The three main tools affect the money supply in opposite directions: open■market purchases **increase** it, a lower reserve requirement **increases** it, and a higher policy rate **decreases** it.

### *Step■by■step picture of the policy mix*

- **Open■Market Operations (OMO)** - the bank buys government bonds from banks.

When the bank pays cash for the bonds, banks get more reserves, which they can lend out.

This expands the money supply by the *money multiplier*.

- **Reserve Requirement** - the fraction of deposits that banks must keep and cannot lend.

Lowering this fraction frees up more of each bank's deposits for lending, also expanding the money supply.

- **Policy Interest Rate (or Discount Rate)** - the rate at which banks can borrow from the central bank.

Raising this rate makes borrowing more expensive, so banks lend less and the money supply contracts.

The central bank must weigh the expanding effects of (1) and (2) against the contracting effect of (3).

### *Worked Example 6 (Hard): Combining the three tools*

**Question:** The economy has 4 % inflation and 2 % unemployment. The central bank decides to (a) buy \20 million of bonds, (b) lower the reserve requirement to 5 %, and (c) raise the policy rate by 0.75 %. Estimate the combined effect on the money supply and discuss the likely impact on inflation.

### *1. Effect of the bond purchase (OMO)*

Assume the banking system's *reserve■to■deposit ratio* before the change is 10 % (so the money multiplier m = 1/0.10 = 10).

When the central bank buys \20 million of bonds, banks receive that amount as new reserves.

\[

\Delta M_{\text{OMO}} = \text{new reserves} \times m = 20\text{ million} \times 10 = 200\text{ million}

\]

So the money supply would rise by **\200 million** if nothing else changed.

### *2. Effect of lowering the reserve requirement*

The reserve requirement falls from 10 % to 5 %.

The new money multiplier becomes m' = 1/0.05 = 20.

The existing total deposits before the policy are unknown, but we can estimate the *additional* lending capacity created by the lower ratio.

If the banking system held the same amount of reserves after the OMO (now \20 million extra), the extra lending capacity is

\[

\Delta M_{\text{RR}} = \text{new reserves} \times (m' - m) = 20\text{ million} \times (20 - 10) = 20\text{ million} \times 10 = 200\text{ million}

\]

Thus the reserve■requirement cut adds another **\200 million** to the money supply.

### *3. Effect of raising the policy rate*

A higher policy rate discourages borrowing. A common rule of thumb is that a 1 % increase in the policy rate reduces the money supply by about 5 % of the current supply (the exact number depends on the economy, but this simple approximation works for a classroom illustration).

Assume the pre■policy money supply was \2 billion.

A 0.75 % rate increase would cut the supply by

\[

\Delta M_{\text{rate}} = -0.05 \times 0.75 \times 2\text{ billion}

= -0.075 \times 2\text{ billion}

= -0.15\text{ billion}

= -150\text{ million}

\]

So the policy rate change **reduces** the money supply by **\150 million**.

#### *4. Combine the three effects*

Add the three changes:

\[

\Delta M_{\text{total}} = \Delta M_{\text{OMO}} + \Delta M_{\text{RR}} + \Delta M_{\text{rate}}

= 200\text{ million} + 200\text{ million} - 150\text{ million}

= 250\text{ million}

\]

**Answer:** The net effect is an increase of roughly **\250 million** in the money supply.

#### *5. What does this mean for inflation?*

Even though the policy rate was raised, the two expansionary moves (bond purchase and lower reserve requirement) dominate, so the overall money supply grows.

A larger money supply tends to push prices up, especially when the economy already has 4 % inflation.

> **Remember:** When the central bank wants to fight high inflation, the *overall* direction of the money supply matters more than any single tool.

In this scenario, the central bank's mixed signal would likely keep inflation near its current level or even let it rise a little, while the higher policy rate may modestly dampen borrowing and help growth a bit. The net outcome depends on how quickly firms and households respond to the cheaper reserves versus the costlier loans.

#### *Putting it all together*

- **Identify the problem** - high inflation, low growth.
- **Choose tools** - decide which levers push the money supply up or down.
- **Quantify each lever** - use the money multiplier for OMO and reserve■requirement changes; use a rule■of■thumb for rate changes.
- **Add the effects** - a simple sum gives the net change in the money supply.
- **Interpret** - compare the sign of the net change with the inflation goal.

> **In daily life:** Think of the central bank like a thermostat. Turning the heat up (buying bonds, lowering reserves) adds warmth (money) to the room, while opening a window (raising rates) lets some heat escape. The goal is to keep the room comfortable (stable prices) without making it too cold (recession).

By practicing this step■by■step approach, you can see how the same set of tools can be combined in many ways, and why central bankers must be careful to balance them to achieve their dual mandate of price stability and sustainable growth.

## Terminology at a glance

| Term | Meaning |
|---|---|
| Central bank | A public institution that controls the money supply, issues the national currency and supervises commercial banks. |
| Monetary policy | The set of actions a central bank uses to control the amount of money and credit in the economy. |
| Open■market operations (OMO) | Buying or selling government securities to add or withdraw reserves from the banking system. |
| Reserve requirement | The legally required fraction of deposits that banks must keep as reserves and cannot lend out. |
| Policy interest rate | The rate the central bank sets for short■term loans to commercial banks, influencing other interest rates. |
| Inflation | A general rise in the price level of goods and services over time, expressed as an annual percentage. |

| Term | Meaning |
|---|---|
| Money multiplier | The process by which an initial deposit leads to a larger increase in the total money supply through repeated lending. |
| Discount rate | The interest rate charged by the central bank when commercial banks borrow directly from it. |

### Practical application

Exercise 9.1:

Exercise 9.2:

Exercise 9.3:

Exercise 9.4:

Exercise 9.5:

Exercise 9.6:

### Checklist

- [ ] Explain the main functions of a central bank
- [ ] Describe how open market operations affect the money supply
- [ ] Illustrate the link between interest rates and inflation
- [ ] Calculate the effect of a change in the reserve requirement on bank lending
- [ ] Analyze a simple scenario of monetary policy to predict its impact on prices

LESSON 10

# 10. International Trade and Finance

### Lesson overview

This lesson explains why countries trade, how comparative advantage creates gains, and how exchange rates and trade balances affect economies. It also looks at the broader impacts of globalization.

### Learning outcomes

- Explain the concept of comparative advantage
- Identify a country's comparative advantage using simple data
- Interpret a trade balance and what it indicates about an economy
- Describe how exchange rates are determined and how they influence trade
- Discuss the main economic effects of globalization

### 10.1 1. Comparative Advantage Basics

Imagine two countries, **Country A** and **Country B**, each able to make two products: **shirts** and **computers**.

If both countries try to produce everything themselves, they will waste resources.

The idea of *comparative advantage* tells us that each country should focus on the product it can make **relatively cheaper**, measured by the **opportunity cost** of producing it.

> **Definition:** *Opportunity cost* is what you give up when you choose one activity instead of another.

> For example, if a farmer uses a field to grow wheat, the opportunity cost is the amount of corn he could have grown on that same field.

Why does this matter?

When a country produces the good with the lower opportunity cost, it uses its resources more efficiently. Then it can trade the surplus for the other good, and both trading partners end up with more of each product than if they tried to do everything alone.

#### *Step■by■step illustration*

- **List the production possibilities** - how many shirts or computers each country can make with all its resources.

Suppose Country A can produce either **100 shirts** *or* **50 computers**. Country B can produce either **80 shirts** *or* **40 computers**.

- **Calculate the opportunity cost** for each good in each country.

- For Country A, making 1 computer means giving up the chance to make 2 shirts (because 50 computers = 100 shirts).

Opportunity cost of 1 computer in A = frac100 shirts50 computers = 2 shirts

- For Country A, making 1 shirt costs 0.5 computers.
- For Country B, making 1 computer costs 2 shirts (40 computers = 80 shirts).

Opportunity cost of 1 computer in B = frac80 shirts40 computers = 2 shirts

- For Country B, making 1 shirt costs 0.5 computers.
- **Compare the opportunity costs**.
- Country A gives up **2 shirts** for each computer, while Country B also gives up **2 shirts** for each computer. In this simple numbers they look equal, so we adjust the example slightly: let Country B be able to produce **90 shirts** or **40 computers**. Then the opportunity cost of a computer in B becomes frac9040=2.25 shirts.
- Now Country A's computer costs **2 shirts**, which is **lower** than B's 2.25 shirts. Therefore **Country A has a comparative advantage in computers**.
- Conversely, the opportunity cost of a shirt in A is 0.5 computers, while in B it is frac4090approx0.44 computers, so **Country B has a comparative advantage in shirts**.

> **In daily life:** Think of two friends, Maya and Leo, who both bake cookies and mow lawns. Maya can bake 10 cookies in the time it takes her to mow one lawn; Leo can bake 6 cookies in the same time. Maya's *opportunity cost* of one cookie is 0.1 lawns, while Leo's is about 0.17 lawns. Maya should bake cookies and trade some for Leo's lawn■mowing services, because she gives up less lawn work per cookie.

- **Specialize and trade**.
- Country A devotes most of its factories to computers, producing, say, **45 computers** (leaving a small amount of capacity for shirts).
- Country B focuses on shirts, making **80 shirts**.
- They agree to trade: A sends 30 computers to B, and B sends 60 shirts to A. After trade, each country ends up with both goods, but in larger quantities than before specialization.

> **Remember:** Comparative advantage does **not** require a country to be the best at anything; it only needs to be relatively better (lower opportunity cost) at one product.

### *Why the concept works*

- **Efficiency**: Resources (labor, capital, land) are allocated where they generate the most output per unit of input.
- **Mutual gain**: Both parties can consume beyond their own production possibilities frontier - the line that shows the maximum output they could achieve without trade.
- **Flexibility**: Even if technology changes, the comparison of opportunity costs updates automatically, guiding new patterns of specialization.

> **Common mistake:** Thinking that a country should produce only the good it can make *absolutely* cheaper (i.e., using fewer total resources). Absolute advantage matters for overall output, but comparative advantage determines the *pattern* of trade that yields the biggest gains for everyone.

By understanding opportunity cost and comparative advantage, students can see why nations trade, why specialization is beneficial, and how simple numbers reveal powerful economic insights.

[Figure to be added: Shows Country A and Country B each producing shirts and computers with opportunity costs.]

Figure 10.1. Shows Country A and Country B each producing shirts and computers with opportunity costs.

[Figure to be added: Illustrates major export and import routes between countries.]

Figure 10.2. Illustrates major export and import routes between countries.

[Figure to be added: Depicts how exchange rates convert one currency into another.]

Figure 10.3. Depicts how exchange rates convert one currency into another.

[Figure to be added: Opportunity Cost in Country A vs Country B]

Figure 10.4. Opportunity Cost in Country A vs Country B

[Figure to be added: Finding Comparative Advantage]

Figure 10.5. Finding Comparative Advantage

[Figure to be added: Gains from Trade Process]

Figure 10.6. Gains from Trade Process

[Figure to be added: Trade Balance Terms]

Figure 10.7. Trade Balance Terms

[Figure to be added: Nominal vs Real Exchange Rate]

Figure 10.8. Nominal vs Real Exchange Rate

## 10.2 2. Finding Comparative Advantage with Numbers

When we talk about **comparative advantage**, we are looking for the good that a country can make *relatively cheaper* than another country.

"Cheaper" here means giving up less of another good. The key number is the **opportunity cost** - the amount of one product that must be forgone to produce one unit of another product.

> **Definition:** *Opportunity cost* is the quantity of the next best alternative that is sacrificed when a choice is made.

Why do we need this idea?

If a country knows where its opportunity cost is lower, it can specialize in that product, trade with others, and both sides end up with more than they could produce on their own. This is the core reason why international trade can make everyone better off.

Think of it like a kitchen. Imagine you can either bake a cake or make a sandwich. If you need to give up only a tiny bite of sandwich to bake a cake, then baking is relatively cheap for you. If a friend has to give up a whole sandwich for the same cake, you have the comparative advantage in cake■making.

### *Step 1: Write down the production possibilities*

First list how many units of each good each country can produce if it uses all its resources for that good alone. This table is called a **production possibilities table**.

- Country - Good 1 (e.g., shirts) - Good 2 (e.g., computers)

- A - 10 - 5
- B - 6 - 6

### *Step 2: Compute the opportunity cost of one unit of each good*

The opportunity cost of one shirt equals "how many computers are given up to make one shirt".

Mathematically:

OC of 1 shirt = fracComputers given upShirts produced

Because the country can produce either all shirts *or* all computers, the "computers given up" is simply the total computers it could have made divided by the total shirts it could have made.

Do the same for the other good.

### *Step 3: Compare the opportunity costs between countries*

The country with the **lower** opportunity cost for a good has the comparative advantage in that good.

**Example 1 (Easy): Opportunity cost of shirts**

Country A can produce 10 shirts **or** 5 computers. Country B can produce 6 shirts **or** 6 computers. Find the opportunity cost of one shirt in each country and state who has the comparative advantage in shirts.

- **Calculate A's opportunity cost of one shirt**

OC_A, shirt= frac5 computers10 shirts = 0.5 computers per shirt

- **Calculate B's opportunity cost of one shirt**

OC_B, shirt= frac6 computers6 shirts = 1 computer per shirt

- **Compare** - 0.5 < 1, so A gives up fewer computers to make a shirt.

**Answer:** Country A has the comparative advantage in shirts because its opportunity cost (0.5 computers per shirt) is lower than Country B's (1 computer per shirt).

> **Remember:** The country with the *smaller* opportunity■cost number for a good holds the comparative advantage in that good.

### *Step 4: Find the opportunity cost of the other good (optional but useful)*

You can also compute the opportunity cost of a computer in each country. It will be the reciprocal of the shirt cost because the trade■off is the same line.

For Country A:

OC_A, computer= frac10 shirts5 computers = 2 shirts per computer

For Country B:

OC_B, computer= frac6 shirts6 computers = 1 shirt per computer

Now B has the lower cost for computers, so B has the comparative advantage in computers. This shows how each country can specialize in the good where its cost is lower.

**Example 2 (Medium): Wheat vs. cars**

Country X can produce 20 units of wheat **or** 10 cars. Country Y can produce 15 units of wheat **or** 15 cars. Determine comparative advantage for wheat and for cars.

- **Opportunity cost of 1 unit of wheat in X**

OC_X, wheat= frac10 cars20 wheat = 0.5 cars per wheat

- **Opportunity cost of 1 unit of wheat in Y**

OC_Y, wheat= frac15 cars15 wheat = 1 car per wheat

- **Compare wheat costs** - 0.5 < 1, so X has the comparative advantage in wheat.
- **Now compute the opportunity cost of 1 car** (reciprocals)
- For X:

OC_X, car= frac20 wheat10 cars = 2 wheat per car

- For Y:

OC_Y, car= frac15 wheat15 cars = 1 wheat per car

- **Compare car costs** - 1 < 2, so Y has the comparative advantage in cars.

**Answer:** Country X has the comparative advantage in wheat (0.5 cars per wheat), while Country Y has the comparative advantage in cars (1 wheat per car).

> **In daily life:** Think of two friends sharing chores. If Alex gives up only 5 minutes of video■games to wash dishes, while Sam must give up 15 minutes, Alex has the comparative advantage in washing dishes.

#### *Putting it all together*

- **List the maximum output of each good for every country.**
- **Divide the alternative good's maximum output by the good's maximum output** to get the opportunity cost.
- **Compare the numbers** - the smaller opportunity cost wins the comparative advantage.

When you follow these steps, you can quickly see which country should specialize in which product, setting the stage for beneficial trade.

> **Note:** Opportunity cost is always a *ratio*; it does not have units of "goods" alone, but "units of good B per unit of good A". This helps keep the comparison fair.

## 10.3 3. Gains from Trade

Imagine two friends, Alice and Ben, each can make two things: **shirts** and **computers**.

If they stay alone (no trade), each must use all of their own time to produce both goods.

When they trade, they can each focus on the good they are relatively better at - this is called **comparative advantage**.

> **Definition:** *Comparative advantage* means a country (or person) can produce a good at a lower **opportunity cost** than another.

> **Opportunity cost** is what you give up of one product to make one more unit of another product.

Why does this matter?

Because by specializing and then swapping some of what they make, both can end up with more shirts **and** more computers than they could ever have on their own. The extra amount is called the **gains from trade**.

#### *Step■by■step picture*

- **Find each country's opportunity cost.**

Suppose Alice can produce either 6 shirts **or** 3 computers in a week.

- If she makes one computer, she gives up $\frac{6\text{ shirts}}{3\text{ computers}} = 2$ shirts.
- So Alice's opportunity cost of a computer is 2 shirts.

Ben can produce either 4 shirts **or** 4 computers in a week.

- If he makes one computer, he gives up $\frac{4\text{ shirts}}{4\text{ computers}} = 1$ shirt.
- Ben's opportunity cost of a computer is 1 shirt.
- **Identify who has the lower opportunity cost for each good.**
- Shirts: Alice gives up $\frac{3}{6}=0.5$ computers per shirt, Ben gives up $\frac{4}{4}=1$ computer per shirt. Alice's cost is lower, so she has the comparative advantage in shirts.
- Computers: Ben's cost (1 shirt) is lower than Alice's (2 shirts), so Ben has the comparative advantage in computers.
- **Let each specialize.**
- Alice spends the whole week making shirts: 6 shirts, 0 computers.
- Ben spends the whole week making computers: 4 computers, 0 shirts.
- **Trade some of the output.**

They agree on a trade rate that lies between the two opportunity costs, for example **2 shirts for 1 computer**. This rate is fair because Alice would need to give up 2 shirts to get a computer on her own, while Ben would need to give up only 1 shirt.

- **Both end up with more of each good.**

After trade, Alice might give 4 shirts to Ben and receive 2 computers. She now has 2 shirts (6■4) and 2 computers.

Ben receives 4 shirts and keeps 2 computers (4■2). He now has 4 shirts and 2 computers.

Compare this with the **autarky** (no■trade) situation:

- Alice alone could have at most 6 shirts **or** 3 computers, never both.
- Ben alone could have at most 4 shirts **or** 4 computers.

Both are strictly better off after trade: they each have **both** goods, and the total amount of each good in the world (10 shirts, 4 computers) is the same, but the distribution is more useful to the individuals.

> **In daily life:** Think of a family where one parent is great at cooking and the other at fixing cars. If each only does their own chores, the house runs slowly. By letting the good cook prepare all meals and the mechanic handle all repairs, then swapping a few meals for a few car■fixes, the whole family enjoys more food and a working car.

### *Worked Example - Using the same numbers*

**Example 3 (Hard): Trade 4 shirts for 2 computers**

*Question:* Using the data from the previous illustration, assume Alice and Ben agree to trade **4 shirts for 2 computers**. Calculate the total output each country consumes after trade and compare it to their autarky consumption.

- **Determine each country's production before trade.**
- Alice: 6 shirts, 0 computers.
- Ben: 0 shirts, 4 computers.

Alice production = (6 shirts, 0 computers)

Ben production = (0 shirts, 4 computers)

- **Apply the trade agreement.**
- Alice gives 4 shirts to Ben and receives 2 computers.
- Ben gives 2 computers to Alice and receives 4 shirts.

Alice after trade = (6-4 shirts, 0+2 computers) = (2 shirts, 2 computers)

Ben after trade = (0+4 shirts, 4-2 computers) = (4 shirts, 2 computers)

- **Compare with autarky consumption.**
- Autarky for Alice: maximum 6 shirts **or** 3 computers, never both.
- Autarky for Ben: maximum 4 shirts **or** 4 computers, never both.

After trade, both have **both** goods:

- Alice: 2 shirts **and** 2 computers.
- Ben: 4 shirts **and** 2 computers.

Each country now consumes a bundle that was impossible without trade.

**Answer:**

Alice consumes 2 shirts and 2 computers; Ben consumes 4 shirts and 2 computers. Both are better off than in autarky because they enjoy positive amounts of both goods instead of being forced to specialize completely.

> **Remember:** Gains from trade arise only when the trade price lies **between** the two countries' opportunity costs. If the price is outside that range, one side would be worse off.

### *Why the world gains*

When every country follows this rule - specialize where they have comparative advantage and trade at mutually acceptable rates - the total world output stays the same, but the **allocation** becomes more efficient. People can enjoy a larger variety of goods, and overall welfare rises.

> **Note:** The total number of shirts and computers does not increase just by trading; the *distribution* changes so that each participant can reach a higher satisfaction level.

In summary, the **gains from trade** are the extra amount of each good that both parties can enjoy after they specialize according to comparative advantage and exchange at a fair rate. This simple principle underlies much of modern international economics and explains why countries engage in global markets.

## 10.4 4. Trade Balance and Current Account

Trade Balance and Current Account

When a country buys and sells goods and services with the rest of the world, it keeps a simple record: **exports** (what it sells abroad) and **imports** (what it buys from abroad). The difference between the two numbers tells us whether the country is a net seller or a net buyer.

> **Definition: Trade surplus** - when a country's exports are larger than its imports, giving a positive difference.

> **Definition: Trade deficit** - when imports exceed exports, giving a negative difference.

> **Definition: Trade balance** - the numerical result of "exports minus imports." It can be positive (surplus), negative (deficit), or zero (balanced).

Why does this matter? A trade surplus means the country is earning more foreign currency than it is spending, which can be a sign of strong production capacity. A trade deficit shows the country is spending more on foreign goods than it is earning, which may raise questions about debt or future currency pressure. In everyday life, think of a household budget: if you earn 3,000 a month and spend2,500 on groceries, rent, etc., you have a **budget surplus** of 500. If you spend3,500, you run a **budget deficit** of 500. The same idea applies to whole nations, just with dollars, euros, yen, etc., instead of personal income.

### *Recording Exports and Imports*

Every time a good leaves the country's border, customs officials record its value as an export. Conversely, every time a good enters, its value is recorded as an import. These values are usually expressed in the country's own currency or in a common reference currency such as US dollars. The national statistical agency adds up all export values for a given period (usually a year) and does the same for imports.

The **trade balance** formula is therefore

Trade Balance = Exports - Imports

If the result is positive, we call it a **surplus**; if negative, a **deficit**.

### *Worked Example 4 (Easy): Simple trade balance*

**Example 4 (Easy): Country C's trade balance**

Country C exported 150 million worth of goods and imported120 million. What is its trade balance?

- Write down the formula: Trade Balance = Exports - Imports
- Substitute the numbers: Trade Balance = 150 million - 120 million
- Perform the subtraction: Trade Balance = 30 million

**Answer:** Country C has a trade surplus of 30 million.

### *From Trade Balance to Current Account*

The **current account** is a broader measure. It adds the trade balance to two other components:

- **Services balance** - exports and imports of services such as tourism, banking, and consulting.
- **Net income** - earnings from investments abroad minus payments to foreign investors.

Thus,

Current■Account Balance = Trade Balance + Services Balance + Net Income

If the sum is positive, the country is a net lender to the world; if negative, a net borrower.

> **Note:** The trade balance is only one part of the current account. A country can have a trade deficit but still a current■account surplus if it earns a lot from services or investments.

### *Worked Example 5 (Medium): Adding services to the current account*

**Example 5 (Medium): Country D's overall current■account balance**

Country D's trade balance is -30 million (a deficit) and its services surplus is10 million. What is its overall current■account balance?

- List the known components: trade balance = -30 million, services balance = +10 million. Assume net income is 0 for this example.
- Write the current■account formula: Current■Account Balance = Trade Balance + Services Balance + Net Income
- Substitute the numbers: Current■Account Balance = (-30 million) + 10 million + 0
- Add the values: Current■Account Balance = -20 million

**Answer:** Country D has a current■account deficit of 20 million.

### *Interpreting the Numbers*

A positive current■account balance means the country is sending more money abroad than it receives, which can lead to an accumulation of foreign reserves. A negative balance may require financing through borrowing or attracting foreign investment.

> **In daily life:** Imagine you run a small online shop that sells handmade crafts abroad. If you earn 2,000 from overseas sales (exports) but spend1,200 on imported materials (imports), you have a 800 trade surplus. If you also earn300 from offering online design services (services balance) and pay 100 in fees to a foreign payment processor (net income), your overall current■account balance is1,000.

### *Common Pitfalls*

> **Common mistake:** Thinking that a trade deficit is always bad. In reality, a deficit can reflect strong consumer demand and the ability to borrow cheaply. The key is whether the deficit is sustainable and financed by healthy capital inflows.

### *Quick Recap*

- **Exports** = goods/services sold abroad.
- **Imports** = goods/services bought from abroad.
- **Trade balance** = exports - imports.
- **Trade surplus** = positive trade balance; **trade deficit** = negative trade balance.
- **Current account** = trade balance + services balance + net income.

> **Remember:** The trade balance is a snapshot of the flow of goods; the current account adds services and investment income to give a fuller picture of a country's economic interactions with the world.

## 10.5 5. Exchange Rates - How They Are Set

When we talk about the price of one country's money in terms of another country's money, we are talking about an **exchange rate**.

Think of it like the price you see on a shop sign: the sign tells you how many units of one good you must give up to get one unit of another good. In a currency market the "goods" are different currencies.

> **Definition:** *Nominal exchange rate* - the number of units of foreign currency that one unit of domestic currency can buy, or vice■versa, without adjusting for price changes inside the countries.

> **Definition:** *Real exchange rate* - the nominal rate adjusted for the relative price levels (inflation) of the two countries. It tells us how many foreign■country goods a domestic■country good can buy after we take inflation into account.

Why do we need both? The nominal rate is what you see on a news ticker or on your bank statement. The real rate is what matters for trade decisions, because it reflects the true purchasing power of money across borders.

### *The supply■and■demand picture*

Currency markets work like any other market where buyers and sellers meet.

- **Demand for a foreign currency** comes from people who want to buy that foreign good, travel abroad, or invest in that country.
- **Supply of a foreign currency** comes from people who are selling that currency to buy something else, such as domestic goods or foreign assets.

When demand is high relative to supply, the price of the foreign currency rises (its value goes up). When supply is high relative to demand, the price falls.

The basic equation that captures this relationship is simply the equality of quantity demanded and quantity supplied at the market■clearing exchange rate:

Q^D_foreign(E) = Q^S_foreign(E)

where \(E\) is the nominal exchange rate (e.g., dollars per euro). The functions \(Q^{D}\) and \(Q^{S}\) depend on many factors - interest rates, expectations about future rates, and the relative price of goods.

### *Worked Example 6 (Easy): Converting a quoted rate*

**Question:** If 1 US dollar buys 0.85 euros, what is the euro price of 1 dollar?

- Identify the quoted rate: 1 USD = 0.85 EUR.
- To find the price of 1 USD in euros, we already have it directly: the euro price is 0.85 EUR per USD.
- Write the answer as a simple statement.

Euro price of 1 USD = 0.85 EUR/USD

**Answer:** 1 USD = **0.85 EUR**.

> **In daily life:** When you travel, the amount of foreign cash you receive for each of your home dollars is exactly this kind of nominal exchange rate.

### *From nominal to real: adjusting for inflation*

Suppose the nominal rate stays the same, but prices in the United States rise faster than in Europe. Then each dollar buys fewer euros *and* each euro can buy relatively more US goods. The real exchange rate falls, meaning domestic goods become less competitive abroad.

The formula for the real exchange rate (RER) is:

RER = E times frac{P_foreign}{P_domestic}

where \(P_{\text{foreign}}\) and \(P_{\text{domestic}}\) are the price indices (inflation■adjusted) of the two countries.

### *Worked Example 7 (Hard): Using purchasing■power parity*

**Question:** The spot exchange rate is 1.2 dollars per pound. If inflation in the US is 3 % and in the UK is 1 %, what should the exchange rate be one year later according to purchasing■power parity (PPP)?

- Write down the initial nominal rate: \(E_0 = 1.2\ \text{USD/GBP}\).
- PPP says that the future nominal rate \(E_1\) should adjust by the relative inflation rates:

E_1 = E_0 times frac{1 + pi_US}{1 + pi_UK}

where \(\pi_{\text{US}} = 0.03\) and \(\pi_{\text{UK}} = 0.01\).

- Plug the numbers in:

E_1 = 1.2 \times \frac{1 + 0.03}{1 + 0.01}

= 1.2 \times \frac{1.03}{1.01}

= 1.2 \times 1.0198

\approx 1.224 \ \text{USD/GBP}

**Answer:** One year later the exchange rate should be about **1.22 USD per pound** if PPP holds.

> **Note:** PPP is a long■run idea. In the short run, exchange rates can move for many other reasons - interest■rate changes, political news, or speculation.

### *Why the model matters*

Understanding that exchange rates are set by supply and demand helps you see why a country's central bank sometimes steps in. If a currency is appreciating too fast, the bank may sell its own currency to increase supply and bring the rate down, protecting exporters. Conversely, to boost exports, a bank might buy its own currency, reducing supply and letting the rate fall.

> **Remember:** The nominal exchange rate tells you the "price" of one currency in another; the real exchange rate tells you the "price" of goods after accounting for inflation. Both are shaped by the forces of supply and demand, just like any other market.

By keeping these ideas clear - what the rate is, how it is determined, and how inflation changes its meaning - you will be able to follow news about currency moves, understand trade statistics, and evaluate the impact of policy decisions on everyday prices.

## 10.6 6. Globalization Effects

Globalization means that countries are more connected through trade, investment, and the flow of ideas.

When borders become less of a barrier, firms can sell their products to many more customers, and people can buy goods that are made far away. This creates both good things and new problems. Let's look at the two sides step by step.

> **Definition:** *Globalization* - the process by which economies, societies, and cultures become integrated through international trade, investment, and information exchange.

### *Why lower prices are a benefit*

If a company in Country A can produce a widget cheaper than a company in Country B, the world market will tend to buy the cheaper widget. Consumers in both countries enjoy the lower price because the cheap producer can sell more units and still make a profit.

Think of it like a school cafeteria that decides to buy fruit from a nearby farm instead of importing it from far away. The farm's fruit costs less because it does not have to pay for a long truck ride. The cafeteria can charge students less for a banana, and everyone is happier.

### *Technology transfer - learning from each other*

When firms operate in many countries, they bring the tools and methods they use at home. This "technology transfer" spreads new ideas, such as more efficient machines or better ways to manage a supply chain.

Imagine a basketball coach who moves from one school to another. He brings new drills that his former team used to win games. The new school's players learn those drills and improve, even though the coach never invented them. In the same way, a car manufacturer that opens a plant in a new country may introduce advanced robotics that local factories had never used before.

> **In daily life:** *Technology transfer* is like borrowing a friend's recipe and then using it to cook a better meal at home.

### *Challenges: job displacement*

When production moves to a place where labor is cheaper, some workers in the original country lose their jobs. This is called *job displacement*. It does not mean that the whole economy is worse off; the displaced workers can later find new jobs in other sectors, but the transition can be painful.

A simple analogy is a school that decides to replace its old computer lab with tablets. The computer technicians who maintained the old PCs may no longer be needed, at least for that task. They must learn new skills to stay employed.

### *Inequality - the gap can widen*

Globalization can increase the income gap between high■skill workers (who often benefit from new technologies) and low■skill workers (who may face competition from cheaper foreign labor). This is called *inequality*.

Think of a class where some students get a new set of textbooks that make learning faster, while others keep the old books. Over time, the students with the new books may pull ahead, creating a larger gap in grades.

> **Note:** Inequality does not automatically rise with globalization; government policies such as education and social safety nets can reduce the gap.

### *Worked example - weighing the benefit of lower prices against job loss*

**Example 8 (Intermediate):** *A country imports cheap shoes from abroad, lowering the price for consumers but causing a local shoe factory to close.*

- **Calculate the consumer savings.**

The domestic price of a pair of shoes was 80, and the imported price is50. If 1 million pairs are bought each year, the total saving is

Saving= (80-50)times 1,000,000 = 30,000,000 dollars.

- **Estimate the cost of job loss.**

The local factory employed 2 000 workers, each earning 30 000 per year. The total wage cost that disappears is

Wage loss= 2,000 times 30,000 = 60,000,000 dollars.

- **Compare the two numbers to see the net effect.**

Subtract the consumer saving from the wage loss:

Net loss= 60,000,000 - 30,000,000 = 30,000,000 dollars.

**Answer:** In this simplified scenario, the economy experiences a net loss of 30 million because the cost of job displacement exceeds the consumer savings from lower shoe prices. Policymakers would need to consider retraining programs or other measures to offset the loss.

### *How to think about the trade■off*

The example shows that the **benefits** (lower prices) and **challenges** (job loss) must be weighed together. If the country can quickly retrain the 2 000 workers for higher■skill jobs, the net loss could disappear. This is why many governments invest in education and vocational training when they open up to global markets.

> **Remember:** The overall impact of globalization depends on how well a society manages the transition from old jobs to new opportunities.

### *Summing up the effects*

- **Lower prices** increase consumer purchasing power, allowing people to buy more with the same income.
- **Technology transfer** spreads innovation, raising productivity across many sectors.
- **Job displacement** can hurt workers in industries that move abroad, but it also frees resources for new, potentially higher■value activities.
- **Inequality** may grow if the gains are captured mainly by those with the skills to use new technology, unless policies intervene.

By understanding each piece, students can see why globalization is a complex phenomenon: it is not simply "good" or "bad," but a mix of forces that shape economies and societies. Careful policy design - such as investing in education, providing safety nets, and encouraging fair trade - helps capture the benefits while reducing the downsides.

## Terminology at a glance

| Term | Meaning |
|---|---|
| Comparative advantage | The ability of a country to produce a good at a lower opportunity cost than another country. |

| Term | Meaning |
|---|---|
| Opportunity cost | The amount of the next best alternative that must be given up to produce one more unit of a good. |
| Trade surplus | When a country's exports exceed its imports, resulting in a positive trade balance. |
| Trade deficit | When a country's imports exceed its exports, resulting in a negative trade balance. |
| Nominal exchange rate | The rate at which one currency can be exchanged for another without adjusting for price level differences. |
| Real exchange rate | The nominal exchange rate adjusted for inflation, showing the relative purchasing power of two currencies. |
| Gains from trade | The increase in total output and consumption that results when countries specialize according to comparative advantage and trade. |
| Globalization | The process of increasing economic, cultural, and political integration across countries through trade, investment, and information flow. |

## Practical application

Exercise 10.1:

Exercise 10.2:

Exercise 10.3:

Exercise 10.4:

Exercise 10.5:

Exercise 10.6:

## Checklist

- [ ] Explain the concept of comparative advantage
- [ ] Identify a country's comparative advantage using simple data
- [ ] Interpret a trade balance and what it indicates about an economy
- [ ] Describe how exchange rates are determined and how they influence trade
- [ ] Discuss the main economic effects of globalization

> Educational note: this book is a study aid. Check important facts with your lecturer or textbook.